Algebra: Chapter 0 - Self Study

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Abstract

This text contains my notes on Algebra: Chapter 0, by Paolo Aluffi. I read this book independently, starting over winter break 2024. My notes largely paraphrase the text, and include few examples. I have also completed every exercise in an external document. I also completed every problem set from this Algebra Course at Johns Hopkins University.

Contents

Chapter 1

Preliminaries: Set Theory and Categories

1.1 Naive Set Theory

1.1.1 Sets

A set is a collection of objects. Sets contain elements, which are claimed to exist. Equality between sets is defined by equivalent elements, meaning repeating elements or re-ordering elements does not change the content of the set. Some sets are written by listing all elements they contain, such as $A = \{1, 2, 3\}$, but this is not possible in general.

Definition 1.1: Set-Builder

Let S be a set, and let P be a statement. Then $\{s \in S \mid P(s)\}$ is the set of all $s \in S$ satisfying $P(s)$.

For example,

$$\{n \in \mathbb{Z} \mid \exists a \in \mathbb{Z}\, (n = 2a)\}$$

would be the set of all even integers. In general, we will write this set in the simpler notation $\{2n \mid n \in \mathbb{Z}\}$, and define these notations to be equivalent.

Definition 1.2: Singleton Set

A Singleton set is a set containing exactly one element.

For example, $\{1\}$, $\{x\}$, and $\{\varnothing\}$ are singleton sets.

Since we will make extensive use of quantifiers in this book, it's prudent to recall their definitions as well.

Definition 1.3: Quantifiers

The quantifier $\forall$ means "for all", the quantifier $\exists$ means "there exists", and the quantifier $\exists!$ means "there exists exactly one", or "there exists a unique". I will notate statements formed with quantifiers as follows:

$$\forall x \in S\, (P(x))\,.$$

This reads "for all $x \in S$, $P(x)$ is true", and can be either true or false. I will also sometimes use the notation

$$\forall\, (x \in S)\, (P(x))\,.$$

1.1.2 Inclusion of Sets

Definition 1.4: Subset

Let S, T be sets. We say that S is a subset of T, denoted $S \subseteq T$, if all elements of S are elements of T. Formally,

$$S \subseteq T \iff \forall\,(s \in S)\,(s \in T)\,.$$

A proper subset S of T is a set $S \subseteq T$ such that there exists some $t \in T$ such that $t \notin S$. This is denoted $S \subsetneq T$ (or $S \subset T$, but I won't use this notation here since the book doesn't), and is formally expressed as

$$S \subsetneq T \iff \forall s \in S(s \in T) \wedge \exists t \in T\,(t \notin S)\,.$$

Definition 1.5: Cardinality

The notation $|S|$ denotes the cardinality of the set S, or the number of things in S. For example, $|\{1, 2\}| = 2$. If S is infinite, we will write $|S| = \infty$.

I'm unsure if we will introduce infinite cardinalities later on, but I'm going to keep with the book's notation and write $|S| = \infty$ if a set is infinite. One thing to note (without proof) is that if S, T are finite, then

$$S \subseteq T \implies |S| \leq |T|\,.$$

Definition 1.6: Power Set

The power set $\mathscr{P}(S)$ of a set S is the set of all subsets of S. It has cardinality $|\mathscr{P}(S)| = 2^{|S|}$ if S is finite.

1.1.3 Operations Between Sets

Now that we have some understanding of what sets are, we can begin considering operations between sets. The important ones we will consider are

- the union $\cup$,
- the intersection $\cap$,
- the difference $\setminus$,
- the disjoint union $\amalg$,
- the Cartesian product $\times$,
- and the quotient by an equivalence relation.

I already know the first 3 and the Cartesian product, so I won't even bother. I will note the following definition, however.

Definition 1.7: Compliment

The compliment of a set S in another set T is simply $T \setminus S$.

The operation $\amalg$ is very strange, and we will motivate it further before formally defining it later. We will also do the same with $\times$ for some reason.

1.1.4 Disjoint Unions, Products

The idea behind the disjoint union is that $S \amalg T$ should be produced by the regular union by disjoint "copies" of S and T. This definition will be made more formal with the introduction of the cartesian product, but note that for finite S, T, that $|S \amalg T| = |S| + |T|$.

Definition 1.8: Cartesian Product

A cartesian product $S \times T$ of two sets S, T is defined as the set of all ordered pairs (s, t) such that $s \in S$ and $t \in T$.

The book notes, rather interestingly, that an ordered pair (s, t) can be defined rigorously as the set $\{s, \{s, t\}\}$. One thing to note about the Cartesian product is that for S, T finite sets, $|S \times T| = |S|\,|T|$.

Using the Cartesian product, we can define the $\amalg$ operator more rigorously. Let $S' = \{0\} \times S$, and $T' = \{1\} \times T$. Clearly, $S' \cap T' = \varnothing$, but S' and T' are pretty much copies of S and T. We then define $S \amalg T := S' \cup T'$. We will make this definition more precise later, but this suffices for now.

It makes sense to generalize $\cup$, $\cap$, $\times$, and $d\cup$ to families of sets. For example, given sets $S_1, \ldots, S_n$, we write

$$\bigcap_{i=1}^{n} S_i = S_1 \cap \cdots \cap S_n.$$

However, these operations are necessarily binary, and it's not apparent that $S_1 \times (S_2 \times S_3) = (S_1 \times S_2) \times S_3$, for example. The language of functions will help us show this does hold, later on, so for now we can gloss over this fact.

More generally, for a set of sets $\mathscr{P}$, we can write

$$\bigcup_{S \in \mathscr{P}} S, \qquad \bigcap_{S \in \mathscr{P}} S, \qquad \coprod_{S \in \mathscr{P}} S, \qquad \prod_{S \in \mathscr{P}} S$$

for the respective operation on all sets in $\mathscr{P}$. There are important subtlties along with each definition. For example, if every $S \in \mathscr{P}$ is nonempty, is

$$\prod_{S \in \mathscr{P}} S \neq \varnothing?$$

It turns out the issue is very strangle if $\mathscr{P}$ is infinite, and amounts to the axiom of choice.

1.1.5 Equivalence Relations, Partitions, Quotients

A relation R on a set S is a subset $R \subseteq S \times S$. If $(a, b) \in R$, we say a and b are related by R, and write $a\,R\,b$. Usually, we use symbols such as $\sim$ for relations, instead of R. An equivalence relation is a relation that is symmetric, transitive, and reflexive. An equivalence class $[a]_\sim$ is the set of all $b \in S$ such that $b \sim a$. The set of equivalence classes partitions S. Every partition determines an equivalence relation.

Definition 1.9: Quotient Set

Let S be a set and let $\sim$ be an equivalence relation on S. The quotient of the set S with respect to the equivalence relation $\sim$ is the set

$$S/\!\sim\; := \{[a]_\sim \mid a \in S\}$$

of equivalence classes of S with respect to $\sim$.

1.2 Functions

1.2.1 Definition

In this book, we will often try to understand structures, as well as the ways in which they interact. Sets interact via functions, which can be formalized as a relation.

Definition 1.10: Function

A function is a relation $f \subseteq A \times B$ such that

$$\forall a \in A \exists! b \in B \text{ such that } (a, b) \in f.$$

We write $f : A \to B$ to mean $f \subseteq A \times B$, and $f(a) = b$ or $a \mapsto b$ to mean $(a, b) \in f$. The collection of all functions $f : A \to B$ forms a set, denoted B^A.

Every set comes with a function $\mathrm{id}_A : A \to A$ defined by $a \mapsto a$. More generally, for any set $S \subseteq A$ there is an inclusion map $S \to A$ that simply maps $s \mapsto s$.

If $S \subseteq A$, we write $f(S) := \{b \in B \mid \exists a \in S(b = f(a))\}$, or more commonly, $f(S) = \{f(s) \mid s \in S\}$. The largest subset $f(A)$ is called *image* of f, and is also denoted $\operatorname{im} f$.

Definition 1.11: Restriction

Let $f : A \to B$ be a function. The *restriction* of f to some subset $S \subseteq A$, denoted $f|_S$, is the function $f|_S : S \to B$ defined by $s \mapsto f(s)$.

Equiavlently, if $\iota : S \to A$ is the inclusion map, then $f|_S = f \circ \iota$.

1.2.2 Examples: Multisets, Indexed Sets

I don't know if I mentioned it in my notes, but multisets were briefly mentioned earlier as sets that allowed multiple entries of the same element. A rigorous definition of a multiset may be formed from a function $m : A \to \mathbb{Z}^+$. If a multiset consists of the element a, and is repeated n times, then $m : a \mapsto n$. A is defined as the set of all elements in the multiset. This function encodes all information we need about the multiset.

Another example of a function is indices. If we say "let $a_1, \ldots, a_n$ be integers," what we really mean is "consider a function $a : \{1, \ldots, n\} \to \mathbb{Z}$," with the understanding that a_i is shorthand for the value of $a(i)$. Although we still think of an indexed set $\{a_i\}_{i \in I}$ as a set who's elements are indexed by i ranging over I, the proper definition is done by this function. These distinctions will be important later, but in general are not that important.

1.2.3 Composition of Functions

Functions can be composed. If $f : A \to B$ and $g : B \to C$ are functions, then $g \circ f$, defined by

$$\forall a \in A\,((g \circ f)(a) := g(f(a)))$$

is also a function. We will often represent this with a diagram, such as

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g \circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd}$$

A diagram is said to commute, or is said to be a commutative diagram, if all possible paths starting from A travel to C.

Function composition is associative. Graphically,

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & C \arrow[r, "h"] & D \end{tikzcd}$$

The identity map is also an identity under function composition. That is, $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$.

1.2.4 Injections, Surjections, Bijections

Definition 1.12: Injection, Surjection, Bijection

A function $f : A \to B$ is *injective* if

$$\forall a \in A, \forall b \in A : a \neq b \implies f(a) \neq f(b).$$

A function $f : A \to B$ is said to be *surjective* if

$$\forall b \in B, \exists a \in A : f(a) = b.$$

A function $f : A \to B$ is said to be *bijective* if it is both injective and surjective.

If a function is injective, we write $f : A \hookrightarrow B$. If a function is surjective, we weite $f : A \twoheadrightarrow B$. If a function is bijective, we write $f : A \xrightarrow{\sim} B$, and say $A \cong B$.

Definition 1.13: Isomorphic Sets

Sets A and B are called isomorphic if there exists a bijection $f : A \to B$.

If two sets are isomorphic, then we have $|A| = |B|$. This allows us to make a bit more sense of the disjoint union $A \amalg B$. We want our copies A', B' to be isomorphic to A and B. Our proposal earlier obviously works.

1.2.5 Injections, Surjections, Bijections: Second Viewpoint

Note that if $f : A \to B$ is a bijection, we can define a map $g : B \to A$ as $b \mapsto a$ iff $f(a) = b$. Since f is a bijection, b is a well-defined function. Note that $f \circ g = \mathrm{id}_A$ and $g \circ f = \mathrm{id}_B$. We say that g is the inverse of f, denoted f^{-1}.

Proposition 1.1

Let $A \neq \varnothing$, and let $f : A \to B$ be a function. Then

1. f has a left inverse if and only if it is injective.
2. f has a right inverse if and only if it is surjective.

Proof. (1) If $f : A \to B$ has a left inverse, then there exists $g : B \to A$ such that $g \circ f = \mathrm{id}_A$, Let $a_1 \neq a_2$ be elements of A. We have

$$g(f(a_1)) = \mathrm{id}_A(a_1) = a_1 \neq a_2 = \mathrm{id}_A(a_2) = g(f(a_2)).$$

Therefore, f is injective.

Now let $f : A \to B$ be injective. Fix some arbitrary $s \in A$, and define

$$g(b) := \begin{cases} a, & \exists a \in A : b = f(a) \\ s, & b \notin f(A). \end{cases}$$

Since f is injective, g is a function. Now note that if $a \in A$, then $g(f(a)) = a$ by definition, so $g \circ f = \mathrm{id}_A$.

The proof of (2) is done in the exercises. ■

A corollary of this is that a function $f : A \to B$ has an inverse if and only if it is a bijection. It's interesting that the left and right inverses of f end up being the same, but we explore this more later. Also note that by our proof, injections have many left inverses, and similarly surjections have many right inverses. These are called *sections*.

Proposition 1.1 shows that even if we did not have elements, we could still relate functions by their relationship to other functions via inverses. We will explore this *much* later during the section on categories.

Although the standard notation for the inverse of a bijection is f^{-1}, we will use this notation for other functinos as well. If $f : A \to B$ is any function and $T \subseteq B$, then $f^{-1}(T)$ denotes the subset of A consisting of elements that map to T. That is,

$$f^{-1}(T) := \{a \in A \mid f(a) \in T\}.$$

Definition 1.14: Fiber

Let $f : A \to B$ be a function, and let $T = \{q\} \subseteq B$ be a singleton set. Then $f^{-1}(T)$, abbreviated $f^{-1}(q)$, is called the *fiber* of f over q.

A function f is bijective if for all $b \in B$, the fibre of f over b is nonempty, and these fibers are singletons. In this case, the notation f^{-1} matches with the common notation.

1.2.6 Monomorphisms and Epimorphisms

There is ANOTHER way to express injectivity and surjectivity, and even though it appears more complex, somehow it's more basic.

Definition 1.15: Monomorphism, Epimorphism

A function $f : A \to B$ is called a *monomorphism*, or is said to be *monic*, if for all sets X and all functions $\alpha', \alpha'' : X \to A$,

$$f \circ \alpha' = f \circ \alpha'' \implies \alpha' = \alpha''.$$

A function $f : A \to B$ is called an *epimorphism*, or is said to be *epic*, if for all sets X and all functions $\beta', \beta'' : A \to X$,

$$\beta' \circ f = \beta'' \circ f \implies \beta' = \beta''.$$

Proposition 1.2

A function is injective if and only if it is a monomorphism. A function is surjective if and only if it is an epimorphism.

Proof. The proof that surjective is equivalent to epimorphism is done in the exercises. Let $f : A \to B$ be injective. By proposition 1.1, there exists a left inverse $g : B \to A$. Suppose $\alpha', \alpha'' : X \to A$ are functions and

$$f \circ \alpha' = f \circ \alpha''.$$

We have

$$\alpha' = (g \circ f) \circ \alpha' = g \circ (f \circ \alpha') = g \circ (f \circ \alpha'') = (g \circ f) \circ \alpha'' = \alpha''.$$

Therefore, $\alpha' = \alpha''$.

Now assume f is a monomorphism. Choose $X = \{p\}$ a singleton. Then all maps $\alpha : X \to A$ are determined by a choice of $a \in A$ such that $p \mapsto a$. Moreover, $\alpha' : p \mapsto a'$ and $\alpha'' : p \mapsto a''$ have $\alpha' = \alpha''$ if and only if $a' = a''$. Therefore, if

$$f \circ \alpha'(p) = f \circ \alpha''(p) \implies \alpha' = \alpha'',$$

then we have

$$f \circ \alpha'(p) - f \circ \alpha''(p) \implies a' = a''.$$

By definitionm

$$f(a') = f(a'') \implies a' = a''.$$

This is the contrapositive of definition 1.12, so we are done. ■

1.2.7 Basic Examples

Let A, B be sets. Then there exist *natural projections* π_A, π_B such that

$$\begin{tikzcd} & A \times B \arrow[dl, "\pi_A"'] \arrow[dr, "\pi_B"] & \\ A & & B \end{tikzcd}$$

defined by

$$\pi_A(a, b) := a, \qquad \pi_B(a, b) := b.$$

These maps are clearly surjective. Similarly, there are natural injections

$$\begin{tikzcd} A \arrow[dr, hook] & & B \arrow[dl, hook'] \\ & A \amalg B & \end{tikzcd}$$

obtained by the inclusion maps $A, B \hookrightarrow A \amalg B$.

Another example is as follows. Let $\sim$ be an equivalence relation on A. Then there exists a clearly surjective canonical projection $A \twoheadrightarrow A/\!\sim$ defined by $a \mapsto [a]_\sim$.

1.2.8 Canonical Decomposition

The reason we focus on injective and surjective maps is because any function can be constructed from them. To see this, let $f : A \to B$ be any function, and note that f determines an equivalence relation $\sim$ on A, where for any $a', a'' \in A$,

$$a' \sim a'' \iff f(a') = f(a'').$$

Theorem 1.1

Let $f : A \to B$ be a function, and let $\sim$ be the relation on A defined such that $a \sim b$ if and only if $f(a) = f(b)$. Then f decomposes as follows:

$$\begin{tikzcd} A \arrow[r, two heads] \arrow[rrr, bend left, "f"] & (A/\sim) \arrow[r, "\sim", "\tilde{f}"'] & f(A) \arrow[r, hook] & B \end{tikzcd}$$

where the function $A \twoheadrightarrow A/\sim$ is the canonical projection, the function $f(A) \hookrightarrow B$ is the inclusion map, and the bijection $\tilde{f}$ is defined by

$$\tilde{f}([a]_\sim) \coloneqq f(a).$$

Proof. Since the formula for $\tilde{f}$ immediately shows the diagram commutes, it suffices to show $\tilde{f}$ is a bijection. First, we show $\tilde{f}$ is well defined. Let $[a]_\sim = [b]_\sim$. We then have $a \sim b$, and by definition, we have $f(a) = f(b)$. Now we show $\tilde{f}$ is a bijection. If $\tilde{f}([a]_\sim) = \tilde{f}([b]_\sim)$, then $f(a) = f(b)$ by definition of $\tilde{f}$, and thus $a \sim b$, and thus $[a]_\sim = [b]_\sim$. Therefore, $\tilde{f}$ is injective. Let $b \in f(A)$. It follows by definition that there exists $a \in A$ such that $f(a) = b$. We have $\tilde{f}([a]_\sim) = f(a) = b$. Therefore, $\tilde{f}$ is surjective, and thus a bijection. ■

This is actually pretty much the first isomorphism theorem of sets. The theorem will re-emerge throughout the text, with a new skin each time.

1.2.9 Clarification

Although it is easy to find isomorphic copies A', B' of A and B sets, it's not guarenteed that all copies are suitable for definining the disjoint union $A \amalg B$. The resulting operation $A \amalg B$ is actually not well-defined insofar, but it's possible to see (in the exercises) that it is well-defined *up to isomorphism*. That is, any two choices of the copies A' and B' are isomorphic to each other. This can be seen as well by looking at cartesian products and quotients. What we actually want to explore is the relationship of these sets with other sets, which will be made clear through category theory.

1.3 Categories

Category theory is known as "abstract nonsense", which is just fucking awesome. Categories are less focused on studying sets, as they are about studying structures in general, and specifically maps between structures. Categories may behave like sets in some ways, and in fact there exist functions between categories called functors, but this is imprecise.

1.3.1 Definition

Categories have complex definitions. There are two important aspects of a category, a collection of objects and a set of morphisms between these objects. Note that the collection of objects is NOT a set. We want there to exist a category of all sets, but there does not exist a set of all sets, as seen by Russel's Paradox. However, we can define such a structure as a *class*, which does not need to have the precise structure a set does. There *is* a class of all sets.

Throughout this book, the notion of a class will not behave differently to that of a set.

Definition 1.16: Category

A category $\mathbf{C}$ consists of

- a class $\mathrm{Obj}(\mathbf{C})$ of *objects* in the category, and
- for every two objects A, B of $\mathbf{C}$, a set $\mathrm{Hom}_{\mathbf{C}}(A, B)$ of *morphisms*, with the following properties:
  - For every object A of $\mathbf{C}$, there exists the morphism $\mathbb{1}_A \in \mathrm{Hom}_{\mathbf{C}}(A, A)$, called the *identity* on A.
  - Two morphisms $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ determine a morphism $gf \in \mathrm{Hom}_{\mathbf{C}}(A, C)$. That is, for every triple of objects A, B, C of $\mathbf{C}$, there exists a function of sets[a]
  $$\mathrm{Hom}_{\mathbf{C}}(A, B) \times \mathrm{Hom}_{\mathbf{C}}(B, C) \to \mathrm{Hom}_{\mathbf{C}}(A, C),$$
  and the image of the pair (f, g) under this map is denoted gf.
  - The composition law is associative: if $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$, $g \in \mathrm{Hom}_{\mathbf{C}}(B, C)$, and $h \in \mathrm{Hom}_{\mathbf{C}}(C, D)$, then
  $$(hg)f = h(gf).$$
  - The identity morphisms are identities with respect to composition; that is, for all $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$,
  $$f\mathbb{1}_A = f, \qquad \mathbb{1}_B f = f.$$

[a] A "function of sets" will be the language used to imply a function has a set as its domain and codomain, rather than an arbitrary object. This is different from a set-function.

The intuition behind this definition is $\mathrm{Obj}(\mathbf{C})$ generalizes the notion of a set of things in a consistent way, which allows us to talk about the structure of the objects of $\mathbf{C}$ by considering morphisms $\mathrm{Hom}_{\mathbf{C}}(A, B)$ for A, B in $\mathbf{C}$. The morphisms can be thought of as functions from sets to sets.

For this reason, morphisms $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ will often be denoted $f : A \to B$ when the category $\mathbf{C}$ is known. We will often also drop the subscript $\mathbf{C}$ and write $\mathrm{Hom}(A, B)$ if the category is implied.

Definition 1.17: Endomorphism

An *endomorphism* is a morphism $f \in \mathrm{Hom}_{\mathbf{C}}(A, A)$; that is, a map from an object to itself. The set $\mathrm{Hom}_{\mathbf{C}}(A, A)$ is denoted $\mathrm{End}_{\mathbf{C}}(A)$.

Denoting morphisms using the usual function-set notation $f : A \to B$, even though A, B do not need to be sets, allows us to create diagrams of morphisms in any category. For example, the commutative diagram for the first isomorphism theorem would look like the following:

$$\begin{tikzcd}
G \arrow[r, "\phi"] \arrow[d, "\eta"'] & \phi(G) \\
G/\ker\phi \arrow[ur, "\psi"] &
\end{tikzcd}$$

Diagrams are actually possible objects of categories. The precise definition of such a diagram is a set of objects in a category $\mathbf{C}$, along with morphisms of theses objects. We say the diagram commutes if every method of traversing it leads to the same destination. The specifics of the visual representation are of course irrelevant in the precise definition.

1.3.2 Examples

Note that most of definition 1.16 concerned morphisms, not the objects of $\mathrm{Obj}(\mathbf{C})$. It's fair to say that the morphisms of a category are the important constituents. However, we still want to think of a category in terms of its objects. For example, oftentimes the *category of sets* is often discussed. However, the emphasis should be on determining what kind of morphisms are being considered. For example, what are morphisms of sets going to be in the category of sets? In general, we take the category of sets to have morphisms identified by functions. In many cases, the morphisms are fairly obvious, as they are here. However, this is not always the case.

Definition 1.18: Category of Sets

The category of sets, denoted **Set**, is the category whos objects $\mathrm{Obj}(\mathbf{Set})$ are sets, and whos morphisms $\mathrm{Hom}_{\mathbf{Set}}(A, B)$ are the set of all functions of sets $f : A \to B$.

The notation I am using in my notes for categories, which is boldfaced letters, is commonly used, but is not the same as the notation for the book, which uses the sans-serif font Set for its categories. I kind of like the notation and may start using it.

Here is another example of a category. Suppose S is a set and $\sim$ is a relation on S. We can model this information with a category. Let $\mathbf{C}$ have $\mathrm{Obj}(\mathbf{C}) := S$ and

$$\mathrm{Hom}_{\mathbf{C}}(a, b) = (a, b) \in S \times S \iff a \sim b, \qquad \text{and} \qquad \mathrm{Hom}_{\mathbf{C}}(a, b) = \varnothing \iff a \nsim b.$$

There are very few morphisms in $\mathbf{C}$. However, we can show this encoding of all the information inside of a relation forms a category. Recall that if $\sim$ is a relation, then for all $a \in S$, $a \sim a$. Therefore, $\mathrm{End}(a) \neq \varnothing$, satisfying the first property of definition 1.16. We can define this element as $\mathbb{1}_a$.

To show composition holds, let $a, b, c \in S$. That is, let $a, b, c \in \mathrm{Obj}(\mathbf{C})$. Let $f \in \mathrm{Hom}(a, b)$ and $g \in \mathrm{Hom}(b, c)$. It follows that $\mathrm{Hom}(a, b), \mathrm{Hom}(b, c)$ are nonempty, and thus $a \sim b$ and $b \sim c$. By the transitive property of $\sim$, we have $a \sim c$, so $\mathrm{Hom}(a, c) \neq \varnothing$. Let $gf := (a, c) \in \mathrm{Hom}(a, c)$. That is, let $(b, c) \circ (a, b) := (a, c)$. This satisfies the composition law.

To show the operation is associative, let $f = (a, b)$, $g = (b, c)$, and $h = (c, d)$. Note that $gf = (a, c)$ and $hg = (b, d)$. We have $h(gf) = (a, d) = (hg)f$. Also note that $\mathbb{1}_a$ is obviously an identity, so $\mathbf{C}$ is a category.

There are many constructions of this category, such as $(\mathbb{Z}, =)$ or $(\mathbb{R}, \leq)$. A category where the only morphisms are identities is called a *discrete* category.

Definition 1.19: Discrete Category

A category $\mathbf{C}$ is called discrete if the only morphisms are identity morphisms.

There are two important things to note abuot these categories. Firstly, note that all discrete categories are small, and secondly note that all diagrams in discrete categories are necessarily commutative. Oh yeah I forgot that definition.

Definition 1.20: Small Category

Let $\mathbf{C}$ be a category. If $\mathrm{Obj}(\mathbf{C})$ is a set, then $\mathbf{C}$ is called *small*.

Here is another example of a small category. Let S be a set, and define $\hat{\mathbf{S}}$ by letting $\mathrm{Obj}(\hat{\mathbf{S}}) := \mathscr{P}(S)$, and for any $A, B \in \mathrm{Obj}(\hat{\mathbf{S}})$, let $\mathrm{Hom}_{\hat{\mathbf{S}}}(A, B) := (A, B)$ if $A \subseteq B$, and $\varnothing$ otherwise. Since $\subseteq$ is a relation on any set S, this is clearly a category by our last example.

This next example is very strange. Let $\mathbf{C}$ be a category, and let A be an object in $\mathbf{C}$. We will define a category who's objects are morphisms and who's morphisms are diagrams. Sure. Define

- $\mathrm{Obj}(\mathbf{C}_A) := \{f \mid f \in \mathrm{Hom}_{\mathbf{C}}(-, A)\}$. That is, the objects of $\mathbf{C}_A$ are the collection of all morphisms f from any object in $\mathbf{C}$ to A. We can then see an object as the arrow $Z \xrightarrow{f} A$ for some object Z in $\mathbf{C}$.

These diagrams are often drawn as

$$\begin{tikzcd} Z \arrow[d, "f"'] \\ A \end{tikzcd}$$

- The text says that the "brave reader" may want to stop reading and continue only after coming up with the definiton. I will make a guess as to what the morphisms are. Let $f : X \to A$ and $g : Y \to A$ be morphisms in $\mathbf{C}$, and thus $f, g \in \mathrm{Obj}(\mathbf{C}_A)$. I think the set $\mathrm{Hom}(f, g)$ will be the commutative diagrams starting at X, Y and terminating at A. That is, a morphism is a choice of map $h \in \mathrm{Hom}_{\mathbf{C}}(X, Y)$ such that $f = gh$. Such a map may not exist in general, or many maps may exist in general, and thus $\mathrm{Hom}_{\mathbf{C}_A}(f, g)$ is simply the set of all such maps. Note that since $\mathbb{1}_A \in \mathrm{End}_{\mathbf{C}}(A)$, for any $f \in \mathrm{Hom}_{\mathbf{C}}(X, A)$, we have $\mathbb{1}_X f = f$, and thus $\mathbb{1}_X \in \mathrm{End}_{\mathbf{C}_A}(f)$ for all f. Morphism composition probably follows from morphism composition in $\mathbf{C}$, and associativity definitely follows from that. Finally, note that the identity morphisms are clearly morphisms.

- Let Z_1, Z_2 be objects in $\mathbf{C}$, and $f_1 \in \mathrm{Hom}_{\mathbf{C}}(Z_1, A)$ and $f_2 \in \mathrm{Hom}_{\mathbf{C}}(Z_2, A)$ be objects in $\mathbf{C}_A$. That is, two arrows

$$\begin{tikzcd} Z_1 \arrow[d, "f_1"'] & Z_2 \arrow[d, "f_2"] \\ A & B \end{tikzcd}$$

in $\mathbf{C}$. Morphisms $f_1 \to f_2$ are defined to be *commutative diagrams*. That is, if

$$\begin{tikzcd} Z_1 \arrow[rr, "\sigma"] \arrow[dr, "f_1"'] & & Z_2 \arrow[dl, "f_2"] \\ & A & \end{tikzcd},$$

then the morphism $\sigma : Z_1 \to Z_2$ such that $f_1 = f_2\sigma$ is a morphism in $\mathbf{C}_A$.

Wow, this is exactly what I predicted! As I noted, these morphisms fairly trivially follow the laws in definition 1.16. The identity is inherited from $\mathbf{C}$,

$$\begin{tikzcd} Z \arrow[rr, "\mathbb{1}_Z"] \arrow[dr, "f"'] & & Z \arrow[dl, "f"] \\ & A & \end{tikzcd}$$

which commutes because $\mathbf{C}$ is a category. Composition also follows from $\mathbf{C}$ being a category. Two morphisms $f_1 \to f_2 \to f_3$ correspond to the diagram

$$\begin{tikzcd} Z_1 \arrow[r, "\sigma"] \arrow[dr, "f_1"'] & Z_2 \arrow[r, "\tau"] \arrow[d, "f_2"] & Z_3 \arrow[dl, "f_3"] \\ & A & \end{tikzcd}$$

Since $\mathbf{C}$ is a category, this is the same as the diagram

$$\begin{tikzcd} Z_1 \arrow[rr, "\tau\sigma"] \arrow[dr, "f_1"'] & & Z_3 \arrow[dl, "f_3"] \\ & A & \end{tikzcd}$$

which is clearly still a morphism $f_1 \to f_3$. Finally, composition is associative in $\mathbf{C}$, so it is associative in $\mathbf{C}_A$.

Here's another example, similar to the previous one. Let $\mathbf{C}$ be a category, and consider the objects of a new category to be the morphisms in $\mathbf{C}$ from an object A to *all objects* in $\mathbf{C}$. Again, let the morphisms be suitable commutative diagrams. These are *coslice categories*, which will be explored in the exercises.

We will give another example, which will lead to an important definition. Consider the category **Set**, and let $A = \{*\}$ be a fixed singleton. The resulting category may be called $\mathbf{Set}^*$. An object in this category is a morphism $f : \{*\} \to S$ for any set S. The information in $\mathbf{Set}^*$ is then just the choice of a nonempty set S and an element s such that $* \overset{f}{\mapsto} s$, and thus is determined by f. We can then denote objects of $\mathbf{Set}^*$ as pairs (S, s), corresponding to the map $f : \{*\} \to S$, $* \mapsto s$. A morphism between two objects $(S, s) \to (T, t)$ is any function of sets $\sigma : S \to T$ such that $\sigma(s) = t$. Objects of $\mathbf{Set}^*$ are called pointed sets, and many objects we will study will in fact be pointed sets. For example, group homomorphisms will be seen to be morphisms of pointed sets.

Definition 1.21: Pointed Set

Let $\{*\}$ be a fixed singleton set. Let $\mathbf{Set}^*$ be the category whos objects are pairs (S, s) corresponding to morphisms $f : \{*\} \to S$, $* \mapsto s$, and where morphisms $(S, s) \to (T, t)$ are functions of sets $\sigma : S \to T$ such that $\sigma(s) = t$. Then any object of $\mathbf{Set}^*$ is called a pointed set.

Another example is a more abstract variation of our previous examples. These will be essential when we revisit the definitions of $\times$ and II. Let $\mathbf{C}$ be a category and let A, B be objects of $\mathbf{C}$. Define the category $\mathbf{C}_{A,B}$ using the same procedure in which we define $\mathbf{C}_A$:

- $\mathrm{Obj}(\mathbf{C}_{A,B})$ consists of diagrams

$$\begin{tikzcd} & A \\ Z \arrow[ur, "f"] \arrow[dr, "g"'] & \\ & B \end{tikzcd}$$

  in $\mathbf{C}$, and

- Morphisms

$$\begin{tikzcd} & A & & & A \\ Z_1 \arrow[ur, "f_1"] \arrow[dr, "g_1"'] & & \arrow[r] & Z_2 \arrow[ur, "f_2"] \arrow[dr, "g_2"'] & \\ & B & & & B \end{tikzcd}$$

  are *commutative* diagrams

$$\begin{tikzcd} & & A \\ Z_1 \arrow[r, "\sigma"] \arrow[urr, bend left, "f_1"] \arrow[drr, bend right, "g_1"'] & Z_2 \arrow[ur, "f_2"] \arrow[dr, "g_2"'] & \\ & & B \end{tikzcd}$$

This means a morphism $(f_1, g_1) \to (f_2, g_2)$ is really a map $\sigma : Z_1 \to Z_2$ such that $f_1 = f_2\sigma$ and $g_1 = g_2\sigma$. Flipping the arrows so that we start from A, B and go to an arbitrary set produces an example more similar to those of the pointed sets, which we denote $\mathbf{C}^{A,B}$. The details are completely left to the reader here. I will note that there does not seem, according to Aluffi, to be an established notation for these categories $\mathbf{C}^{A,B}$, so we will use this.

Our final example is fairly complicated. "Experts would tell you that this looks fairly sophisticated for students just learning categories", but I can totally get it. I don't know if this will be useful, but since I haven't actually read this part yet, "recall" that if $f : A \to B$ is a function of sets, and $T = \{x\} \subseteq B$ is a singleton set, then $f^{-1}T = f^{-1}(x) = \{y \in A \mid f(y) = x\}$ is called the *fiber* of f over x. Now I will present the example.

Let $\mathbf{C}$ be a category, and choose two fixed morphisms $\alpha : A \to C$, $\beta : B \to C$ in $\mathbf{C}$, with the same target C. We can define the category $\mathbf{C}_{\alpha,\beta}$ as follows.

- The objects $\mathrm{Obj}(\mathbf{C}_{\alpha,\beta})$ are commutative diagrams

$$\begin{tikzcd} & A \arrow[dr, "\alpha"] & \\ Z \arrow[ur, "f"] \arrow[dr, "g"'] & & C \\ & B \arrow[ur, "\beta"'] & \end{tikzcd}$$

  in $\mathbf{C}$, and

- morphisms also correspond to commutative diagrams

$$\begin{tikzcd} & & A \arrow[dr, "\alpha"] & \\ Z_1 \arrow[r, "\sigma"] \arrow[urr, bend left, "f_1"] \arrow[drr, bend right, "g_1"'] & Z_2 \arrow[ur, "f_2"] \arrow[dr, "g_2"] & & C \\ & & B \arrow[ur, "\beta"] & \end{tikzcd}$$

I will now try to understand this. An object corresponds to an object Z in $\mathrm{Obj}(\mathbf{C})$, and a pair of morphisms $f : Z \to A$ and $g : Z \to B$ such that $\alpha f = \beta g$. We can represent this as a pair (f, g), although I will represent it as a triple (Z, f, g) to emphasize the object the morphisms originate from. A morphism $(Z_1, f_1, g_1) \to (Z_2, f_2, g_2)$ is a morphism $\sigma : Z_1 \to Z_2$ such that $f_1 = f_2\sigma$ and $g_1 = g_2\sigma$, which implies that $\alpha f_1 = \alpha f_2 \sigma$ and $\beta g_1 = \beta g_2 \sigma$. A similar construction can be done $\mathbf{C}^{\alpha,\beta}$, where $\alpha : C \to A$ and $\beta : C \to B$. An object would be a diagram

$$\begin{tikzcd} & A \arrow[dr, "f"] & \\ C \arrow[ur, "\alpha"] \arrow[dr, "\beta"'] & & Z \\ & B \arrow[ur, "g"'] & \end{tikzcd}$$

and a morphism would also be a diagram

$$\begin{tikzcd} & A \arrow[dr, "f_2"] \arrow[drr, bend left, "f_1"] & & \\ C \arrow[ur, "\alpha"] \arrow[dr, "\beta"'] & & Z_2 \arrow[r, "\sigma"] & Z_1 \\ & B \arrow[ur, "g_2"] \arrow[urr, bend right, "g_1"'] & & \end{tikzcd}$$

In this case, an object consists of a pair of morphisms $f : A \to Z$, $g : B \to Z$ of $\mathbf{C}$ such that $f\alpha = g\beta$, and a morphism in $\mathbf{C}^{\alpha,\beta}$ is a morphism $\sigma : Z_2 \to Z_1$ in $\mathbf{C}$ such that $\sigma f_2 \alpha = f_1 \alpha$ and $\sigma g_2 \beta = g_2 \beta$.

Definition 1.22: Fibered

Let $\alpha : A \to C$, $\beta : B \to C$ be morphisms in $\mathbf{C}$. Then $\mathbf{C}_{\alpha,\beta}$ is called the *fibered* version of $\mathbf{C}_{A,B}$. Similarly, for $\alpha : C \to A$ and $\beta : C \to B$ in $\mathbf{C}$, the category $\mathbf{C}^{\alpha,\beta}$ is called the *fibered* version of $\mathbf{C}^{A,B}$.

1.4 Morphisms

In section 2, we highlighted various typess of functions for **Set**. It's useful to do the same for morphisms in an arbitrary category. However, it's not possible to make these definitions based on the elements of objects in a category, since objects need not have elements. This is why we analyzed these properties from different viewpoints, since these viewpoints translate nicely.

1.4.1 Isomorphisms

Definition 1.23: Isomorphism

A morphism $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ is an *isomorphism* if it has a two-sided inverse under composition. That is, if $\exists g \in \mathrm{Hom}_{\mathbf{C}}(B, A)$ such that

$$gf = \mathbb{1}_A, \qquad fg = \mathbb{1}_B,$$

then f is an isomorphism.

We are able to prove the identity of an inverse function is unique using sets, but we must re-verify this in the new case.

Proposition 1.3

The inverse of an isomorphism is unique.

Proof. We must verify that if $g_1, g_2 : B \to A$ act as inverses to $f : A \to B$ an isomorphism, then $g_1 = g_2$. We can show this easily.

$$g_1 = g_1 \mathbb{1}_B = g_1 (f g_2) = (g_1 f) g_2 = \mathbb{1}_A g_2 = g_2.$$

■

What this argument *actually* proves is that a morphism f with a left **and** a right inverse implies the left and right inverses are equal, and f is an isomorphism. We will now denote the inverse of an isomorphism f^{-1}, since there is no ambiguity here.

Proposition 1.4

The following are true.

- Each identity $\mathbb{1}_A$ is an isomorphism and is its own inverse.
- If f is an isomorphism, then f^{-1} is an isomorphism, and $\left(f^{-1}\right)^{-1} = f$.
- If $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathbf{C}}(B, C)$ are isomorphisms, them gf is an isomorphism and $(gf)^{-1} = f^{-1} g^{-1}$.

Proof. (1) We must show $\mathbb{1}_A^{-1} = \mathbb{1}_A$. Note that $\mathbb{1}_A \mathbb{1}_A = \mathbb{1}_A$ and $\mathbb{1}_A \mathbb{1}_A = \mathbb{1}_A$ by definition 1.16. Therefore, $\mathbb{1}_A$ is an isomorphism and $\mathbb{1}_A^{-1} = \mathbb{1}_A$.

(2) We must show that f^{-1} has an inverse, namely f. Note that since f is an isomorphism,

$$f^{-1} f = \mathbb{1}_A, \qquad f f^{-1} = \mathbb{1}_B.$$

Therefore, $\left(f^{-1}\right)^{-1} = f$, and f^{-1} is an isomorphism.

(3) Note that

$$\left(f^{-1}g^{-1}\right)(gf) = f^{-1}\left(g^{-1}g\right)f = f^{-1}(\mathbb{1}_B f) = f^{-1}f = \mathbb{1}_A.$$

Additionally,

$$(gf)\left(f^{-1}g^{-1}\right) = g\left(ff^{-1}\right)g^{-1} = (g\mathbb{1}_A)\,g^{-1} = gg^{-1} = \mathbb{1}_B.$$

Therefore, $(gf)^{-1} = f^{-1}g^{-1}$. Since gf has an inverse, gf is an isomorphism. ■

Definition 1.24: Isomorphic

Two objects A, B of a category are *isomorphic* if there exists an isomorphism $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$. We write $A \cong B$ to say A is isomorphic to B.

It's easy to check that $\cong$ is an equivalence relation on any category. The proof was already done in the exercises for **Set**, but it can be generalized by the same argument. In fact, I will do it right now.

Proposition 1.5

Isomorphism is an equivalence relation on any category.

Proof. Let $\mathbf{C}$ be a category. Note that $A \cong A$ by proposition 1.4. Note that if $A \cong B$, then by proposition 1.4, $f^{-1} : B \to A$ is an isomorphism, so $B \cong A$. Finally, if $A \cong B$ and $B \cong C$, then there exist isomorphisms $f : A \to B$ and $g : B \to C$. By propsosition 1.4, $gf : A \to C$ is an isomorphism, so $A \cong C$. Therefore, $\cong$ is an equivalence relation on $\mathbf{C}$. ■

We noted that identities are isomorphisms. However, in some cases, they are the *only* isomorphisms in a category. For example, the category $\mathbf{C}$ obtained by the relation $\leq$ on $\mathbb{Z}$ pretty obviously only has identities as isomorphisms. If $f : a \to b$ and $g : b \to a$ are morphisms, then $a \leq b$ and $b \leq a$, implying that $a = b$.

Conversely, consider a category where all morphisms are isomorphisms. Such a category is called a *groupoid*.

Definition 1.25: Groupoid

A category $\mathbf{C}$ is called a *groupoid* if all morphisms are isomorphisms.

A natural extension of the concept of isomorphisms is an automorphism.

Definition 1.26: Automorphism

An automorphism of an object A of a category $\mathbf{C}$ is an isomorphism $A \to A$. The set of automorphisms is denoted $\mathrm{Aut}_{\mathbf{C}}(A)$, and it is a subset of $\mathrm{End}_{\mathbf{C}}(A)$.

There's a few interesting things about the structure of $\mathrm{Aut}_{\mathbf{C}}(A)$. By our definitions and ropositions, we have

- for all $f, g \in \mathrm{Aut}_{\mathbf{C}}(A)$, their composition is an element $gf \in \mathrm{Aut}_{\mathbf{C}}(A)$;
- composition is associative;
- $\mathrm{Aut}_{\mathbf{C}}(A)$ contains the element $\mathbb{1}_A$, which is an identity under composition ($f\mathbb{1}_A = \mathbb{1}_A f = f$);
- every element $f \in \mathrm{Aut}_{\mathbf{C}}(A)$ has an inverse $f^{-1} \in \mathrm{Aut}_{\mathbf{C}}(A)$.

The set $\mathrm{Aut}_{\mathbf{C}}(A)$ is a group under function composition, for all objects A in all categories $\mathbf{C}$.

1.4.2 Monomorphisms and Epimorphisms

As noted above, since objects need not have elements, we cannot define injective and surjective morphisms in the same way we define them for functions of sets. We can, however, define *monomorphisms and epimorphisms* in an arbitrary category:

Definition 1.27: Monomorphism

Let $\mathbf{C}$ be a category. A morphism $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ is a *monomorphism* if for all objects Z of $\mathbf{C}$ and all morphisms $\alpha', \alpha'' \in \mathrm{Hom}_{\mathbf{C}}(Z, A)$,

$$f \circ \alpha' = f \circ \alpha'' \implies \alpha' = \alpha''.$$

Definition 1.28: Epimorphism

Let $\mathbf{C}$ be a category. A morphism $f \in \mathrm{Hom}_{\mathbf{C}}(A, B)$ is an *epimorphism* if for all objects Z of $\mathbf{C}$ and all morphisms $\beta', \beta'' \in \mathrm{Hom}_{\mathbf{C}}(B, Z)$,

$$\beta' \circ f = \beta'' \circ f \implies \beta' = \beta''.$$

It's been proven previously that in the category **Set**, the monomorphisms are preciely the injective functions, and the epimorphisms are precisely the surjective functions. These functions are thus the category-theoretical generalizations of surjective and injective functions.

In categories defined by a set and a relation, every morphism is both a monomorphism and an epimorphism. Since there is at most one morphism between any two objects, the conditions defininig these morphisms are vacuous.

This example breaks some intuition we have from set theory. In **Set**, a function is an isomorphism if it is both surjective and injective, and equivalently if it is both a monomorphism and an epimorphism. However, in the category defined by $\leq$ on $\mathbb{Z}$, every morphism is both a monomorphism and an epimorphism, and yet only the identity morphisms are isomorphisms. This property is thus a special feature of **Set**, and it will not hold in every category. For example, it will not hold in **Ring**. It will be shown in the distant future, however, that it does hold in abelian categories. Similarly, an epimorphism need not have a right inverse, unlike in **Set**.

1.5 Universal Properties

Categories are extremely important in math, giving us a bird's eye view of many constructions, including those in algebra. This will be apparent in the steady appearance of constructions satisfying suitable *universal properties*. We will see shortly that disjoint unions and Cartesian products can be characterized by universal properties related to the categories $\mathbf{C}^{A,B}$ and $\mathbf{C}_{A,B}$. Many of the things we will see will have a (often seemily arbitrary) explicit definition, and then an accompanying one in terms of universal properties, which will not have arbitrary choices. Finally, deeper relationships between objects become apparent when viewing them in terms of their universal properties. For example, $\times$ and $\amalg$ are really just mirror constructions of each other, which is NOT AT ALL apparent from the previous definitions.

1.5.1 Initial and Final Objects

Definition 1.29: Initial and Final

Let $\mathbf{C}$ be a category. We say an object I of $\mathbf{C}$ is *initial* in $\mathbf{C}$ if for all objects A in $\mathbf{C}$, there exists *exactly one* morphism $I \to A$ in $\mathbf{C}$. That is,

$$\forall A \in \operatorname{Obj}(\mathbf{C})\,(\operatorname{Hom}_{\mathbf{C}}(I, A)) \text{ is a singleton set.}$$

Similarly, we say an object $F \in \operatorname{Obj}(\mathbf{C})$ is *final* in $\mathbf{C}$ if for all $A \in \operatorname{Obj}(\mathbf{C})$, there exists *exactly one* morphism $A \to F$ in $\mathbf{C}$. That is,

$$\forall A \in \operatorname{Obj}(\mathbf{C})\,(\operatorname{Hom}_{\mathbf{C}}(A, F)) \text{ is a singleton set.}$$

We will call an object $A \in \operatorname{Obj}(\mathbf{C})$ *terminal* if it is either initial or final.

Initial and final objects don't need to exist. For example, the category defined by the $\leq$ relation on $\mathbb{Z}$ has no initial or final objects. Additionally, initial and final objects need not be unique. For example, every singleton set in **Set** is final in **Set**. However, we can claim that if they exist, then they are unique *up to a unique isomorphism.*

Proposition 1.6

Let $\mathbf{C}$ be a category. If $I_1, I_2 \in \operatorname{Obj}(\mathbf{C})$ are both initial, then $I_1 \cong I_2$. Additionally, if $F_1, F_2 \in \operatorname{Obj}(\mathbf{C})$ are both final, then $F_1 \cong F_2$.

Proof. By definition 1.16, for all $A \in \operatorname{Obj}(\mathbf{C})$, there exists at least one morphism $\mathbb{1}_A \in \operatorname{Hom}_{\mathbf{C}}(A, A)$. If I is initial, then by definition 1.29 there exists a *unique* morphism $\mathbb{1}_I$. Now suppose I_1, I_2 are initial. By definition 1.29, there exists exactly one morphism $f \in \operatorname{Hom}_{\mathbf{C}}(I_1, I_2)$, and likewise, there exists exactly one morphism $g \in \operatorname{Hom}_{\mathbf{C}}(I_2, I_1)$. Now note that $gf \in \operatorname{Hom}_{\mathbf{C}}(I_1, I_1)$, and thus $gf = \mathbb{1}_{I_1}$. Similarly, $fg = \mathbb{1}_{I_2}$. Therefore, $g - f^{-1}$, and by proposition 1.4, f is an isomorphism. The proof for final objects is the same and has been done in the exercises. ■

This proposition shows why all singletons in **Set** are isomorphic; although some may *look* more special than others, they are really all the same.

1.5.2 Universal Properties

Although it is best to introduce universal properties using the language of functors, this will not be introduced until much later. For now, and for most of the text, the following working definition will suffice.

Definition 1.30: Universal Property

We say a construction satisfies a universal property when it may be viewed as a terminal object of a category.

This definition is strange and requires motivation. In general, we will not name the category it is viewed as terminal in, but rather will describe it as a property. For example, $\{\varnothing\}$ is universal with respect to the property of mapping to sets. This is the same as saying $\{\varnothing\}$ is initial in **Set**.

This is a simple example, but oftentimes the example is more complex. We will often say something along the lines of "object X is universal with respect to the following property: for any Y such that $P(Y)$, there exists a unique morphism $Y \to X$ such that ...". Note that this implies that in the category consisting of all Y satisfying $P(Y)$ and some accompanying collection of morphisms, the set X is final. We will now illustrate this strange definition with a lot of examples.

1.5.3 Quotients

Let $\sim$ be an equivalence relation on a set A. Consider the following statement. *The quotient $A/\sim$ is universal with respect to the property of mapping A to a set in such a way that equivalent elements have the same image.*

This assertion is talking about maps $\phi : A \to Z$ for any set Z, such that $a' \sim a'' \implies \phi(a') = \phi(a'')$. Thee morphisms are objects of a category, very similar to $\mathbf{C}^A$. Although an object needs only be specified by a morphism, we will specify it with an ordered pair (ϕ, Z) for convenience. The only reasonable definition of morphisms $(\phi_1, Z_1) \to (\phi_2, Z_2)$ is by commutative diagrams

$$\begin{tikzcd} Z_1 \arrow[rr, "\sigma"] & & Z_2 \\ & A \arrow[ul, "\phi_1"] \arrow[ur, "\phi_2"'] & \end{tikzcd}$$

This is the same as the example $\mathbf{C}^A$. The next natural question, given our assertion, is whether this category has initial objects. We claim that if π is the canonical projection $A \twoheadrightarrow A/\sim$ given by $a \mapsto [a]_\sim$, then $(\pi, A/\sim)$ is initial in this category.

Proof. Consider any (ϕ, Z) in the category. We wish to show there exists a unique morphism $(\pi, A/\sim) \to (\phi, Z)$. That is, we wish to show there exists a diagram

$$\begin{tikzcd} A/\sim \arrow[rr, "\overline{\phi}"] & & Z \\ & A \arrow[ul, "\pi"] \arrow[ur, "\phi"'] & \end{tikzcd}$$

Let $[a]_\sim \in A/\sim$. If we want the diagram to commute, then $\overline{\phi}([a]_\sim) = \phi(a)$. Therefore, ϕ fully specifies the behavior of $\overline{\phi}$, and thus $\overline{\phi}$ is unique, if it exists. Now we must show it is well-defined. That is, if $[a_1]_\sim = [a_2]_\sim$, then $\phi(a_1) = \phi(a_2)$. Since it follows that $a_1 \sim a_2$, and since ϕ is an object in our category, $\phi(a_1) = \phi(a_2)$. Therefore, $(\pi, A/\sim)$ is initial in this category. ■

Note that the above assertion is very bad; it does not specify what the category to be considered is, nor does it specify what the solution to the universal problem is (the morphism π). Abuse of language like this is common, and being able to translate it into precise statements is important. For example, in this statement there is only one obvious choice for a morphism which can form the initial object, and the final object is very uninteresting, so the lengthy language can be discarded.

What did we learn by viewing this statement (which looks a lot like the first isomorphism theorem, but for sets) in terms of its universal property? Suppose $\sim$ is the equivalence relation starting from a function $f : A \to B$, as seen previously. Then f also satisfies the universal property for $A/\sim$, and thus $f(A) \cong A/\sim$ by proposition 1.6. This is the same as theorem 1.1, so the universal property really describes the canonical decomposition of sets.

1.5.4 Products

We will now look at the Cartesian product, and identify the universal property behind it. The book says to try it yourself, and I tried, but I can't think of anything beyond using $\mathbf{C}^{A,B}$ or $\mathbf{C}_{A,B}$ to define the cartesian product, and I don't know how it follows or which to use.

Proposition 1.7 (Cartesian Product)

Let $A, B \in \mathrm{Obj}(\mathbf{Set})$, and consider $A \times B$, with the natural projections

$$\begin{tikzcd} & A \\ A \times B \arrow[ur, "\pi_A"] \arrow[dr, "\pi_B"'] & \\ & B \end{tikzcd}$$

Then for every set Z and morphisms

$$\begin{tikzcd} & A \\ Z \arrow[ur, "f_1"] \arrow[dr, "f_2"'] & \\ & B \end{tikzcd}$$

there exists a unique morphism $\sigma : Z \to A \times B$ such that the diagram

$$\begin{tikzcd} & & A \\ Z \arrow[r, "\sigma"] \arrow[urr, bend left, "f_A"] \arrow[drr, bend right, "f_B"'] & A \times B \arrow[ur, "\pi_A"] \arrow[dr, "\pi_B"] & \\ & & B \end{tikzcd}$$

commutes. We often denote σ by $f_A \times f_B$.

Proof. For all $z \in Z$, define

$$\sigma(z) = (f_A(z), f_B(z)).$$

Then for all $z \in Z$,

$$\pi_A \sigma(z) = \pi_A(f_A(z), f_B(z)) = f_A(z),$$
$$\pi_B \sigma(z) = \pi_B(f_A(z), f_B(z)) = f_B(z).$$

Therefore, $\pi_A \sigma = f_A$ and $\pi_A \sigma = f_B$. The definition is also required due to the commutativity of the diagram, so σ is specified by f_A and f_B, and is thus unique. ■

We can then see $A \times B$ as a final object in the category $\mathbf{Set}_{A,B}$. The main advantage of seeing $\times$ in this way is that the universal property may be stated in *any* category, whereas the definition of $A \times B$ seen earlier only makes sense in **Set**. We can say that a category $\mathbf{C}$ has (finite) products, or is a category with (finite) products, if for all $A, B \in \mathrm{Obj}(\mathbf{C})$, the category $\mathbf{C}_{A,B}$ has final objects. These final objects then carry the data of a single object in $\mathbf{C}$, which we denote $A \times B$, and of two morphisms $A \times B \to A, B$, which can be seen as the canonical projections.

One example of this is that in the category determined by $\leq$ on $\mathbb{Z}$, the categorical product $a \times b$ must satisfy for all $z \in \mathbb{Z}$ such that $z \leq a$ and $z \leq b$, we have $z \leq a \times b$. This universal problem has the solution $\min(a, b)$. We can then see that there is a relationship between the cartesian product of two sets and the minimum of two integers, since they both satisfy the same universal property.

Definition 1.31: Product

Let $A, B \in \text{Obj}(\mathbf{C})$ for $\mathbf{C}$ a category. A product $A \times B$ of A and B is an object of $\mathbf{C}$ with two morphisms $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$, satisfying the following universal property: for all objects $Z \in \text{Obj}(\mathbf{C})$ and morphisms

$$\begin{tikzcd} & A \\ Z \arrow[ur, "f_A"] \arrow[dr, "f_B"'] & \\ & B \end{tikzcd}$$

there exists a unique morphism $\sigma : Z \to A \times B$ such that the diagram

$$\begin{tikzcd} & & A \\ Z \arrow[r, "\sigma"] \arrow[urr, bend left, "f_A"] \arrow[drr, bend right, "f_B"'] & A \times B \arrow[ur, "\pi_A"] \arrow[dr, "\pi_B"] & \\ & & B \end{tikzcd}$$

commutes.

1.5.5 Coproducts

The prefix "co" generally means the reversing of all arrows. In this case, since products are final objects in $\mathbf{C}_{A,B}$, who's objects are pairs of morphisms with a common source and targets A, B, coproducts would be initial objects in the category $\mathbf{C}^{A,B}$, which are pairs of morphisms with common target and sources A, B. The book says to predict the universal property, so I will. I was close, and have edited my answer to be correct. I got a few things backwards though, so I should probably do more exercises.

Definition 1.32: Coproduct

Let $A, B \in \text{Obj}(\mathbf{C})$ for $\mathbf{C}$ a category. A coproduct $A \amalg B$ of A and B is an object of $\mathbf{C}$ with two morphisms $i_A : A \to A \amalg B$ and $i_B : B \to A \amalg B$ satisfying the universal property: for every set Z and morphisms $f_A : A \to Z$, $f_B : B \to Z$, there exists a unique morphism $\sigma : A \amalg B \to Z$ such that the diagram

$$\begin{tikzcd} A \arrow[dr, "i_A"] \arrow[drr, bend left, "f_A"] & & \\ & A \amalg B \arrow[r, "\sigma"] & Z \\ B \arrow[ur, "i_B"'] \arrow[urr, bend right, "f_B"'] & & \end{tikzcd}$$

commutes.

It should be very obvious that this is simply the reverse of definition 1.31. We say that a category has coproducts if this universal problem has a solution for all pairs of objects A, B. For example, consider $A \amalg B$ in **Set**.

Proposition 1.8

The disjoint union is a coproduct in **Set**.

Proof. Recall that the disjoint union $A \amalg B$ is defined as the union of disjoint "copies" of A and B, such as $\{0\} \times A =: A'$ and $\{1\} \times B =: B'$. We can define $i_A(a) = (0, a)$ and $i_B(b) = (1, b)$, where

$$a, b \in (\{0\} \times A) \cup (\{1\} \times B) .$$

Let $f_A : A \to Z$ and $f_B : B \to Z$ be arbitrary morphiisms. Define

$$\sigma : A \amalg B \to Z$$

as the map

$$\sigma(c) = \begin{cases} f_A(a) & \text{if } c = (0, a) \in \{0\} \times A, \\ f_B(b) & \text{if } c = (1, b) \in \{1\} \times B. \end{cases}$$

This definition makes the diagram commute rather easily, and is forced by requiring commutativity. Therefore, σ is unique and exists. ■

This shows us that the arbitrarity of choosing $\{0\}$ and $\{1\}$ in disjoint unions was simply one definition satisfying the universal property, and the universal property shows that all such choices are isomorphic. This sheds a lot of light on the weird disjoint union stuff, by showing all defintions are equivalent. The book also notes that in $\mathbb{Z}, \leq$, we have $a \amalg b = \max(a, b)$, which is extremely beautiful and follows from the incredible symmetry between the definitions.

1.6 Note from the Reader

Although I *understood* most of the stuff in this chapter, I didn't understand all of it well, and I also frequently got the definitions for $\mathbf{C}^A$ vs $\mathbf{C}_A$, as well as related categories, mixed up. I have added this section to give some clarification.

Chapter 2

Groups, First Encounter

This chapter introduces groups, the category **Grp**, and general features of the category, such as identifying monomorphisms, epimorphisms, equivalence relations, and quotients in **Grp**. A more object-oriented treatment of **Grp** is left for Chapter 4. I will also transition for the remainder of this chapter to the serif font C for categories, because I want to compare it to the boldface font.

2.1 Definition of a Group

2.1.1 Groups and Groupoids

Recall our previous definition of a groupoid. Definition 1.25 says a groupoid is a category $\mathbf{C}$ where all morphisms are isomorphisms. The author begins with the joke that "a group is a groupoid with only one element." In fact, this makes sense. Consider a groupoid G with $\mathrm{Obj}(\mathsf{G}) = \{*\}$. We have $\mathrm{Hom}_{\mathsf{G}}(*,*) = \mathrm{Aut}_{\mathsf{G}}(*)$, which we already noted was a group. This set carries all the information about G, since it only has one element. We can thus represent a group as morphisms in G. The group operation is morphism composition, since we have an identity $\mathbb{1}_*$, associativity, and all morphisms are invertible since they are isomorphisms.

2.1.2 Definition

The official definition of a group is not as a groupoid, for obvious reasons. This was simply some introduction. The official definition is as follows.

Definition 2.1: Group

Let G be a set with an operation $\cdot : G \times G \to G$. The set endowed with this operation, written as $(G, \cdot)$, is a *group* if

- the operation is associative:

$$\forall g, h, k \in G, (g \cdot h) \cdot k = g \cdot (h \cdot k).$$

- there exists an identity element e_G:

$$\exists (e_G \in G) \text{ s.t. } \forall (g \in G), e_G g = g e_G G = g.$$

- every element has an inverse:

$$\forall g \in G, \exists g^{-1} \in G \text{ s.t. } gg^{-1} = g^{-1}g = e_G.$$

The second axiom requires G be nonempty. We then find that the most trivial group is a singleton $G = \{e\}$, where $e \cdot e = e$. We call this the *trivial group*. Various other standard constructions, such as $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, $(\mathbb{Q}, +)$, and $(\mathbb{Z}, +)$ form groups fairly obviously. However, all of these cases are *commutative*. Groups need not be commutative, although we will see commutative groups behave more nicely than normal groups.

An example of a noncommutative group is the set the set of $n \times n$ invertible matrices with real entries. This is generally denoted $\mathrm{GL}_n(\mathbb{R})$. Since all matrices are invertible, this forms a group under matrix multiplication. However, as we know, matrix multiplication is *not commutative*. We will work with linear algebra more extensively in chapter 6.

2.1.3 Basic Properties

From the groupoid point of view, the identity e_G would instead be given as $\mathbb{1}_*$. We usually omit the G subscript when the group e belongs to is understood. Additionally, sometimes other symbols are used for the identity, such as 0 and 1. One important fact about the identity is the following:

Proposition 2.1

If $g, h \in G$ are identities of G, then $g = h = e_G$.

Proof. Since h is an identity, $gh = g$. Since g is an identity, $gh = h$. Therefore, $g = gh = h$. ∎

Note that this proof requires only that g be a left identity and h be a right identity. This means if there exists both a left and right identity in a group, then they are th same, and are a total identity. Additionally, this makes groups into pointed sets. Group homomorphisms are also morphisms of pointed sets, since identities must map to identities.

Proposition 2.2

Inverses in a group are also unique.

Proof. Let $g, h \in G$ be inverses of some element k. Then $e = gk = gh$. We can left multiply by k and apply associativity to obtain

$$(kg)k = (kg)h \implies ek = eh \implies k = h.$$

∎

Since the inverse is unique, we generally denote the inverse of some $g \in G$ as g^{-1}.

Finally, note that although the group operation is binary and can only apply to two items at the time, by associativity we have

$$g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3.$$

This allows us to simply write

$$g_1 \cdot g_2 \cdot g_3,$$

since all possible interpretations of it are equivalent. This gives rise to exponential notation in groups. Given a positive integer n, we denote

$$g^n := \underbrace{g \cdot \dots \cdot g}_{n \text{ times}}.$$

Additionally,

$$g^{-n} := \underbrace{g^{-1} \cdot \dots \cdot g^{-1}}_{n \text{ times}},$$

and

$$g^0 = e.$$

It's easy to check that many usual properties of exponents are satisfied. For example,

$$g^n g^m = g^{n+m}.$$

2.1.4 Cancellation

Cancellation properties hold in groups.

Proposition 2.3

Let G be a group. Then $\forall a, g, h \in G$,

$$ga = ha \implies g = h \quad \wedge \quad ag = ah \implies g = h.$$

Proof. Since G is a group, $\exists g^{-1} \in G$. We have

$$ga = ha \implies gaa^{-1} = haa^{-1} \implies g = h.$$

Additionally,

$$ag = ah \implies a^{-1}ag = a^{-1}ah \implies g = h.$$

∎

Some notation - we write $\mathbb{R}^* := \mathbb{R} \setminus \{0\}$.

2.1.5 Commutative Groups

Note that definition 2.1 does not require commtuativity. However, commutative groups arise naturally in many contexts, often involving the integers, as we will see throughout the text. We use the additive notation here to emphasize this fact, denote the identity as 0_A, the inverse as $-a$, and exponential notation is replaced by multiplication. Commutative groups are called *abelian* groups.

Definition 2.2: Abelian Group

Let G be a group. We say G is *abelian* if $\forall a, b \in G$,

$$ab = ba.$$

Also note that na is *notation*; it denotes $a + \cdots + a$. We cannot always reasonably multiply a group element by a random integer.

2.1.6 Order

Definition 2.3: Order

An element g of a group G has finite order if there exists $n \in \mathbb{Z}^+$ such that $g^n = e$. We define the order $|g|$ as the smallest such positive n. If no such n exists, we often write $|g| = \infty$.

Lemma 2.1

If $g^n = e$ for some $n \in \mathbb{Z}^+$, then $|g|$ divides n.

Proof. By definition, $n \geq |g|$, so $n - |g| \geq 0$. Note that there exists some $m \in \mathbb{Z}$ such that $r = n - m\,|g| \geq 0$ and $n - m(|g| + 1) < 0$. Note here that $r < |g|$ by this argument. We have

$$g^r = g^{n-|g|m} = g^n \left(g^{|g|}\right)^{-m} = e.$$

By definition, $|g|$ is the smallest *positive* integer such that $g^{|g|} = e$, so we have $r = 0$. Thus, $n = m\,|g|$, and thus $|g|$ divides n. ■

Corollary 2.1

Let $g \in G$ be an element of finite order, and let $N \in \mathbb{Z}$. Then

$$g^N = e \iff N \text{ is a multiple of } |g|.$$

Proof. By the previous lemma, if $g^N = e$, then N is a multiple of $|g|$. Conversely, if N is a multiple of $|g|$, we have

$$g^N = g^{m|g|} = \left(g^{|g|}\right)^m = e^m = e.$$

■

Definition 2.4: Group Order

Let G be a group. The order of G, denoted $|G|$, is the number of elements of G. If G is infinite, we write $|G| = \infty$.

Consider the $|G| + 1$ powers of some element g:

$$e g^1 g^2 \cdots g^{|G|}.$$

Since the number of powers exceeds the number of elements of G, there exist powers $i \neq j$ such that $g^i = g^j$. We then have $g^{j-i} = e$, and thus $|g| \leq j - i \leq |G|$. We will be able to say more once we reach Lagrange's theorem.

In infinite groups, odd things can happen. You can have g, h with finite order, yet $|gh| = \infty$. In abelian groups, there is much more structure.

Proposition 2.4

Let $g \in G$ have finite order. Then g^m has finite order for all $m \geq 0$, and

$$|g^m| = \frac{\operatorname{lcm}(m, |g|)}{m} = \frac{|g|}{\gcd(m, |g|)}.$$

Proof. The second equality follows from basic facts of gcd and lcm, so we prove only the first. The order of g^m is the least $d \in \mathbb{Z}^+$ such that $g^{md} = e$. By corollary 2.1, md is a multiple of $|g|$. We then require d be the smallest positive integer such that this is true. So $m\,|g^m|$ is the smalest multiple of m is the smallest multiple of $|g|$:

$$m\,|g^m| = \operatorname{lcm}(m, |g|).$$

■

Proposition 2.5

If $gh = hg$, then $|gh|$ divides $\text{lcm}(|g|, |h|)$.

Proof. If N is a multiple of $|g|$ and $|h|$, then by induction,

$$(gh)^N = g^N h^N = e.$$

■

2.2 Examples of Groups

2.3 The Category Grp

Obviously, groups will be the objects in the category Grp. This section defines the morphisms in this category and explores basic properties.

2.3.1 Group Homomorphisms

A group consists of two pieces of information, a set G, and an operation $\cdot : G \times G \to G$ satisfying definition 2.1. A morphism of groups $(G, m_G) \to (H, m_H)$, known as a *group homomorphism*, should be a function

$$\phi : G \to H$$

that also carries information about the operations on these sets. The question is then what is the most natural way to do this?

Consider the function

$$(\phi \times \phi) : G \times G \to H \times H$$

defined naturally by

$$(\phi \times \phi)(a, b) = (\phi(a), \phi(b)).$$

This satisfies the universal property for products, interestingly. We then have the diagram

$$\begin{tikzcd}
G \times G \arrow[r, "\phi\times\phi"] \arrow[d, "m_G"'] & H \times H \arrow[d, "m_H"] \\
G \arrow[r, "\phi"'] & H
\end{tikzcd}$$

The most natural thing to ask for be that this diagram commute; that is,

$$m_H((\phi \times \phi)(a, b)) = \phi(m_G(a, b)).$$

In more standard notation,

$$\phi(a) \cdot_H \phi(b) = \phi(a \cdot_G b).$$

Definition 2.5: Group Homomorphism

A function $\phi : G \to H$ is a group homomorphism if $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$. Equivalently, ϕ is a group homomorphism if the diagram above commutes.

2.3.2 Grp: Definition

For groups G, H, we define

$$\mathrm{Hom}_{\mathsf{Grp}}(G, H)$$

to be the set of group homomorphisms $G \to H$. If $\phi : G \to H$ and $\psi : H \to K$ are homomorphisms, we define composition $\psi \circ \phi$ in the usual way, such that $(\psi \circ \phi)(a) = \psi(\phi(a))$ for all $a \in G$. This can be represented with the commutative diagram

$$\begin{tikzcd}
G \times G \arrow[r, "\phi\times\phi"] \arrow[d, "m_G"] \arrow[rr, bend left, "(\psi\circ\phi)\times(\psi\circ\phi)"] & H \times H \arrow[r, "\psi\times\psi"] \arrow[d, "m_H"] & K \times K \arrow[d, "m_K"] \\
G \arrow[r, "\phi"] \arrow[rr, bend right, "\psi\circ\phi"'] & H \arrow[r, "\psi"] & K
\end{tikzcd}$$

Since function composition is associative, homomorphisms are associative, and moreover id_G is a group homomorphism, so Grp is a category.

2.3.3 Pause for Reflection

Definition 2.1 gives groups more structure than just multiplication, so why don't we require group homomorphisms to preserve this structure as well? It turns out the definition we have given does, in fact, preserve these properties.

Proposition 2.6

Let $\phi : G \to H$ be a group homomorphism. Then

- $\phi(e_G) = e_H$.
- $\forall g \in G, \phi\left(g^{-1}\right) = \phi(g)^{-1}$.

Equivalently, we require that the diagram

$$\begin{tikzcd}
G \arrow[r, "\phi"] \arrow[d, "g\mapsto g^{-1}"'] & H \arrow[d, "h\mapsto h^{-1}"] \\
G \arrow[r, "\phi"'] & H
\end{tikzcd}$$

commute.

Proof. The first item is fairly trivial.

$$e_H \cdot \phi(e_G) = \phi(e_G) = \phi(e_G \cdot e_G) = \phi(e_G) \cdot \phi(e_G).$$

It follows that $e_H \cdot \phi(e_G) = \phi(e_G) \cdot \phi(e_G)$, and thus, $e_H = \phi(e_G)$. Similarly,

$$\phi(g)\phi\left(g^{-1}\right) = \phi\left(gg^{-1}\right) = \phi(e_G) = e_H.$$

Therefore, $\phi(g^{-1}) = \phi(g)^{-1}$. ■

This proposition guarentees that these morphisms preserve the important properties of groups.

2.3.4 Products et al.

Grp and Set look very similar, as a group is simply a set with more structure. In fact, there exists a forgetful functor Grp $\rightsquigarrow$ Set (functors will be discussed later). However, the intitial objects in Set are different from those in Grp.

Proposition 2.7

Trivial groups are initial and final in Grp.

Proof. Since singletons are final in Set, and since the constant function is fairly obviously a homomorphism, singletons are final in Grp. Singletons are also intial since, by proposition 2.6, the only homomorphism $\{e\} \to G$ is the map $e \mapsto e_G$. Since only one morphism exists for all groups G, singletons are both initial and final. ■

There are, however, some similarities between the categories. For example, products in Grp are similar to those in Set.

Proposition 2.8

With operations defined componentwise, $G \times H$ as a set product is a product in Grp.

Proof. Recall that a product must come equipped with natural projections π_G, π_H that satisfy the universal property that for all groups A and pairs of morphisms $\phi_G : A \to G$, $\phi_H : A \to H$, there exists a unique morphism $\phi_G \times \phi_H$ such that the diagram

$$\begin{tikzcd} & & G \\ A \arrow[r, "\phi_G \times \phi_H"] \arrow[urr, bend left, "\phi_G"] \arrow[drr, bend right, "\phi_H"'] & G \times H \arrow[ur, "\pi_G"] \arrow[dr, "\pi_H"] & \\ & & H \end{tikzcd}$$

commutes. Using the same definitions as in Set, it's trivial to see that π_G and π_H are group homomorphisms, so we show only that the function $\phi_G \times \phi_H$ is a homomorphism, since its relation with Set guarentees it is unique.

$$\phi_G \times \phi_H(ab) = (\phi_G(ab), \phi_H(ab)) = (\phi_G(a), \phi_H(a))(\phi_G(b), \phi_H(b)) = (\phi_G \times \phi_H(a))(\phi_G \times \phi_H(b)).$$

■

Coproducts also exist in Grp, but are not the same as the disjoint union in set, since there is no way to equip that with a group structure. The coproduct of groups, known as the *free product* and denoted $G * H$, will be explored in the exercises and not used in the book.

2.3.5 Abelian Groups

The category Ab is the category of abelian groups, with morphisms given by group homomorphisms. We will see later that Ab is an instance of a class of categories of modules over a commutative ring, all of which are *abelian categories*, which will be explored in the final chapter somewhat.

Products in Ab are the same as in Grp, but coproducts in Ab are the same as products in Ab. We refer to the coproduct in Ab as the *direct sum*, denoted $A \oplus B$. It's important to note that even though the product $A \times B$ satisfies the universal property for coproducts in Ab, it *doesn't necessarily* satisfy it in Grp. The coproduct in Grp may be different than that in Ab.

2.4 Group Homomorphisms

2.4.1 Examples

The set $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is always nonempty; we can always give the homomorphism $g \mapsto e_H$. Thus, $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is a "pointed set". This morphism is called the trivial morphism.

Group actions are an important class of homomorphism. An action of a group G on an object A of a category C is a group homomorphism $G \to \mathrm{Aut}_{\mathsf{C}}(A)$. That is, a group acts on an object if the elements of the group determine isomorphisms $A \to A$ in a way that preserves the group structure. An example of this is when $\mathsf{C} = \mathsf{Set}$, each element of G determines a permutation of A. For example, $D_6 \to S_3$ is a group action on the set $\{0, 1, 2\}$. We will expand on group actions much more later.

Most of the rest of the examples I have seen before and don't care much about, so I will quickly list some of them. The exponential map $\epsilon_g : \mathbb{Z} \to G$, $a \mapsto g^a$; the quotient map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $a \mapsto a \cdot [1]_n = [a]_n$; if $m|n$ there is a homomorphism $\phi_n^m : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ defined by $[a]_n \mapsto [a]_m$.

2.4.2 Homomorphisms and Order

Group homomorphisms preserve many parts of the underlying group structure, somewhat including order.

Proposition 2.9

Let $\varphi : G \to H$ be a group homomorphism, and let g have finite order. Then $|\phi(g)|$ divides $|g|$.

Proof. Since $\phi(g)^{|g|} = \phi\left(g^{|g|}\right)$ by induction, applying lemma 2.1, we obtain the statement. ∎

The order is in general, preserved. The map $\pi_n : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ does not preserve order, except for the identity element 0. However, the order is *always* preserved by isomorphisms.

2.4.3 Isomorphisms

An isomorphism in Grp is an invertible homomorphism, and since all homomorphisms are functions of sets, we might assume that an isomorphism is a bijective homomorphism, as this is the case of isomorphisms in Set.

Proposition 2.10

Let $\phi : G \to H$ be a group homomorphism. Then ϕ is an isomorphism if and only if it is a bijection.

Proof. Suppose $\phi : G \to H$ is an isomorphism. Then it must have an inverse. Using our knowledge of functions in Set, a function has an inverse if and only if it is a bijection, so ϕ is a bijection.

Now suppose $\phi : G \to H$ is a bijective homomorphism. We must show the inverse function ϕ^{-1} is a group homomorphism. Let $h_1, h_2 \in H$ and $\phi^{-1}(h_1) = g_1$, $\phi^{-1}(h_2) = g_2$. Then

$$\phi^{-1}(h_1 h_2) = \phi^{-1}(\phi(g_1)\phi(g_2)) = \phi^{-1}(\phi(g_1 g_2)) = g_1 g_2 = \phi^{-1}(h_1)\phi^{-1}(h_2).$$

Therefore, bijections are precisely the isomorphisms in Grp. ■

Definition 2.6: Isomorphic Groups

Two groups G and H are isomorphic if they are isomorphic in Grp. We write $G \cong H$ to state that G is isomorphic to H.

Automorphisms of a group are isomorphisms $G \to G$, and as explored previously, $\mathrm{Aut}_{\mathsf{Grp}}(G)$ forms a group.

Definition 2.7: Cyclic

A group is *cyclic* if it is isomorphic to either $\mathbb{Z}$ or $C_n \coloneqq \mathbb{Z}/n\mathbb{Z}$ for some $n \in \mathbb{N}$.

Isomorphic objects in a category are fairly indistinguishable within that category. Isomorphic *groups* generally share the exact same structure. For example, we can prove the following.

Proposition 2.11

Let $\phi : G \to H$ be an isomorphism.

- For all $g \in G$, $|\phi(g)| = |g|$.
- G is abelian if and only if H is abelian.

Proof. The first follows from proposition 2.9; since $|\phi(g)|$ divides $|g|$ and vice versa, we have the statement. The second is left to the reader, but is fairly trivial. I will show G abelian implies H abelian, since the reverse is completed by proposition 1.4. Let G be abelian. Since ϕ is bijective, for all $h \in H$ there exists $g \in G$ such that $\phi(g) = h$. Let $h_1, h_2 \in H$ and let $\phi(g_1) = h_1$, $\phi(g_2) = h_2$. We have

$$h_1 h_2 = \phi(g_1)\phi(g_2) = \phi(g_1 g_2) = \phi(g_2 g_1) = \phi(g_2)\phi(g_1) = h_2 h_1.$$

■

The idea that isomorphic groups share basically all aspects of their structure will be applied without proof at many times throughout this book. One interesting note that will be proven in a later exercise is that two finite abelian groups are isomorphic if and only if all of their elements are of the same order.

2.4.4 Homomorphisms of Abelian Groups

Here is another example of how Ab is more well-behaved than Grp. We've seen already that $\mathrm{Hom}_{\mathsf{Grp}}(A, B)$ is a pointed set, but in Ab, we can say more. $\mathrm{Hom}_{\mathsf{Ab}}(A, B)$ is an abelian group, but *not* under function composition, since inverses need not exist. Rather, for $\phi, \psi \in \mathrm{Hom}_{\mathsf{Ab}}(G, H)$, we define $\phi + \psi$ as the map

$$(\phi + \psi)(a) = \phi(a) + \psi(a).$$

Note that

$$(\phi + \psi)(a + b) = \phi(a + b) + \psi(a + b) = \phi(a) + \phi(b) + \psi(a) + \psi(b).$$

Since H is commutative, we have

$$\phi(a) + \psi(a) + \phi(b) + \psi(b) = (\phi + \psi)(a) + (\phi + \psi)(b).$$

The identity then becomes the trivial homomorphism, associativity is inherited from H, and inverses are defined as $(-\phi)(a) = \phi(-a)$. In fact, our proof did not require G to be commutative, or to even be a group. We simply require that G be a set and H be a commutative group, and we have $\text{Hom}_{\mathsf{Set}}(G, H)$ an abelian group.

2.5 Free Groups

2.5.1 Motivation

Free groups are a more advanced type of group that we are now ready to tackle. Given a set A with no special group properties, we want to construct a group $F(A)$ containing A.

We start with a singleton. Letting $\{a\}$ be the trivial group would be odd, since that requires a have a group-theoretic property, so instead we define $F(A) = \langle a \rangle$, and we take each power of a to be a distinct group element. We define multiplication such that the exponential map $\epsilon_a(g) = a^g$ is an isomorphism $\mathbb{Z} \to \langle a \rangle$.

The author asks me to now contemplate how we are going to do this for all sets. I think there will be a universal property involved. I had an idea at one point, but I forget what it is now. I will skip to the answer.

2.5.2 Universal Property

Given a set A, it's natural to consider the category $\mathscr{F}^A$ of pairs (j, G) where G is a group and $j : A \to G$ is a function of sets. Morphisms

$$(j, G) \to (k, H)$$

are commutative diagrams

$$\begin{tikzcd} G \arrow[r, "\phi"] & H \\ A \arrow[u, "j"] \arrow[ur, "k"'] & \end{tikzcd}$$

where ϕ is a group homomorphism. A free group $F(A)$ will then be the group component of an initial object in $\mathscr{F}^A$. This implements the idea that A should map to $F(A)$ in the "most efficient way", as any other contruction can be recovered from the initial objects via group homomorphism. This universal property defines $F(A)$ up to isomorphism, but the question is whether $F(A)$ exists.

Before we construct $F(A)$, we will show the set $\langle a \rangle$ satisfies this universal property for $A = \{a\}$. The function $j : A \to \mathbb{Z}$ will send $a^n \mapsto n$. A set function $A \to G$ then amounts to choosing some group element $g \in G$ such that $g = f(a)$. This forces the unique homomorphism $\phi : \mathbb{Z} \to G$ defined by $\phi(1) = g$ for the diagram to commute.

2.5.3 Concrete Construction

Given a set A, we will treat elements of A as an alphabet, and construct words from the elements of A. Let A' be isomorphic but disjoint from A, and for all $a_i \in A$, denote $a_i^{-1} \in A'$. A word on the set A is the ordered list $(a_1, \ldots, a_n)$, denoted $w = a_1 \ldots a_n$, where $a_i \in A \cup A'$ for all i. The set of words on A will be denoted $W(A)$; the number n of letters of w is the length of w, and the empty word $w = () \in W(A)$.

Words on a singleton may look like

$$aa^{-1}a^{-1}aaa^{-1}aaaa^{-1}a^{-1}a.$$

Note that some of these letters are redundant; we would expect aa^{-1} and $a^{-1}a$ to annihilate each other. We would like to find a reduction process for these words. Define $r : W(A) \to W(A)$ as the "elementary reduction" that searches for the first occurance of a pair aa^{-1} or $a^{-1}a$, and removes it. For example,

$$aa^{-1}aaa^{-1}a \mapsto aaa^{-1}a.$$

If no reduction is possible, meaning $r(w) = w$, we say w is a reduced word.

Lemma 2.2

If $w \in W(A)$ has length n, then $r^{\lfloor \frac{n}{2} \rfloor}(w)$ is a reduced word.

Proof. Since each application of r reducesd the length by 2, one cannot reduce the word by more than $\frac{n}{2}$ times. ■

Define the 'reduction' $R : W(A) \to W(A)$ as $R(w) = r^{\lfloor \frac{n}{2} \rfloor}(w)$, where n is the length of w. Let $F(A) = R(W(A))$; that is, the set of reduced words on A. We ae ready to define a group structure on $F(A)$. For two words $w, w' \in F(A)$, define $w \cdot w'$ as the reduced juxtaposition of w and w',

$$w \cdot w' = R(ww').$$

This operation is clearly associative, the empty word $e = ()$ is clearly an identity, and the inverse of a reduced word is the word obtained by reversing the letters and inverting them.

Note that there exists a function $j : A \to F(A)$ defined by a mapping to the word a.

Proposition 2.12

The pair $(j, F(A))$ satisfies the universal property for free groups.

Proof. I will prove this myself before looking at the book proof, since I need practice with universal properties. Let $f : A \to G$, and let G be a group. For all $a \in A$, there exists $g \in G$ such that $f(a) = g$. For the diagram to commute, we require that $\phi(j(a)) = \phi(a) = g$. Applying the condition that ϕ is a homomorphism makes ϕ unique: for any composite word, since ϕ is a homomorphism we can break it down into singleton words, and the mappings of each singleton word is uniquely determined by f in this way, so the mapping of the composite word is as well. We also let $\phi(a^{-1}) = f(a)^{-1}$. ■

2.5.4 Free *Abelian* Groups

What if we ask for free groups in Ab? We wish to find a free abelian group $F^{ab}(A)$ that most efficiently contains the set A. We will use the same universal property, but require that $F^{ab}(A)$ be abelian. That is, we want $F^{ab}(A)$ to be the group satisfying the universal property that for all abelian groups G and functions $f : A \to G$, there exists a unique group homomorphism $\phi : F^{ab}(A) \to G$ such that the diagram

$$\begin{tikzcd}
F^{ab}(A) \arrow[r, "\phi"] & G \\
A \arrow[u, "j"] \arrow[ur, "f"'] &
\end{tikzcd}$$

commutes.

First, some notation. Let $\mathbb{Z}^{\oplus n}$ denote

$$\underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{n \text{ times}}.$$

Since this is the same as a product, it is often to denote it $\mathbb{Z}^n$, but we will use this notation instead.

Proposition 2.13

Let $A = \{1, \ldots, n\}$. Then $\mathbb{Z}^{\oplus n}$ is a free abelian group on A.

Proof. Let $i \mapsto (0, \ldots, 1, \ldots, 0)$, where 1 occurs at the ith position. Then every element of $\mathbb{Z}^{\oplus n}$ can be written uniquely of the form

$$(m_1, \ldots, m_n) = \sum_{i=1}^{n} m_i j(i).$$

Let $f : A \to G$ be a function of sets. To make the diagram commute, we must have

$$\phi\left(\sum_{i=1}^{n} m_i j(i)\right) = \sum_{i=1}^{n} m_i f(i).$$

We just need to show ϕ is a homomorphism. Since G is abelian, we have

$$\begin{aligned} \phi\left(\sum_{i=1}^{n} m_i j(i)\right) + \phi\left(\sum_{i=1}^{n} k_i j(i)\right) &= \sum_{i=1}^{n} m_i f(i) + \sum_{i=1}^{n} k_i f(i) = (m_i + k_i) f(i) \\ &= \phi\left(\sum_{i=1}^{n} (m_i + k_i) j(i)\right) = \phi\left(\sum_{i=1}^{n} m_i j(i) + \sum_{i=1}^{n} k_i j(i)\right). \end{aligned}$$

■

Proposition 2.13 defines free abelian groups for all finite sets, since all finite sets are isomorphic to some set $\{1, \ldots, n\}$, and we can redefine j given this isomorphism. Now we consider the general case. Recall that $H^A = \mathrm{Hom}_{\mathsf{Set}}(A, H)$ has abelian structure given that H is abelian. We can define a subset $H^{\oplus A}$ of H^A as

$$H^{\oplus A} \coloneqq \{\alpha : A \to H \mid \alpha(a) \neq e_H \text{ for only finitely many } a \in A\}.$$

My personal formulation of this would read something like

$$H^{\oplus A} \coloneqq \{\alpha \in \mathrm{Hom}_{\mathsf{Set}}(A, H) \mid |\{a \in A \mid \alpha(a) = e_H\}| < \infty\}.$$

The operation of composition in H^A induces an operation in $H^{\oplus A}$, under which $H^{\oplus A}$ is a group. Note that if A is finite and $H = \mathbb{Z}$, we have

$$\mathbb{Z}^{\oplus A} \cong \mathbb{Z}^{\oplus n},$$

since we can represent A as $\{1, \ldots, n\}$, and identify each element $(m_1, \ldots, m_n)$ with a function that maps i to m_i for all i.

We can generalize this idea by setting $H = \mathbb{Z}$, and noting there is a natural function $j : A \to \mathbb{Z}^{\oplus A}$ obtained by letting $a \mapsto j_a$, where

$$j_a(x) = \begin{cases} 1, & x = a \\ 0, & x \neq a \end{cases}.$$

We then have the following representation, which is a reskinned version of what we saw before.

Proposition 2.14

For every set A, $F^{ab}(A) \cong \mathbb{Z}^{\oplus A}$.

Proof. Note that due to the finiteness condition, every element of $\mathbb{Z}^{\oplus A}$ can be written as the finite sum

$$\sum_{a \in A} m_a j(a), \qquad m_a \neq 0 \text{ for only finitely many } a.$$

The argument that this satisfies the universal property is the same as proposition 2.13. ■

2.6 Subgroups

2.6.1 Definition

Definition 2.8: Subgroup

Let $(G, \cdot)$ and $(H, *)$ be groups such that $H \subseteq G$. Then H is a subgroup of G if the inclusion map $\iota : H \hookrightarrow G$ is a homomorphism.

This definition is different than the one usually seen in undergraduate level classes. Note that this implies that for all $h_1, h_2 \in H$, we have

$$h_1 * h_2 \cong \iota(h_1 * h_2) = \iota(h_1) \cdot \iota(h_2) = h_1 \cdot h_2.$$

We then say the operation $*$ is induced by the group G, since it must necessarily be the same or an isomorphic operation, and we are then able to omit mention of the inclusion map and the alternate function without any ambiguity. This is then equivalent to the standard presentation of subgroups, being that a set H is a subgroup of a group $(G, \cdot)$ if $(H, \cdot)$ is a group and $H \subseteq G$. This can be streamlined even further, in fact.

Proposition 2.15

A nonempty subset $H \subseteq G$ is a subgroup if and only if $\forall a, b \in H$,

$$ab^{-1} \in H.$$

Proof. If H is a subgroup, then the stataement is trivial, so we show only the reverse. Let $ab^{-1} \in H$ for all $a, b \in H$. Since $a \in H$, we have $aa^{-1} = e_G \in H$. Since we now have $e_G \in H$, we have $ae_G^{-1} \in H$ for all $a \in H$. Associativity is inhereted from G, so H is a subgroup of G. ■

We can show some fairly basic statements about subgroups using this fact.

Lemma 2.3

If $\{H_\alpha\}_{\alpha \in A}$ is some family of subgroups of a group G, then

$$H := \bigcap_{\alpha \in A} H_\alpha$$

is a subgroup of G.

Proof. Since $e \in H_\alpha$ for all $\alpha \in A$, $H \neq \varnothing$. Now for all $a, b \in H$, we have $a, b \in H_\alpha$ for all $\alpha \in A$, implying that $ab^{-1} \in H_\alpha$ for all $\alpha \in A$, implying that $ab^{-1} \in H$ for all $a, b \in H$. By proposition 2.15, H is a subgroup of G. ■

Lemma 2.4

Let $\phi : G \to G'$ be a group homomorphism, and let H' be a subgroup of G'. Then $\phi^{-1}(H')$ is a subgroup of G.

Proof. Since $e_{G'} \in H'$, we have $e_G \in \phi^{-1}(H')$, and the set is nonempty. Let $a, b \in \phi^{-1}(H)$. It follows that $\phi(a), \phi(b) \in H'$. Thus,

$$\phi(a)\phi(b)^{-1} = \phi\left(ab^{-1}\right) \in H'.$$

Therefore, $ab^{-1} \in \phi^{-1}(H')$, and by proposition 2.15, $\phi^{-1}(H')$ is a subgroup of G. ■

2.6.2 Examples: Kernel and Image

In this subsection we discuss two interesting subgroups determined by homomorphisms, the kernel and image.

Definition 2.9: Kernel

Let $\phi : G \to G'$ be a group homomorphism. The *kernel* of ϕ is the subset

$$\ker \phi := \{g \in G \mid \phi(g) = e_{G'}\} = \phi^{-1}(e_{G'}).$$

Proposition 2.16

The kernel of a homomorphism $\phi : G \to G'$ is a subgroup. Additionally, the image $\phi(G)$ is a subgroup.

Proof. Since $\{e_{G'}\}$ is a subgroup, by lemma 2.4, $\ker \phi$ is a subgroup. The second proof was left as an exercise, along with a short extension. Let $a, b \in \phi(G)$. Then there exist at least one $g, h \in G$ such that $\phi(g) = a$ and $\phi(h) = b$. By the axiom of choice, we can fix two such g, h. Since G is a group, $gh^{-1} \in G$. It follows that

$$\phi(gh^{-1}) = \phi(g)\phi(h)^{-1} \in \phi(G).$$

By proposition 2.15, $\phi(G)$ is a group homomorphism.

Now the extension - note that this argument requires only that G be a group, and thus contains gh^{-1}. If $H \subseteq G$ is a subgroup, the same argument can be applied to show the image of any subgroup of G under ϕ is a subgroup of G'. ■

Kernels are special subgroups, in that they satisfy the following universal property.

Proposition 2.17

Let $\phi : G \to G'$ be a group homomorphism. Then the inclusion $\iota : \ker\phi \hookrightarrow G$ is final in the category of group homomorphisms $\alpha : K \to G$ such that $\phi \circ \alpha$ is the trivial map.

In other words, every group homomorphism $\alpha : K \to G$ such that $\phi \circ a$ is the trivial homomorphism $\mathbf{0}$ factors uniquely through $\ker\phi$:

$$\begin{tikzcd} K \arrow[r, "\alpha"] \arrow[rr, bend left, "\mathbf{0}"] \arrow[dr, "\exists!\overline{\alpha}"'] & G \arrow[r, "\phi"] & G' \\ & \ker\phi \arrow[u, hook] & \end{tikzcd}$$

Proof. I will attempt to prove this myself, once again. Note that morphisms in this category $\alpha \to \beta$ are group homomorphisms $\overline{\alpha}$ such that $\beta\overline{\alpha} = \alpha$. A final object is thus a morphism $\iota : K \to G$ with $\phi \circ \iota = \mathbf{0}$ such that for all K' and all morphisms $\alpha : K \to G$ with $\phi \circ \alpha = \mathbf{0}$, there exists exactly one morphism $\overline{\alpha} : \alpha \to \iota$. This explains why the diagram as shown must commute for the inclusion map to be final. This category is similar to the category C^G.

First, note that since $\phi \circ \alpha = \mathbf{0}$, we know $\alpha(K) \subseteq \ker\phi$. We can then define $\overline{\alpha} : K \to \ker\phi$ as the map $\overline{\alpha}(x) = \alpha(x)$ for all $x \in K$. It then follows that $\iota \circ \overline{\alpha} = \alpha$, and the diagram commutes. Furthermore, note that $\overline{\alpha}$ is unique, since if there exists $x \in K$ such that $\overline{\alpha}(x) \neq \alpha(x)$, we have $(\iota \circ \overline{\alpha})(x) \neq \alpha(x)$. Thus, the commutative diagram forces $\overline{\alpha}$ to be unique. ∎

Proposition 2.17 will be used later to define kernels in more general areas. Furthermore, note that this argument did not require K be a group, nor that α be a group homomorphism. We can then state that for all sets K and functions of sets α, the morphism $\iota : \ker\phi \hookrightarrow G$ is still final.

2.6.3 Example: Subgroups Generated by a Subset

By the universal property of free groups, if $A \subseteq G$ is a set and G is a group, then there exists a unique morphism

$$\phi_A : (j, F(A)) \to (\iota, G)$$

which is also a group homomorphism

$$\phi_A : F(A) \to G$$

extending the inclusion map. The image of this homomorphism is called the subgroup $\langle A\rangle$ in G, or the subgroup generated by A. If G is abelian, we replace $F(A)$ by $F^{ab}(A)$. Using our description of free groups, we know that $j(a) = a$ as a word, and thus $\phi(j(a)) = \iota(a)$, and thus $\phi(a) = a$. We then replace string concatenation with multiplication, and note that all words collapse into repeated products with elements in A, A', or the identity. That is, $\langle A\rangle$ is the set of all products of elements in A, inverses of those elements, or the identity. This is the most economical way to find a subgroup containing A in G.

Definition 2.10: Subgroup Generated by A

Let G be a group and $A \subseteq G$ a subset. Let $\phi_A : F(A) \to G$ be the unique group homomorphism $a \mapsto a$ guarenteed by the universal property of free groups. Then $\phi_A(F(A)) = \langle A\rangle$ is called the subgroup generated by A. It is a subgroup of G consisting of all products of elements in A, inverses of elements in A, and the identity. This is equivalent to all integer powers of all elements of A.

If A is finite, we generally write $\langle a_1, \dots, a_n \rangle$ rather than A. An alternative definition is as follows. Let $\mathcal{H}$ be the set of all subgroups of G that contain A. Then

$$\langle A \rangle = \bigcap_{H \in \mathcal{H}} H.$$

Both representations describe the *smallest group containing* A, so these are clearly equivalent.

If $A = \{g\}$ is a singleton, then as we have seen before $F(A) = \mathbb{Z}$ and $\phi_A : \mathbb{Z} \to G$ is the exponential map ϵ_g. Then

$$\langle A \rangle = \langle g \rangle = \epsilon_g(\mathbb{Z}) = \left\{ \dots, g^{-2}, g^{-1}, e, g^1, g^2, \dots \right\}.$$

This is the cyclic subgroup generated by g.

Definition 2.11: Finitely-Generated

A group G is *finitely generated* if there exists a finite subset $A \subseteq G$ such that $G = \langle A \rangle$.

Equivalently, a group is finitely generated if and only if there is a surjective homomorphism

$$F(A) \twoheadrightarrow G$$

for some $A = \{1, \dots, n\} \subseteq \mathbb{N}$. We will in fact classify all finitely generated abelian groups later on, which is an extremely important result in group theory. We will show each is a direct product of cyclic groups. Abelian groups turn out to be relatively easy to classify. Classifying finite simple groups, for example, required thousands of research articles. Moreover, there are 42 abelian groups of order 1024, while there are allegedly 49,487,365,402 groups in general of this order. Wow.

2.6.4 Example: Subgroups of Cyclic Groups

We can now identify all subgroups of $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$. First, we start with $\mathbb{Z}$. Let $d\mathbb{Z} = \langle d \rangle$.

Proposition 2.18

Let $G \subseteq \mathbb{Z}$ be a subgroup. Then $G = d\mathbb{Z}$ for some $d \in \mathbb{N}$.

Proof. If $G = \{0\}$, then $G = 0\mathbb{Z}$. Otherwise, note that G must contain positive integers, and must thus contain a smallest positive integer by the well-ordering principle. Let d be the smallest such positive integer. Clearly, $d\mathbb{Z} \subseteq G$, since G must be closed under addition. To show the converse, fix some $m \in G$. By the division algorithm, there exist q, r with $0 \leq r < d$ such that

$$m = dq + r.$$

It follows that $r = m - dq$, and since $dq, m \in G$, we have $r \in G$. However, $r < d$ implies that r is not positive, since d is the smallest positive integer in G. We then have $r = 0$, and $m = dq$ for all $m \in G$. Therefore, $G = d\mathbb{Z}$. ■

This shows us that all nontrivial subgroups of $\mathbb{Z}$ are isomorphic to $\mathbb{Z}$. Equivalently, this says that every subgroup of the free group with one generator is free.

Proposition 2.19

Let $n > 0$ and let $G \subseteq \mathbb{Z}/n\mathbb{Z}$ be a subgroup. Then $G = \langle [d]_n \rangle$ for some divisor d of n.

Proof. Let $\pi_n : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ be the quotient map $x \mapsto [x]_n$, and consider $G' = \pi_n^{-1}(G)$. By lemma 2.3. G' is a subgroup of $\mathbb{Z}$, and by proposition 2.18 G' is cyclic, generated by some d. It follows that

$$G = \pi_n(G') = \pi_n(\langle d \rangle) = \langle [d]_n \rangle \,.$$

Therefore, G is cyclic. Since $n \in G'$ because $\pi_n(n) = [0]_n \in G$, we have that n is a multiple of d, and thus $d|n$. ■

It follows fairly easily that the divisors of n are in bijection with the subgroups of $\mathbb{Z}/n\mathbb{Z}$.

2.6.5 Monomorphisms

If H is a subgroup of G, the inclusion $H \hookrightarrow G$ is a monomorphism in Grp in the sense of definition 1.27. We can actually characterize all monomorphisms in Grp.

Proposition 2.20

Let $\phi : G \to G'$ be a group homomorphism. Then the following are equivalent.

- ϕ is a monomorphism,
- $\ker \phi = \{e_G\}$,
- $\phi : G \to G'$ is injective.

Proof. This proof is fairly trivial, so I will do it myself. ($1 \implies 2$) Let ϕ be a monomorphism. By definition 1.27, for all groups H and morphisms $\alpha, \alpha' \in \mathrm{Hom}_{\mathsf{Grp}}(H, G)$,

$$\phi \circ \alpha = \phi \circ \alpha' \implies \alpha = \alpha'.$$

Let $H = \ker \phi$, let α be the trivial map, and let α' be the inclusion map. We have obviously that $\phi \circ \alpha = \phi \circ \alpha'$, so $\alpha = \alpha'$. Therefore, $\alpha'(\ker \phi) = \ker \phi = \{e_G\}$.

($2 \implies 3$) Let $\ker \phi = \{e_G\}$. Suppose $\phi(g) = \phi(h)$. It follows that

$$\phi(g)\phi(h^{-1}) = e_{G'} \implies gh^{-1} \in \ker \phi \implies gh^{-1} = e_G \implies g = h.$$

($3 \implies 1$) If ϕ is injective, then it is a monomorphism in Set. Since all morphisms in Grp are morphisms in Set, we have that ϕ is a monomorphism in Grp. ■

It's important to note that not all monomorphisms necessarily have left inverses, as they do in Set.

2.7 Quotient Groups

2.7.1 Normal Subgroups

Definition 2.12: Normal Subgroup

Let $N \subseteq G$ be a subgroup. Then N is *normal* if $\forall g \in G$ and $\forall n \in N$,

$$gng^{-1} \in N.$$

Clearly every subgroup of an abelian group is normal, and there are in fact some nonabelian groups with only normal subgroups, but these are rare, and not all subgroups are normal in general.

Lemma 2.5

Let $\phi : G \to G'$ be a homomorphism. Then $\ker\phi$ is a normal subgroup of G.

Proof. By proposition 2.16, $\ker\phi$ is a subgroup. To show it's normal, let $g \in G$. It follows that for all $n \in \ker\phi$,

$$\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g)^{-1} = \phi(g)\phi(g)^{-1} = e_{G'} \implies gng^{-1} \in \ker\phi.$$

■

There is a convenient shorthand to express relevant sets. If $A \subseteq G$ is a subset and $g \in G$, then

$$gA \coloneqq \{ga \mid a \in A\} = \{h \in G \mid \exists a \in A : h = ga\}.$$

Similarly,

$$Ag \coloneqq \{ag \mid a \in A\}.$$

We then have normality if

$$gNg^{-1} \subseteq N.$$

As will be shown in the exercises, this is equivalent to the following:

$$gNg^{-1} = N, \qquad gN \subseteq Ng^{-1}, \qquad gN = Ng.$$

Note here that $gN = Ng$ does not imply that $gn = ng$ for all $n \in N$, it simply means $\forall n \in N \exists n' \in N : gn = n'g$.

2.7.2 Quotient Group

Recall that we could quotient a set A by a relation $\sim$, and that this construction satisfied the universal property that the quotient $A/\sim$ is universal with respect to the property of mapping A to a set in such a way that equivalent elements have the same image. This is equivalent to showing the canonical projection $\pi : A \to A/\sim$ is initial in the category of maps $\phi : A \to Z$ such that $a' \sim a'' \implies \phi(a') = \phi(a'')$. We then wish to find a relation such that the canonical map $a \mapsto [a]_\sim$ satisfies this universal property.

In the category Grp, we seek a relation $\sim$ and a group homomorphism called the canonical map $\pi : G \to G/\sim$ that is initial with respect to group homomorphisms $\phi : G \to G'$ such that $a \sim b \implies \phi(a) = \phi(b)$. The group homomorphism requirement restricts what relations satisfy this property. For π to be a group homomorphism, we require

$$[a]_\sim \bullet [b]_\sim = \pi(a) \bullet \pi(b) = \pi(ab) = [ab]_\sim.$$

The question is whether this operation $a \mapsto [a]_\sim$ is well defined. For this to be true, we require that if $[a]_\sim = [a]'_\sim$, then $\forall g \in G$, $[ag]_\sim = [a'g]_\sim$.

Proposition 2.21

The operation $[a] \bullet [b] \coloneqq [ab]$ defines a group structure on $G/\sim$ if and only if $\forall a, a', g \in G$,

$$[a]_\sim = [a']_\sim \implies [ag]_\sim = [a'g]_\sim \wedge [ga]_\sim = [ga']_\sim.$$

In this case the quotient function $\pi : G \to G/\sim$ is a homomorphism and is universal with respect to homomorphisms $\phi : G \to G'$ such that $a \sim a' \implies \phi(a) = \phi(a')$.

Proof. We have noted the condition is necessary, so we show it is sufficient. Assume that $a \sim a' \implies ag \sim a'g \wedge ga \sim ga'$. Then the group operation is well defined, so we need only check it is a group. Associativity is inhereted:

$$([a] \bullet [b]) \bullet [c] = [ab] \bullet [c] = [(ab)c] = [a(bc)].$$

The class $[e_G]$ is an identity:

$$[e_G] \bullet [a] = [e_G a] = [a], \qquad [a] \bullet [e_G] = [a e_G] = [a].$$

The class $[g^{-1}]$ is the inverse of g:

$$[g] \bullet [g^{-1}] = [gg^{-1}] = [e_G].$$

Therefore, $G/{\sim}$ is a group, and we have already seen that π is a homomorphism.

I will do the universal property on my own. We want to show π is initial. That is, we want to show the diagram

$$\begin{tikzcd} & G \arrow[dl, "\phi"'] \arrow[dr, "\pi"] & \\ G/{\sim} \arrow[rr, "\exists!\psi"'] & & G' \end{tikzcd}$$

commutes. For the diagram to commute, we require $\psi\pi = \phi$, so $a \underset{\pi}{\mapsto} [a]_\sim \underset{\psi}{\mapsto} \phi(a)$. We then define $\psi : G/{\sim} \to G'$ as the map $\psi([a]) = \phi(a)$. Uniqueness is guarenteed by commutativity of the diagram, and $\psi\pi = \phi$ because $\forall [a] \forall a, b \in [a]$, we know $\phi(a) = \phi(b)$ so $\phi([a]) = \{\phi(a)\}$. We need only show ψ exists; that is, show it is a group homomorphism. This follows from the fact that ϕ is a group homomorphism. For all $[a], [b] \in G/{\sim}$,

$$\psi([a] \bullet [b]) = \phi(ab) = \phi(a) \cdot \phi(b) = \psi([a]) \cdot \psi([b]).$$

■

If $\sim$ is a relation satisfying the universal property, we say $\sim$ is compatable with the group structure of G, and call $G/{\sim}$ the quotient group of G by $\sim$.

2.7.3 Cosets

The conditions obtained in proposition 2.21 characterize all compatable relations on G. These conditions are that

$$a \sim b \implies ag \sim bg,$$

$$a \sim b \implies ga \sim gb.$$

Using this fact, we can explicitly define the compatable equivalence relations. It's prudent to show what happens if only one of these conditions is satisfied before completing our description.

Proposition 2.22

Let $\sim$ be an equivalence relation on a group G satisfying $a \sim b \implies ga \sim gb$. Then

- The equivalence class H of e_G is a subgroup of G,
- $a \sim b \iff a^{-1}b \in H \iff aH = bH$.

Proof. Let $[e_G]_\sim = H \subseteq G$. Let $a \sim e_G$ and $b \sim e_G$. We have

$$b \sim e_G \implies e_G \sim b^{-1} \implies a \sim ab^{-1} \sim e_G,$$

where the last step follows by transitivity and the symmetric property of equivalence relations. Therefore, by proposition 2.15, $[e_G]_\sim$ is a subgroup of G.

Now let $a \sim b$. It follows that $e_G \sim a^{-1}b$, and thus $a^{-1}b \in H$. For the converse, let $a^{-1}b \in H$. It follows that $a^{-1}b \sim e_G \implies b \sim a$.

Now let $a^{-1}b \in H$. Since H is closed, we have $a^{-1}bH \subseteq H$ and thus $bH \subseteq aH$. Now let $ah \in aH$. Since $a \sim b \iff a^{-1}b \in H$, we have

$$ah \sim a \sim b \implies ah \in bH \implies aH \subseteq bH.$$

Therefore, $aH = bH$. Now for the converse, suppose $aH = bH$. It follows that $a \in bH$, and thus $a^{-1}a = e_G \in a^{-1}bH$. ■

This shows all the equivalence classes of an equivalence relation satisfying a single property are of the form aH for a fixed subgroup $H = [e]_\sim$.

Definition 2.13: Cosets

The left cosets of a subgroup H in a group G are the sets aH for $a \in G$. The right cosets are analogously Ha.

We now have the following

Proposition 2.23

Let $H \subseteq G$ be a subgroup, and let $\sim_L$ be the relation

$$a \sim_L b \iff a^{-1}b \in H.$$

Then $\sim_L$ satisfies $a \sim_L b \implies ga \sim_L gb$.

Proof.

$$a \sim b \implies a^{-1}b \in H \implies a^{-1}(g^{-1}g)b \in H \implies (ga)^{-1}(gb) \in H \implies ga \sim_L gb.$$

■

Proposition 2.24

There is a one-to-one correspondence between subgroups of G and equivalence relations on G satisfying

$$a \sim b \implies ga \sim gb.$$

The set $G/\sim_L$ can be described as the set of left cosets of H in G.

Proof. Follows from propositions 2.22 and 2.21. ■

The mirror statements using the relation $a \sim_R b \implies ag \sim_R bg$ all follow equivalent proofs. It's important to note that left and right cosets can be different; $aH \neq Ha$ in general. Recall again that for $G/\sim$ to be a group, we require $\sim_L = \sim_R$. That is, we require $aH = Ha$.

2.7.4 Quotient by Normal Subgroups

Proposition 2.25

The relations $\sim_L$ and $\sim_R$ corresponding to a subgroup H of G coencide if and only if H is normal in G.

Proof. Obviously if they coencide, then by proposition 2.22 we have $aH = Ha$. By proposition 2.22, if the cosets coencide, then the relations coencide. ∎

This means the relation $\sim$ related to a group H is compatible with the main group G if and only if H is normal in G.

Definition 2.14: Quotient Group

Let H be a normal subgroup of G. Then the *quotient group of G modulo H*, denoted G/H or $\frac{G}{H}$, is the group $G/\sim$ where $\sim$ is the relation as defined above, and

$$(aH)(bH) \coloneqq (ab)H.$$

The identity $e_{G/H} = e_G H = H$.

Then, by proposition 2.21, $\pi : G \to G/H$ defined by $g \mapsto gH$ is a group homomorphism and is universal with respect to group homomorphisms $\phi : G \to G'$ such that $aH = bH \implies \phi(a) = \phi(b)$. This universal property is suprisingly useful.

Theorem 2.1

Let H be a normal subgroup of G. Then for every group homomorphism $\phi : G \to G'$ such that $H \subseteq \ker \phi$ there exists a unique group homomorphism $\widetilde{\phi} : G/H \to G'$ such that the diagram

$$\begin{tikzcd} G \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & G' \\ & G/H \arrow[ur, "\widetilde{\phi}"'] & \end{tikzcd}$$

commutes.

Proof. We need only show $H \subseteq \ker \phi$ implies that $a \sim b \implies \phi(a) = \phi(b)$. That is, we need to show π satisfies the universal property given earlier. This follows immediately. Let $a \sim b$, and we have $ab^{-1} \in H \implies \phi(ab^{-1}) = e_{G'}$, and thus $\phi(a) = \phi(b)$. ∎

2.7.5 Kernel $\iff$ Normal

If H is a normal subgroup, then note taht $\pi : G \to G/H$ has the kernel $\ker \pi = H$. Since we already observed that kernels are normal, we now have that normal subgroups are kernels. This also says every subgroup of an abelian group is the kernel of some homomorphism.

2.8 Canonical Decomposition and Lagrange's Theorem

2.8.1 Canonical Decomposition

In theorem 1.1, we showed that a function omits a canonical decomposition into a surjection, bijection, and injection. We are now ready to show this decomposition also exists in Grp.

Theorem 2.2 (First Isomorphism Theorem)

Every group homomorphism $\phi : G \to G'$ may be decomposed as follows:

$$\begin{tikzcd} G \arrow[r, two heads] \arrow[rrr, bend left, "\phi"] & G/\ker\phi \arrow[r, "\sim", "\widetilde{\phi}"'] & \phi(G) \arrow[r, hook] & G' \end{tikzcd}$$

where the isomorphism $\widetilde{\phi}$ is the homomorphism induced by ϕ in theorem 2.1.

Proof. We have shown already that this decomposition exists, by way of the universal property $G/\ker\phi$ satisfies - that is, each element $a\ker\phi \in G/\ker\phi$ consists of all elements g such that $\phi(g) = a$. Therefore, we have already shown everything here. ■

Corollary 2.2

Let $\phi : G \to G'$ be a *surjective* group homomorphism. Then

$$G' \cong \frac{G}{\ker\phi}.$$

Proof. $\phi(G) = G'$, so by theorem 2.2, $G/\ker\phi \cong G'$. ■

The canonical decomposition of a homomorphism is very powerful, as it lets us fully characterize the image of the homomorphism. We can also prove various isomorphisms with this result.

Proposition 2.26

Let $H_1 \subseteq G_1$ and $H_2 \subseteq G_2$ be normal subgroups. Then $H_1 \times H_2$ is normal in $G_1 \times G_2$ and

$$\frac{G_1 \times G_2}{H_1 \times H_2} \cong \frac{G_1}{H_1} \times \frac{G_2}{H_2}.$$

Proof. Composing the projections with the quotient maps gives the surjective homomorphisms

$$\pi_1 : G_1 \times G_2 \to \frac{G_1}{H_1}, \qquad \pi_2 : G_1 \times G_2 \to \frac{G_2}{H_2}.$$

They are surjective because the quotient map and projections are surjective by definition. By the universal property of products, $\pi_1 \times \pi_2 =: \pi$ is a morphism

$$\pi : G_1 \times G_2 \to \frac{G_1}{H_1} \times \frac{G_2}{H_2}$$

defined by $\pi(g_1, g_2) = (g_1 H_1, g_2 H_2)$. Moreover, $\ker \pi = H_1 \times H_2$ fairly trivially. Since π is surjective, by corollary 2.2 we have

$$\frac{G_1 \times G_2}{H_1 \times H_2} \cong \frac{G_1}{H_1} \times \frac{G_2}{H_2}.$$

■

The above proposition extends to longer factors, and a general proof of this nature can be done via induction.

One interesting example is the circle group. A unit circle is often denoted S^1, and we can give it a group structure by associating to each point on the circle a rotation of a plane by the angle at that point. The function $\rho : \mathbb{R} \to S^1$ defined by r mapping to a rotation by $2\pi r$ is then a surjective group homomorphism. Obviously the kernel of this map is $\mathbb{Z}$, so we have by the above corollary

$$S^1 \cong \mathbb{R}/\mathbb{Z}.$$

Geometrically, this amounts to 'wrapping $\mathbb{R}$ around the circle', and $\mathbb{Z}$ is the fundamental group of S^1.

2.8.2 Presentations

Every group is a quotient of a free group, and every abelian group is a quotient of a free abelian group. We find this by means of corollary 2.2, using a surjective homomorphism.

A presentation of a group G is an explicit isomorphism

$$G \cong \frac{F(A)}{R}$$

where A is a set and R is a subgroup of relations. Equivalently, a presentation of a group is a surjective homomorphism $\rho : F(A) \to G$ such that $\ker \rho = R$. This is a useful concept when A is small and R can be described explicitly. That is, by listing a set $\mathscr{R}$ of words such that R is the smallest set containing $\mathscr{R}$; $R = \langle \mathscr{R} \rangle$.

A presentation of a group G is then usually given of the form $(A|\mathscr{R})$, where definitions are as above. A group is finitely presented if $A, \mathscr{R}$ are finite. Finitely presented groups are not necessarily finite; for example for any finite set A, the group $F(A)$ is very well infinite, however it admits the finite presentation $(A|\varnothing)$. It is also clear that groups admitting the same presentation are isomorphic, since isomorphism is an equivalence relation on any category.

Now that we understand presentations, finding coproducts in Grp is possible, and will be done in the exercises.

2.8.3 Subgroups of Quotients

The lattice of subgroups of a quotient group G/H can be described by the lattice of subgroups in G; keeping the part of the lattice that corresponds to subgroups containing H gives the lattice for G/H. This works because of the following.

Proposition 2.27

Let H be a normal subgroup of G. Then for every subgroup K of G containing H, K/H can be identified with a subgroup of G/H. More precisely,

$$u : \{K \in \mathrm{Obj}(\mathsf{Grp}) \mid U \subseteq K \subseteq G\} \to \{K \in \mathrm{Obj}(\mathsf{Grp}) \mid K \subseteq G/H\}$$

defined by $u(K) = K/H$ is a bijection.

Proof. The group K/H consists of cosets $aH \in G/H$ such that $a \in K$. It is very obvious that K/H is thus a subgroup of G/H, and moreover if $H \subseteq K \subseteq L$, then $u(K) = K/H \subseteq L/H = u(L)$, so inclusion is preserved.

Now we show u has an inverse. Let v be the obvious inverse function. Let K' be a subgroup of G/H, and let

$$v(K') = \left\{a \in G \mid aH \in K'\right\}.$$

This is clearly an inverse of u, so we have the statement. ■

The correspondence also preserves normality. The following statement is not so much about the isomorphism, but rather about normality of the subgroups.

Proposition 2.28 (Third Isomorphism Theorem)

Let $H \subseteq N \subseteq G$ be subgroups, and let$H$ be normal. Then N/H is normal in G/H if and only if N is normal in G, and

$$\frac{G/H}{N/H} \cong \frac{G}{N}.$$

Proof. Let N be normal, and consider the projection $G \to G/N$. By the universal property of quotients (theorem 2.1), since $H \subseteq \ker \pi = N$, there exists an induced homomorphism $G/H \to G/N$. It's easy to show the kernel of this homomorphism is N/H, so by theorem 2.2, we have

$$\frac{G/H}{N/H} \cong \frac{G}{N}.$$

Conversely, let N/H be normal in G/H. We then have the composition of the two projections

$$G \to \frac{G}{H} \to \frac{G/H}{N/H}.$$

It's easy to show the kernel of this homomorphism is N. Also note that since these are the canonical maps, this homomorphism is surjective, so we have the isomorphism. ■

2.8.4 HK/H vs. $K/(H \cap K)$

The notation for cosets extends to sets. We say $AB \coloneqq \{ab \mid a \in A \wedge b \in B\}$. Unfortunately H, K subgroups does not imply HK is a subgroup in general, especially in noncommutative groups. However, this does in fact happen if H and K are normal. This is called the second isomorphism theorem.

Proposition 2.29 (Second Isomorphism Theorem)

Let H, K be subgroups of G and let H be normal in G. Then HK is a subgroup of G, H is normal in HK, $H \cap K$ is normal in K, and
$$\frac{HK}{H} \cong \frac{K}{H \cap K}.$$

Proof. Note that HK is the union of all cosets Hk; that is, if $\pi : G \to G/H$ is the canonical projection, then $HK = \pi^{-1}(\pi(K))$. This deserves more justification. The image $\pi(K)$ is a collection of all cosets of H by elements of K, and so the inverse is all elements in these cosets. By previous lemmas icba citing, $\pi(K)$ is a subgroup and thus $\pi^{-1}(\pi(K))$ is a subgroup. It's very obvious that H is normal in HK because $K \subseteq G$. Now consider the homomorphism
$$\phi : K \to \frac{HK}{H}.$$
defined by $k \mapsto Hk$. That is, the inclusion $K \hookrightarrow HK$ followed by the canonical projection onto the quotient. Each of these maps is surjective, so we have immediately that
$$\frac{HK}{H} \cong \frac{K}{\ker \phi}.$$
The kernel of ϕ is the set of all $k \in K$ such that $Hk = H$; that is, $k \in H$. Therefore, $\ker \phi = H \cap K$. ■

2.8.5 The Index and Lagrange's Theorem

We use the notation G/H even when H is not normal to denote the set of cosets of H in G

Definition 2.15: Index

The index of H in G, denoted $[G : H]$, is defined as
$$[G : H] = \left|\frac{G}{H}\right|.$$

Lemma 2.6

Let H be a subgroup of G. Then the maps
$$H \to gH, \quad h \mapsto gh,$$
$$H \to Hg, \quad h \mapsto hg$$
are bijections.

Proof. Both functions are surjective by definition of coset, and cancellation shows they are invertible. ■

Corollary 2.3 (Lagrange's Theorem)

If G is a finite group and $H \subseteq G$ is a subgroup, then $|G| \,/\, |H| = [G : H]$. In particular, $|H|$ divides $|G|$.

Proof. By the previous lemma, and since cosets are disjoint, G is the disjoint union of the cosets gH with $|gH| = |H|$, and then we have

$$|G| = [G : H] \cdot |H| \,.$$

■

Lagrange's theorem is an extremely important result in finite group theory. One important corollary is that since $|g| = |\langle g \rangle|$ for any $g \in G$, we have $|g|$ divides $|G|$. You can also show Fermat's Little theorem with it; that for any prime integer p, $a^p \equiv a \pmod{p}$.

2.8.6 Epimorphisms and Cokernels

We should expect a mirror statement of proposition 2.20 for epimorphisms. We almost have one; epimorphisms in Grp are surjections, but the proofs are suprisingly difficult for one direction. The proof is easier in Ab because there is a notion of a cokernel.

The universal property for cokernels is the opposite of that of kernels. Given a homomorphism $\phi : G \to G'$ of *abelian groups*, the cokernel of ϕ is the set such that the map $\pi : G' \to \operatorname{coker} \phi$ is initial with respect to all morphisms α such that $\alpha\phi = 0$. That is,

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[rr, bend left, "0"] & G' \arrow[r, "\alpha"] \arrow[d, two heads, "\pi"] & L \\ & \operatorname{coker}\phi \arrow[ur, "\exists!\overline{\alpha}"] & \end{tikzcd}$$

must commute. Cokernels exist in Ab because the image of ϕ is *normal* in G', so the condition that $\alpha\phi = 0$ implies $\phi(G) \subseteq \ker \alpha$, so by theorem 2.1, $G'/\phi(G)$ is initial in this category. Therefore,

$$\operatorname{coker} \phi \cong \frac{G'}{\phi(G)}.$$

In Grp, subgroups are not guarenteed to be normal, so cokernels are more complex. Also note that in the abelian case, the requirement that α be a homomorphism is not required; it can simply be a set function that is constant on cosets of $\phi(G)$. We can now make our statement about epimorphisms of abelian groups.

Proposition 2.30

Let $\phi : G \to G'$ be a homomorphism of abelian groups. Then the following are equivalent.

- ϕ is an epimorphism;
- $\operatorname{coker} \phi$ is trivial;
- $\phi : G \to G'$ is surjective.

Proof. I can prove this on my own.

($a \implies b$) Let ϕ be an epimorphism. Recall definition 1.28: for all groups H and all group homomorphisms $\beta', \beta'' : G' \to H$,

$$\beta'\phi = \beta''\phi \implies \beta' = \beta''.$$

Consider the projections

$$G \xrightarrow{\phi} G' \overset{\pi}{\underset{0}{\rightrightarrows}} \operatorname{coker}\phi$$

where π is the canonical projection $G' \to G'/\phi(G) \hookrightarrow \operatorname{coker}\phi$, and 0 is the trivial map $g \mapsto 0$. Since ϕ is an epimorphism, it suffices to show $\pi \circ \phi = 0 \circ \phi$. For all $g \in G$, we have

$$\pi(\phi(g)) \cong \phi(g)\phi(G) = \phi(G) \cong e_{\operatorname{coker}\phi}.$$

Note here that the isomorphisms come from the fact that $\operatorname{coker}\phi \cong G'/\phi(G)$. Therefore, $\pi \circ \phi = 0 \circ \phi$.

($b \implies c$) Let $\operatorname{coker}\phi$ be trivial. It follows that $G'/\phi(G)$ is trivial by definition, and thus there must be precisely one coset, which is guarenteed to be $\phi(G)$. Therefore, $\phi(G) = G'$, and ϕ is surjective.

($c \implies a$) Let ϕ be surjective. It follows that ϕ is an epimorphism in Set, and since all morphisms in Grp are morphisms in Set, the epimorphism condition will hold in Grp. ■

A cokernel can still be defined in Grp as G'/N, where N is the smallest normal subgroup containing $\phi(G)$. However, the above proposition fails, as $\operatorname{coker}\phi$ trivial does not imply ϕ is surjective. There are non-surjective morphisms with trivial cokernels in Grp.

2.9 Group Actions

2.9.1 Actions

Definition 2.16: Group Action

An action of a group G on an object A in a category C is a group homomorphism

$$\sigma : G \to \operatorname{Aut}_{\mathsf{C}}(A).$$

That is, each element $g \in G$ determines an automorphism of A, such that composition is compatable with the underlying group structure in G. Group actions are thus extremely important tools in studying geometric and algebraic entities, since we can study actions on these objects, no matter what category they're in. Group actions are also useful in studying groups themselves. For example, letting D_6 act on a triangle showed that D_6 is a permutation of the vertices of a triangle, viewed as a rigid body.

Definition 2.17: Faithful/Effective

An action of a group G on an object A in a category C is called *faithful* or *effective* if the map $\sigma : G \to \operatorname{Aut}_{\mathsf{C}}(A)$ is injective.

2.9.2 Actions on Sets

We will focus on the category Set this chapter, since the theory is quite rich. In this case, $\operatorname{Aut}_{\mathsf{Set}}(A) = S_A$. The book also provides this second definition for a group action, which I personally dislike but oh well.

An action of a group G on a set A is a function $\rho : G \times A \to A$ such that $\rho(e_G, a) = a$ for all $a \in A$ and $\rho(gh, a) = \rho(g, \rho(h, a))$ for all $g, h \in G$. This definition is equivalent to definition 2.16, when you define

$$\rho(g, a) = \sigma(g)(a),$$

and note that this is sufficient and necessary for the two requirements of ρ to be satisfied. Proof of this is fairly trivial and I will skip it. We generally write ga to denote $\sigma(g)(a)$.

One important example of an application of group actions is Cayley's theorem.

Theorem 2.3 (Cayley)

Every group acts faithfully on some set. That is, every group is isomorphic to a subgroup of a permutation group.

Proof. Note that the action of any group G on itself by left multiplication is faithful. Proving that the action of left multipliation is a permutation is fairly trivial, and proving permutations are distinct is simple and can be done considering the action on the identity element. From this, the isomorphism follows. I've done this proof 10 times im not writing it again. ■

Note that we are only considering actions on the *left*, due to our notation ga, and right actions exist such that associativity looks like $(ag)h = a(gh)$. All right actions can be turned into left actions, as shown in the exercises, so this notation is fine.

2.9.3 Transitive Actions and the Category G-Set

Definition 2.18: Transitive

An action of a group G on an nonempty set A is *transitive* if for all $a, b \in A$, there exists $g \in G$ such that $b = ga$.

The importance of this definition is made clear by the following two definitions.

Definition 2.19: Orbit

Let a group G act on a set A. The orbit of $a \in A$ is defined as

$$O_G(a) \coloneqq \{ga \mid g \in G\} .$$

Definition 2.20: Stabilizer

Let a group G act on a set A. The stabilizer of $a \in A$ is defined as

$$\mathrm{Stab}_G(a) \coloneqq \{g \in G \mid ga = a\} .$$

Note that if an action is transitive, then the orbit of any element is the entire set A. Also note that (fairly obviously) orbits partition A, and the action is transitive within each particular orbit. Thus, we need only understand transitive actions to get a good understanding of any given group action. We can also very naturally define a category in this context.

Definition 2.21: G-Set

The category G-Set consists of objects as pairs (ρ, A), where $\rho : G \times A \to A$ is a group action in the sense of the discussion earlier, and morphisms $(\rho_1, A_1) \to (\rho_2, A_2)$ are functions $\phi : A \to A'$ such that the diagram

$$\begin{tikzcd}
G \times A_1 \arrow[r, "\mathbb{1}_G \times \phi"] \arrow[d, "\rho_1"'] & G \times A_2 \arrow[d, "\rho_2"] \\
A_1 \arrow[r, "\phi"'] & A_2
\end{tikzcd}$$

commutes. With the usual shorthand, this means

$$g\phi(a) = \phi(ga)$$

for all $a \in A$ and $g \in G$. Such functions are called equivariant.

Given this categorical formalism, we have a notion of an isomorphism of G-sets, which are simply equivariant bijections.

Proposition 2.31

Every transitive left action of G on a nonempty set A is isomorphic to the action of left multiplication of G on G/H, for $H = \mathrm{Stab}_G(a)$ for any $a \in A$.

Proof. Let G act transitively on some set A, and fix $a \in A$. Let $H = \mathrm{Stab}_G(a)$. We wish to show the map

$$\phi : G/H \to A$$

defined by

$$gH \mapsto ga$$

is an equivariant bijection. I think I can prove this without looking at the book. First, I will show the map is well defined. Let $g, h \in G$ and $gH = hH$. It follows that $g^{-1}h \in H$, and thus

$$g^{-1}ha = a \implies ha = ga.$$

Therefore, the map is well-defined. Now I show it is equivariant. Let $k \in G$. We have

$$\phi(k(gH)) = \phi(kgH) = kga = k(ga) = k\phi(gH).$$

Therefore, the map is equivariant. Now we show it is a bijection by providing an explicit inverse. Let $\psi : A \to G/H$ be the map $ga \mapsto gH$. First we show thismap is well-defined. Let $g_1a = g_2a$ for some $g_1, g_2 \in G$. It follows that

$$g_1^{-1}g_2a = a \implies g_1^{-1}g_2 \in H \implies g_1H = g_2H \implies \phi(g_1a) = \phi(g_2a).$$

The maps are obviously inverses, so we need only show ψ is a morphism in G-Set; that is, it's equivariant. Upon reflection, it's interesting to note the book did not even mention ψ had to be a morphism in G-Set. We have

$$\psi(k(ga)) = k(gH) = k\phi(ga).$$

■

Corollary 2.4 (Orbit-Stabilizer)

If O is the orbit of the action of a finite group G on a set A, then O is finite, and $|O|$ divides $|G|$. In particular, $|G| = |\operatorname{Stab}_G(a)|\,|O_G(a)|$ for any $a \in A$.

Proof. Consider the group action of G on O, instead of on A. By definition 2.19, this is still a group action, and thus is obviously transitive. By proposition 2.31, we then have a bijection from $G/\operatorname{Stab}_G(a) \to O$. Since A is finite, it follows from corollary 2.3 that

$$\left|\frac{G}{\operatorname{Stab}_G(a)}\right| = \frac{|G|}{|\operatorname{Stab}_G(a)|},$$

and thus,

$$|G| = |O|\,|\operatorname{Stab}_G(a)|\,.$$

■

This corollary, called the Orbit-Stabilizer theorem, is extremely important and upgrades Lagrange's theorem to not just subgroups, but orbits of group actions. For example, there are no transitive actions of S_3 on a set with 5 elements, as 6 doesn't divide 5.

We can say more, in fact, about the orders of these sets.

Proposition 2.32

Let a group G act on a set A, and let $a, b \in A$ and $g \in G$ such that $b = ga$. Then

$$\operatorname{Stab}_G(b) = g\operatorname{Stab}_G(a)g^{-1}.$$

Proof. I can do this proof on my own easily. Let $h \in \operatorname{Stab}_G(a)$. We have

$$(ghg^{-1})(b) = (gh)(g^{-1}g)(a) = (gh)(a) = ga = b.$$

Thus, $g\operatorname{Stab}_G(a)g^{-1} \subseteq \operatorname{Stab}_G(b)$. To show the reverse, let $h \in \operatorname{Stab}_G(b)$. We wish to show $h = gkg^{-1}$ for some $k \in \operatorname{Stab}_G(a)$. This implies that

$$g^{-1}hg = k \in \operatorname{Stab}_G(a),$$

so it suffices to show $g^{-1}hg$ fixes a. We have

$$b = ga = hga \implies g^{-1}hga = a.$$

■

This shows also that if the stabilizer is normal, then it is independent of the choice of a within the orbit of a. It will also be shown in the exercises that

$$\frac{G}{H} \cong_{G\text{-}\mathsf{Set}} \frac{G}{gHg^{-1}}.$$

2.10 Group Objects in Categories

2.10.1 Categorical Viewpoint

The definition of a group is grounded in Set. However, it really depends on two functions, multiplication m and the inverse map $\iota : g \mapsto g^{-1}$ such that various properties are satisfied. In fact, we can re-express the definition of a group using only these functions and universal properties, removing the reliance on Set. We can even define homomorphisms purely in terms of commutative diagrams.

Definition 2.22: Group Object

Let C be a category with finite products and with a finite object 1. A *group object* in C consists of an object G of C and morphisms

$$m : G \times G \to G, \qquad e : 1 \to G, \qquad \iota : G \to G$$

in C such that the diagrams

$$\begin{tikzcd}
(G \times G) \times G \arrow[r, "m \times \mathbb{1}_G"] \arrow[d, "\cong"] & G \times G \arrow[r, "m"] & G \arrow[d, equal] \\
G \times (G \times G) \arrow[r, "\mathbb{1}_G \times m"] & G \times G \arrow[r, "m"] & G
\end{tikzcd}$$

$$\begin{tikzcd}
1 \times G \arrow[r, "e \times \mathbb{1}_G"] \arrow[dr, "\cong"] & G \times G \arrow[d, "m"] \\
& G
\end{tikzcd}
\qquad
\begin{tikzcd}
G \times 1 \arrow[r, "\mathbb{1}_G \times e"] \arrow[dr, "\cong"] & G \times G \arrow[d, "m"] \\
& G
\end{tikzcd}$$

$$\begin{tikzcd}
G \arrow[r, "\Delta"] \arrow[d] & G \times G \arrow[r, "\mathbb{1}_G \times \iota"] & G \times G \arrow[d, "m"] \\
1 \arrow[rr, "e"] & & G
\end{tikzcd}
\qquad
\begin{tikzcd}
G \arrow[r, "\Delta"] \arrow[d] & G \times G \arrow[r, "\iota \times \mathbb{1}_G"] & G \times G \arrow[d, "m"] \\
1 \arrow[rr, "e"] & & G
\end{tikzcd}$$

commute, where $\Delta = \mathbb{1}_G \times \mathbb{1}_G$.

This definition deserves extensive commentary. First, we will discuss morphisms that are not named. These morphisms are all uniquely determined by some universal property. For example, $\epsilon : G \to 1$ is unique because 1 is final. The projection $1 \times G \to G$ is the projection guarenteed by the universal property of products. By this same universal property, the composition

$$G \xrightarrow{\epsilon \times \mathbb{1}_G} 1 \times G \xrightarrow{\pi} G$$

is the identity, because by the universal property of products there exists precisely one map $\sigma : G \to 1 \times G$ such that $\epsilon = \pi_1 \sigma$ and $\mathbb{1}_G = \pi_G \sigma$. We can then see that this map must be $\epsilon \times \mathbb{1}_G$. By this same logic,

$$1 \times G \xrightarrow{\pi} G \xrightarrow{\epsilon \times \mathbb{1}_G} 1 \times G$$

is also the identity, so we find that the projection $\pi : 1 \times G \to G$ is an isomorphism.

Now we want to recover group properties from this mess. Note that the first diagram shows commutativity; no matter the order that m is applied in, we still obtain the second result. The second diagram suggests that the map $e : 1 \to G$ constitutes the identity, as the identity in a group can be viewed as the embedding of a singleton (a final object in Set) into the group such that the object is invariant under left-and right-multiplication. The final diagram defines inverses. Illustrating what this means in set is very beneficial. Given an element of the set a, we have

$$a \mapsto (a, a) \mapsto (a, a^{-1}) = a \mapsto e \mapsto e.$$

That is, the inverse map ι sends an element to an inverse of itself, which mutually annihilate into the identity.

Many important categories, such as the categories of topological spaces, differentiable manifolds, schemes, algebraic varieties, and others, have notions of group objects. For example, Lie groups are group objects in the category of differentiable manifolds.

Chapter 3

Rings and Modules

3.1 Definition of a Ring

3.1.1 Definition

Rings, and by extension, modules, are created by "decorating" abelian groups. They are abelian groups with an additional multiplication operation on top of them, and behave similarly to common sets such as $\mathbb{Z}$ and $\mathbb{R}$. An additional, more advanced example of some motivation is that the set $\mathrm{End}_{\mathsf{Ab}}(A)$ not only forms an abelian group under pointwise addition, but in fact using morphism composition as multiplication turns $\mathrm{End}_{\mathsf{Ab}}(A)$ into a ring.

Definition 3.1: Ring

A ring $(R, +, \cdot)$ is an abelian group $(R, +)$ together with a second binary operation $\cdot : R \times R \to R$ such that

- (Associativity) $\forall x, y, z \in R$, $x \cdot (y \cdot z) = (x \cdot y) \cdot z$.
- (Identity) $\exists 1 \in R$ such that $\forall r \in R$, $1r = r1 = r$.
- (Distributivity) $\forall r, s, t \in R$,

$$r(s+t) = rs + rt \text{ and } (r+s)t = rt + st.$$

It's important to note that not all authors include the requirement that rings have 1; they would call this definition a "ring with identity" or a "ring with unity" or a "unital ring". On the other hand, rings without identities are sometimes called rngs. We will use the convention that rings have identity.

Given that a ring has unity, the identity is then clearly unique, by the same argument as given with groups. The identity under addition is often called 0_R or simply 0, and interestingly, it behaves as one would expect.

Lemma 3.1

Let R be a ring. Then for all $r \in R$,

$$0r = r0 = 0.$$

Proof. We have

$$r0 = r(0+0) = r0 + r0 \implies r0 - r0 = r0 + r0 - r0 \implies r0 = 0.$$

The proof for the other side is identical. ■

Additionally, we can show $-1r = -r$. We have

$$r + (-1r) = 1r + (-1r) = (1-1)r = 0r = 0$$

from lemma 3.1.

3.1.2 First Examples and Special Classes of Rings

The zero ring $\{*\}$ is often called the trivial ring. Note that $0 = 1$ in this ring. Other important examples include more familiar rings, such as $\mathbb{R}$ and $\mathbb{Z}$.

Definition 3.2: Commutative

A ring R is commutative if for all $r, s \in R$, $rs = sr$.

Commutative rings with identity are extremely important, and are studied in the subfield of algebra known as *commutative algebra*.

Rings need not be commutative, however. For example, the set of 2×2 matrices with real entries form a ring, yet are not commutative.

One final example is the rings $\mathbb{Z}/n\mathbb{Z}$, where multiplication is defined as

$$[a]_n \cdot [b]_n = [ab]_n.$$

This construction satisfies the ring axioms. This ring is important, because it fails the cancellation property. In this ring, $ab = ac$ does not imply $b = c$. Of course, we assume $a, b, c \neq 0$ here, since in general 0 cannot be canceled, and we can only define a cancellation property for nonzero elements. This property does hold in other rings, such as $\mathbb{R}, \mathbb{Z}, \mathbb{Q}$, and more. So the question is *when* does it hold? To solve this problem, we need to investigate zero divisors. Note that in $\mathbb{Z}/4\mathbb{Z}$, $2 \cdot 2 = 0$, but $2 \neq 0$. This ring has what we call *zero divisors*.

Definition 3.3: Zero Divisor

An element a in a ring R is a *left-zero-divisor* if there exists at least one element $b \in R^*$ such that $ab = 0$. Conversely, a *right-zero-divisor* is an element a such that for $b \neq 0$, $ba = 0$.

Proposition 3.1

In a ring R, an element a is not a left-zero-divisor if and only if left multiplication $r \mapsto ar$ is injective. Similarly, a is not a right-zero-divisor if and only if right multiplication $r \mapsto ra$ is injective.

Proof. We prove only the left statement, as the right statement is analogous. First let $a, b, c \in R$ and suppose a is not a zero divisor, and $ab = ac$. We then have

$$a(b-c) = ab - ac = 0.$$

Since a is not a zero divisor, we have $b - c = 0$. Now suppose $r \mapsto ar$ is injective. Note that $a0 = 0$, so by injectivity, $ar = 0 \implies r = 0$. Therefore, a is not a zero divisor. ■

Rings such as $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are commutative and do not have zero divisors. Since these rings are so important, we give a definition to all rings of this type.

Definition 3.4: Integral Domain

An integral domain is a nonzero commutative ring with no zero divisors. That is,

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

Integral domains are so important that chapter 5 will be devoted entirely to them. One important thing to note here is that $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

We will see later that $\mathbb{Z}$ is also a UFD, PID, and Euclidean domain (chapter 5 stuff), but $\mathbb{R}, \mathbb{Q}, \mathbb{C}$ are even more special than $\mathbb{Z}$. They are what we call fields.

Definition 3.5: Unit

An element u of a ring R is a left-unit if there exists $v \in R$ such that $uv = 1$. Equivalently, if $vu = 1$, then u is a right unit, and if $uv = vu = 1$, then u is a unit.

Proposition 3.2

In a ring R,

- u is a left/right unit if and only if left/right multiplication by u is surjective.
- if u is a left/right unit, then right/left multiplication by u is injective.
- The inverse of a unit is unique.
- Units form a group under multiplication.

Proof. The first statement is trivial. Let u be a left unit with an inverse v. Then $\forall r \in R$, $vr \mapsto uvr = 1r = r$. Proof for right units is analogous. Now let the map be surjective. Then there exists v such that $uv = 1$, so u is a unit.

Let u be a right unit, and let $ur = us$. If the right-inverse of u is v, then

$$vur = vus \implies r = s.$$

Therefore, left multiplication is injective. The other two statements are proven in the exercises. ■

We denote the inverse of a unit as u^{-1}. Left- and right-units are not unique in general, so we only adopt this notation when u is a unit.

Definition 3.6: Division Ring

A division ring is a ring in which all nonzero elements are units.

We are mainly concerned with the commutative case in this book, which has it's own name.

Definition 3.7: Field

A field is a nonzero commutative division ring.

The entirety of chapter 7 will be spent studying fields. By proposition 3.2, all fields are integral domains, but the converse is not true. It is true, however, in the finite case.

Proposition 3.3

Let R be a *finite* ring. Then R is a field if and only if R is an integral domain.

Proof. The one direction is trivial, so let R be an integral domain. Then multiplication by any element is injective, and since R is finite, it is surjective. Therefore, by previous logic, every element is a unit. Since the ring is commutative, we don't specify left vs right. ■

We will show much later that all finite division rings are commutative, so this commutativity postulate is in fact not needed. Note that $\mathbb{Z}/p\mathbb{Z}$ is a field iff p is prime. However, these are not the only finite fields. Later, we will consider fields of the form $\mathbb{Z}/p\mathbb{Z} \times \cdots \times \mathbb{Z}/p\mathbb{Z}$.

3.1.3 Polynomial Rings

Polynomial rings are an important class of rings that we will study.

Definition 3.8: Polynomial Ring

Let R be a ring. A polynomial $f(x)$ in the indeterminant x with coefficients in R is a finite linear combination

$$f(x) = \sum_{i \geq 0} a_i x^i,$$

where $a_i \in R$ for all i. Two polynomials are equal iff the coefficients are equal. The set $R[x]$ is the set of all such polynomials. Polynomial addition and multiplication are defined as

$$f(x) + g(x) = \sum_{i \geq 0} (a_i + b_i) x^i$$

and

$$f(x)g(x) = \sum_{k \geq 0} \sum_{i+j=k} a_i b_j x^{i+j}.$$

The degree of a polynomial, denoted $\deg f(x)$, is the largest integer d such that $a_d \neq 0$. Polynomials of degree 0 are called constants, which form a copy of R in $R[x]$. Polynomials in more indeterminants $R[x, y, z]$ can be formed by iterating this construction

$$R[x, y, z] \coloneqq R[x][y][z].$$

It can be easily checked that the order of the listing is not important, and as such these can be manipulated the same as ordinary polynomials. We will very rarely consider polynomials in infinitely many determinants; $R[x_1, x_2, \ldots]$ will denote countably many indeterminants. We must keep in mind however that we only admit finite linear combinations, even though there are infinite indeterminants. A true definition of this ring requires limits, which will be discussed in chapter 8. Ring of power series

$$\sum_{i=0}^{\infty} a_i x^i$$

are denoted $R[[x]]$. We will only encounter these occasionally, athough they are important.

We will discuss what properties $R[x]$ inherets from R in a later section. However, note that $R[x]$ can never be a field, since the polynomial x will never have an inverse.

3.1.4 Monoid Rings

Polynomial rings are specific instances of a more general construction. A semigroup is a set with an associative operation, and a monoid is a semigroup with an identity. That is, a group is a monoid where all elements are invertible. Given a monoid $(M, \cdot)$ and a ring R, we can define a ring $R[M]$ as formal finite linear combinations

$$\sum_{m \in M} a_m \cdot m,$$

where coefficients are elements in R. Note that as an abelian group, $R[M] = R^{\oplus M}$. Operations are defined in the usual way, and the identity is $1_R 1_M$. Group rings are the result of this construction when M is a group. The Laurent polynomials is the group ring $R[\mathbb{Z}]$ or something.

3.2 The Category Ring

Definition 3.9: Ring Homomorphism

A ring homomorphism $\phi : R \to S$ is a function such that for all $r, s \in R$,

$$\phi(r+s) = \phi(r) + \phi(s),$$

$$\phi(rs) = \phi(r)\phi(s),$$

$$\phi(1_R) = 1_S.$$

It is obvious that rings, together with ring homomorphisms, form a category, denoted Ring. Also note that the zero ring is obviously final in Ring, as it is a singleton and the only map to a singleton is the constant map. However, the requirement that $1 \mapsto 1$ makes it NOT initial. Rather, $\mathbb{Z}$ is initial. For any ring R, we can define a map $\phi : \mathbb{Z} \to R$ by $\phi(n) = n \cdot 1_R$. That is, the exponential map we saw in chapter 2. Note that ϕ is a ring homomorphism:

$$\phi(1) = 1_R, \qquad \phi(n+m) = (n+m)1_R = n1_R + m1_R,$$

and by distributivity,

$$\phi(nm) = (nm)1_R = n1_R \cdot m1_R = \phi(n)\phi(m).$$

This ring homomorphism is also unique, since $1 \mapsto 1_R$ and the fact that ϕ preserves addition determine it fully. This is why we include 1 in the definition of a ring, to make $\mathbb{Z}$ initial. $\mathbb{Z}$ is then a very special ring.

Ring homomorphisms also preserve units. If u is a left/right unit, then $\phi(u)$ is the same:

$$uv = 1_R \implies 1_S = \phi(1_R) = \phi(uv) = \phi(u)\phi(v).$$

However, the image of a non-zero-divisor can be a zero divisor. For example, the canonical projection $\pi : \mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$ is a homomorphism, but $2 \mapsto [2]_6$ is a zero divisor.

3.2.1 Universal Property of Polynomial Rings

Polynomial rings satisfy a universal property similar to that of free groups. We will state the universal property for commutative rings.

Let $A = \{a_1, \ldots, a_n\}$ be a set of order n. Consider the category $\mathscr{R}_A$ where objects are pairs (j, R) where R is a ring and $j : A \to R$ is a function of sets; morphisms

$$(j_1, R_1) \to (j_2, R_2)$$

are commutative diagrams

$$\begin{tikzcd} R_1 \arrow[r, "\phi"] & R_2 \\ A \arrow[u, "j_1"] \arrow[ur, "j_2"'] & \end{tikzcd}$$

in which ϕ is a ring homomorphism. Consider the object $(i, \mathbb{Z}[x_1, \ldots, x_n])$, where

$$i : A \to \mathbb{Z}[x_1, \ldots, x_n]$$

is the map $a_k \mapsto x_k$.

Proposition 3.4

$(i, \mathbb{Z}[x_1, \ldots, x_n])$ is initial in $\mathscr{R}_A$.

Proof. Let (j, R) be an arbitrary object in $\mathscr{R}_A$. If the diagram

$$\begin{tikzcd} \mathbb{Z}[x_1, \ldots, x_n] \arrow[r, "\phi"] & R \\ A \arrow[u, "i"] \arrow[ur, "j"'] & \end{tikzcd}$$

commutes, then we are forced to define $\phi(x_k) = j(a_k)$ for all k. The requirement that ϕ be a ring homomorphism forces

$$\phi\left(\sum m_{i_1 \cdots i_n} x_1^{i_1} \cdots x_n^{i_n}\right) = \sum \phi(m_{i_1 \cdots i_n}) \phi(x_1)^{i_1} \cdots \phi(x_n)^{i_n}$$

$$\sum \iota(m_{i_1 \cdots i_n}) j(a_1)^{i_1} \cdots j(a_n)^{i_n},$$

where ι is the unique ring homomorphism $\mathbb{Z} \to R$. This forces ϕ to be unique. It is clear the identity is preserved, so ϕ is a homomorphism. ■

One corollary of this is that for all (not necessarily commutative) rings S, for all $s \in S$, there exists a unique ring homomorphism $\mathbb{Z}[x] \to S$ mapping $x \mapsto s$.

Another corollary of this, which is to be shown in the exercises, is that for a ring homomorphism $\alpha : R \to S$ and some fixed $s \in S$ commuting with $\alpha(r)$ for all $r \in R$, there exists a unique ring homomorphism $\phi_s : R[x] \to S$ that maps $x \mapsto s$. Applying this to the commutative case, we can define the evaluation of a polynomial at some element $r \in R$. Consider the identity map $\mathrm{id} : R \to R$, with R commutative. Clearly, $\mathrm{id}(r)$ commutes with all r, so we find that for all $r \in R$, there exists a unique $\phi_r : R[x] \to R$ such that $x \mapsto r$. We can then define for all $f(x) \in R[x]$ the evaluation map,

$$f : R \to R$$

defined by

$$f(r) = \phi_r(f(x)).$$

Informally, this map sends

$$a_n x^n \mapsto a_n r^n.$$

We will call these evaluation maps "polynomial functions", and will call elements of $R[x]$ "polynomials".

3.2.2 Monomorphisms and Epimorphisms

The kernel is defined analogously.

Definition 3.10: Kernel

The kernel of a ring homomorphism $\phi : R \to S$ is defined as

$$\ker \phi := \{r \in R \mid \phi(r) = 0\}.$$

This is the same as the kernel of ϕ when viewed as a homomorphism of groups. However, the kernel is more than just a normal subgroup of $(R, +)$, as we will explore in this section. First, we have an analogous proposition for monomorphisms as in Grp.

Proposition 3.5

Let $\phi : R \to S$ be a ring homomorphism. Then the following are equivalent.

- ϕ is a monomorphism.
- $\ker \phi = \{0\}$.
- ϕ is injective.

Proof. The book only proves the first implication. ($a \implies b$). Let $\phi : R \to S$ be a ring homomorphism, and let $r \in \ker \phi$. By previous logic concerning proposition 3.4, there exist unique ring homomorphisms $\mathrm{ev}_r, \mathrm{ev}_0 : \mathbb{Z}[x] \to R$.

$$\mathbb{Z}[x] \underset{\mathrm{ev}_r}{\overset{\mathrm{ev}_0}{\rightrightarrows}} R \xrightarrow{\phi} S$$

Let $x \in \mathbb{Z}[x]$, and note that $\mathbb{Z}$ can be viewed as a subset of $\mathbb{Z}[x]$ by the inclusion map $\mathbb{Z} \hookrightarrow \mathbb{Z}[x]$. There are two cases. If $x \in \mathbb{Z}$, then by proposition 3.4, $\mathrm{ev}_0(x) = \mathrm{ev}_r = \iota(x)$ is the unique ring homomorphism $\mathbb{Z} \to R$. If $x \in \mathbb{Z}[x] \setminus \mathbb{Z}$, then once again the action on the coefficients is the same, and we need to look only at the action on the variable x. Note that $(\phi \circ \mathrm{ev}_0)(x) = 0 = \phi(r) = (\phi \circ \mathrm{ev}_r)(x)$. Therefore, $\phi \circ \mathrm{ev}_0 = \phi \circ \mathrm{ev}_r$. Since ϕ is a monomorphism, we have $\mathrm{ev}_0 = \mathrm{ev}_r$. Therefore, $0 = r$, and $\ker \phi = \{0\}$.

($b \implies c$) Let $\ker \phi = \{0\}$, and let $r, s \in R$ such that $\phi(r) = \phi(s)$. It follows that

$$\phi(r) = \phi(s) \implies \phi(r - s) = 0 \implies r - s \in \ker \phi \implies r - s = 0.$$

Therefore, ϕ is injective.

($c \implies a$) Let ϕ be injective. Then ϕ is a monomorphism in Set. Since all morphisms in Ring are also morphisms in Set, we have that ϕ is a monomorphism in Ring. ■

Definition 3.11: Subring

A subring S of a ring R is a ring S such that $S \subseteq R$ and the inclusion map $S \hookrightarrow R$ is a ring homomorphism.

Equivalently, S is a subring of R if $S \subseteq R$ is a subset containing 1_R, and S satisfies the ring axioms given the operations inhereted from R. Note that under this definition, $\{0\}$ is not a subring of nontrivial rings, since $0 \neq 1_R$ unless in the trivial case.

There is a strange but important distinction between Ring and Grp, concerning epimorphisms. We saw in Set, Grp, and Ab, that epimorphisms are simply surjective morphisms. However, in Ring, epimorphisms don't need to be surjective. Consider the inclusion $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$. Clearly ι is not surjective, and thus is not an epimorphism in Set or Ab, but we can show it is an epimorphism of rings. Let $\alpha_1, \alpha_2 : \mathbb{Z} \to R$ be ring homomorphisms such that $\alpha_1 \iota = \alpha_2 \iota$. Note that these ring homomorphisms must be equivalent on $\mathbb{Z}$, since $\mathbb{Z}$ is initial, and note that for all rational numbers, we have

$$\alpha_i \left(\frac{a}{b} \right) = \alpha(a)\alpha(b)^{-1}.$$

Therefore, $\alpha_1 = \alpha_2$. Therefore, ι is an epimorphism. This means in Ring, a morphism can be both epi and mono, yet not an isomorphism.

3.2.3 Products

Products exist in Ring. The product of two rings can be defined by endowing the direct product of groups $R_1 \times R_2$ with componentwise multiplication. Note that this is *not the only ring structure* that can be defined on the direct product of the underlying groups, as we have mentioned previously you can define a field structure on $\mathbb{Z}/p\mathbb{Z}\times\mathbb{Z}/p\mathbb{Z}$, but interpreting this as a product of rings is not a field, since $(1,0)\cdot(0,1)=(0,0)$ are zero divisors.

Coproducts are difficult to develop in a general sense without tensor products, since defining a canonical projection onto it in the natural way is difficult, and we will have basically forgotten about this question by the time we develop those. We will do simple examples in the exercises, however.

3.2.4 $\mathrm{End}_{\mathsf{Ab}}(G)$

The endomorphism group of an abelian group can be viewed as a ring, by taking function composition to be multiplication. Checking that these satisfy the ring axioms is trivial.

Proposition 3.6

As rings, $\mathrm{End}_{\mathsf{Ab}}(\mathbb{Z})\cong\mathbb{Z}$.

Proof. Consider the map $\phi:\mathrm{End}_{\mathsf{Ab}}(\mathbb{Z})\to\mathbb{Z}$ defined by $\alpha\mapsto\alpha(1)$. By definition of addition in $\mathrm{End}_{\mathsf{Ab}}(\mathbb{Z})$, we have

$$(\alpha+\beta)(1)=\alpha(1)+\beta(1).$$

Since for all $n\in\mathbb{Z}$, $\alpha(n)=n\alpha(1)$, we have

$$\phi(\alpha\beta)=(\alpha\circ\beta)(1)=\alpha(\beta(1))=\alpha(1)\beta(1)=\phi(\alpha)\phi(\beta).$$

Finally, define $\psi:\mathbb{Z}\to\mathrm{End}_{\mathsf{Ab}}(\mathbb{Z})$ as the map $a\mapsto\alpha$, where α is defined by $n\mapsto an$. We have

$$(\phi\circ\psi)(n)=\phi(\alpha)=\alpha(1)=1n=n.$$

Additionally,

$$(\psi\circ\phi)(\alpha)=\psi(\alpha(1))=\beta,$$

where $\beta(n)=n\alpha(1)=\alpha(n)$. ■

Endomorphism rings of various structures, such as groups and vector spaces, are arguably the most important class of rings. In fact, every ring R interacts with the ring of endomorphisms of the underlying abelian group. For all $r\in R$, define left multiplication by r as the map λ_r, and right multiplication μ_r. We can then show a ring analogue of Cayley's theorem.

Proposition 3.7

Let R be a ring. Then the map $\lambda:R\to\mathrm{End}_{\mathsf{Ab}}(R)$ given by $r\mapsto\lambda_r$ is an injective ring homomorphism.

Proof. Fix some $r\in R$. We have

$$\lambda_r(a+b)=r(a+b)=ra+rb=\lambda_r(a)+\lambda_r(b).$$

Therefore, $\lambda_r\in\mathrm{End}_{\mathsf{Ab}}(R)$. The function λ is clearly injective; let $\lambda_r=\lambda_s$. Then $\lambda_r(1)=r=s=\lambda_s(1)$. Moreover, $\lambda_1(x)=x$ is trivially the identity map. We must show λ satisfies the other two properties of ring homomorphisms. For all $a\in R$, we have

$$\lambda(r+s)(a)=\lambda_{r+s}(a)=(r+s)(a)=ra+sa=\lambda_r(a)+\lambda_s(a).$$

Additionally, we have

$$\lambda(rs)(a) = \lambda_{rs}(a) = (rs)(a) = \lambda_r(sa) = (\lambda_r\lambda_s)(a) = (\lambda(r)\lambda(s))(a).$$

Therefore, λ is a ring homomorphism. ■

It will be shown in the exercises that μ is not quite a ring homomorphism, because $\mu_{rs} = \mu_s \circ \mu_r$. It would be a ring homomorphism if the direction of multiplication were reversed.

3.3 Ideals and Quotient Rings

3.3.1 Ideals

In Grp and Set, we found suitable equivalence relations to quotient objects by, and were able to classify surjective morphisms with these objects. That is, we were able to identify each surjective morphism $\phi : G \to G'$ with the group

$$\frac{G}{\ker \phi},$$

which is isomorphic to G'. We will do the same in Ring.

Definition 3.12: Ideal

Let R be a ring. A subgroup I of $(R, +)$ is a left-idea of R if $rI \subseteq I$ for all $r \in R$. Equivalently, for all $r \in R$ and $a \in I$, $ra \in I$. Right ideals are defined analogously, and two-sided ideals are simply called *ideals*.

Even in noncommutative cases, we are generally only interested in two-sided ideals, so we will use the word *ideal* to explicitly refer to these objects.

Note that for an ideal I to be a subring, it must include 1_R, but by definition this means that for all $r \in R$, $r1 = r \in I$. That is, $I = R$. In general then, ideals are not subrings, but rather rngs. However, they are much more important than subrings in developing the theory. Similarly to Grp, ideals are also kernels of ring homomorphisms. Also note that the image of any homomorphism is a subring.

Proposition 3.8

Let $\phi : R \to S$ be a ring homomorphism. Then $\ker \phi$ is an ideal of R.

Proof. For all $r \in R$, $a \in I$, we have

$$\phi(ra) = \phi(r)0 = 0 = 0\phi(r) = \phi(ar).$$

Proving a kernel is a subgroup is trivial. More generally, the inverse image of an ideal is an ideal, and $\{0\}$ is clearly an ideal. ■

3.3.2 Quotients

Since all subgroups of $(R, +)$ are normal, we have the quotient group R/I who's elements are cosets $r+I$. We also have the surjective group homomorphism $\pi : R \to R/I$, $r \mapsto r + I$. We wish to findd the circumstances under which this construction satisfiess the universal property in Ring.

Since we require π be a ring homomorphism, multiplication must be as follows: for all $a+I, b+I \in R/I$,

$$(a + I)(b + I) = \pi(a)\pi(b) = \pi(ab) = ab + I.$$

If this operation is well-defined, then R/I instantly becomes a ring, as associativity and distributivity are inhereted from R. We need only make the operation well-defined.

Assuming that the operation is well-defined, we have that I has to be an ideal; more precisely, we must have $I = \ker \pi$ as $i \in I$ implies $i \mapsto I$. That is, R/I is a ring and $\pi : R \to R/I$ is the canonical projection if and only if I is an ideal.

Definition 3.13: Quotient Ring

Given a ring R and an ideal I of R, the quotient of R modulo I is the ring R/I.

The fact that $\mathbb{Z}$ is initial in Ring now gives the natural definition of characteristic. Note that the ideals of $\mathbb{Z}$ are precisely the sets $n\mathbb{Z}$ for nonnegative integers n.

Definition 3.14: Characteristic

Let R be a ring, and let $f : \mathbb{Z} \to R$ be the unique ring homomorphism, defined by $a \mapsto a \cdot 1_R$. The characteristic of R is the well-defined non-negative integer n such that $\ker f = n\mathbb{Z}$.

Equivalently, the characteristic of R is the order of 1_R in $(R, +)$ if 1_R has finite order, and 0 if it has infinite order. This shows that kernels are ideals, and ideals are kernels. We can now restate some important theorems from Grp.

Theorem 3.1

Let I be an ideal of R. Then for all ring homomorphisms $\phi : R \to S$ such that $I \subseteq \ker \phi$, there exists a unique ring homomorphism $\widetilde{\phi} : R/I \to S$ defined by $\widetilde{\phi}(r + I) = \phi(r)$ such that the diagram

$$\begin{tikzcd} R \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & S \\ & R/I \arrow[ur, "\widetilde{\phi}"'] & \end{tikzcd}$$

commutes.

Proof. The morphism as a group homomorphism exists, is unique, and is trivially a ring homomorphism. ■

3.3.3 Canonical Decomposition and Consequences

Theorem 3.2 (Canonical Decomposition)

Every ring homomorphism $\phi : R \to S$ can be decomposed as

$$\begin{tikzcd} R \arrow[r, two heads] \arrow[rrr, bend left, "\phi"] & R/\ker\phi \arrow[r, "\sim", "\widetilde{\phi}"'] & \phi(R) \arrow[r, hook] & S \end{tikzcd}$$

where $\widetilde{\phi}$ is the isomorphism induced by theorem 3.1.

Proof. It suffices to show the isomorphism is a ring isomorphism, since we have already shown it for groups. We have already done this, so we are done. ■

Corollary 3.1 (First Isomorphism Theorem)

Let $\phi : R \to S$ be a surjective ring homomorphism. Then

$$S \cong \frac{R}{\ker \phi}.$$

We can also show that subgroups of R/I are in correspondence with the ideals in R containing I. We have shown this already in proposition 2.28 for Grp, and it takes minor effort to show the same bijection holds for ideals as well. From this, we get the third isomorphism theorem.

Proposition 3.9 (Third Isomorphism Theorem)

Let I be an ideal of R, and let J be an ideal of R containing I. Then J/I is an ideal of R/I, and

$$\frac{R/I}{J/I} \cong \frac{R}{J}.$$

Proof. The first statement for groups follows from proposition 2.28. To show the set in question is an ideal, let $j + I \in J/I$ and $r + I \in R/I$. We have

$$jr, rj \in J \implies (j + I)(r + I), (r + I)(j + I) \in J/I.$$

Consider the canonical projection $\pi : R \to R/J$. Since $I \subseteq J$, by theorem **??** we have a unique ring homomorphism

$$\phi : R/I \to R/J$$

defined by $r + I \mapsto r + J$. To show the map is surjective, let $r + J \in R/J$, and note that there are two cases. If $r \in J$, we have $\phi(I) = J$. If $r \notin J$, we have $r \notin I$, so $r + I \in R/I$, and thus $\phi(r + I)r + J$. Therefore, ϕ is surjective, and by corollary 3.1, we have

$$\frac{R/I}{\ker \phi} \cong \frac{R}{J}.$$

Note that $\ker \phi = \{r + I \mid r \in J\} = J/I$. Thus, J/I is an ideal since it is a kernel. ■

The second isomorphism theorem *would* state that

$$\frac{I + J}{I} \cong \frac{J}{I \cap J},$$

but according to the conventions in this text, $I + J$ and $I \cap J$ are not rings. Thus, we will wait until we reach modules to discuss this theorem.

3.4 Ideals and Quotients: Remarks and Examples. Prime and Maximal Ideals

3.4.1 Basic Operations

We often define ideals in terms of a set of generators. Let $a \in R$ be any element. Then the subsets Ra and aR are left- and right-ideals, respectively. If the ring is commutative, these sets are clearly the same, and we denote them (a).

Definition 3.15: Principal Ideal

Let R be a ring and let $a \in R$. Then the principal ideal generated by a is the set

$$(a) = aR = Ra.$$

From the exercises, we know that for any collection $\{I_\alpha\}_{\alpha \in A}$ of ideals of R, the sum

$$\sum_{\alpha \in A} I_\alpha$$

is an ideal of R. We have a similar idea for ideals generated by multiple elements. Let $\{a_\alpha\}_{\alpha \in A} \subseteq R$. Then define

$$(a_\alpha)_{\alpha \in A} \coloneqq \sum_{\alpha \in A} (a_\alpha)\,.$$

In the finite case, this is given as

$$(a_1, \ldots, a_n) = (a_1) + \cdots + (a_n).$$

This is the smallest ideal containing $\{a_\alpha\}_{\alpha \in A}$. An ideal I is called *finitely generated* if there exists a finite collection $a_1, \ldots, a_n \in R$ such that

$$I = (a_1, \ldots, a_n)\,.$$

One example that I want to copy down is as follows. Let R be a commutative ring, and let $a, b \in R$. Let $\bar{b}$ denote the equivalence class of b under the relation $R/(a)$. Then

$$\frac{R/(a)}{(\bar{b})} \cong \frac{R}{(a, b)}.$$

Note that this follows from proposition 3.9, as for some reason,

$$(\bar{b}) \cong \frac{(a, b)}{(b)}.$$

I don't really care to prove this honestly.

Principal ideals are very special, so we use special terminology when referring to them.

Definition 3.16: Noetherian

A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Definition 3.17: Principal Ideal Domain

An integral domain R is a PID (Principal Ideal Domain) if every ideal of R is principal.

Proposition 3.10

$\mathbb{Z}$ is a PID.

Proof. Let $I \subseteq \mathbb{Z}$ be an ideal. Since I is a subgroup, $I = n\mathbb{Z} = (n)$ for some $n \in \mathbb{Z}$. ■

Interestingly, the fact that $\mathbb{Z}$ is a PID explains why gcds behave as they do. If $m, n \in \mathbb{Z}$, then (m, n) is principal, meaning there exists $d \in \mathbb{Z}$ such that $(m, n) = (d)$. This d is fairly trivially $\gcd(m, n)$. Additionally, for any field k, the ring $k[x]$ is also a PID, which we will prove soon. By contrast, $\mathbb{Z}[x]$ is not a PID. For example, $(2, x)$ cannot be generated by a single element.

Here are two more operations on ideals. Let $\{I_\alpha\}_{\alpha\in A}$ be a collection of ideals of some ring R. Then

$$\bigcap_{\alpha\in A} I_\alpha$$

is an ideal of R; it is the largest ideal contained in $\{I_\alpha\}_{\alpha\in A}$. Another operation is the following: IJ denotes the ideal generated by all products ij with $i \in I$, $j \in J$. Note that this is not the same as the IJ notation used in group theory, as it is not just the products in these ideals, but rather the ideal it generates. Moreover, note that $ij \in I, J$ since I, J are ideals, so we have $IJ \subseteq I \cap J$.

3.4.2 Quotients of Polynomial Rings

Let R be a ring, and let $f(x) = x^d + \cdots + a_1x^1 + a_0 \in R[x]$. We are assuming for convenience that f is monic; the leading coefficient is the identity 1_R. It will be shown in the exercises that if R is a field, then all principal ideals are generated by unique monic polynomials. In the general case, however, choosing f to be monic can have impacts on the ideal. However, assuming it is monic is convenient, because it is necessarily not a zero divisor, and additionally we have a notion of division in general rings. That is, for all $g(x) \in R[x]$, there exist $q(x), r(x) \in R[x]$ such that

$$g(x) = q(x)p(x) + r(x)$$

and $\deg r < \deg f$, where we take the convention that the polynomial 0 has degree $-\infty$.

Lemma 3.2

Let $f(x)$ be a monic polynomial, and suppose

$$f(x)q_1(x) + r_1(x) = f(x)q_2(x) + r_2(x),$$

where $\deg r_1, \deg r_2 < \deg f$. Then $q_1(x) = q_2(x)$ and $r_1(x) = r_2(x)$.

Proof. We have

$$f(x)(q_1(x) - q_2(x)) = r_1(x) - r_2(x).$$

Suppose $r_2 \neq r_1$. Then $\deg(r_2 - r_1) < \deg f$. However, since $f(x)$ is monic,

$$\deg(f(x)q(x)) = \deg f(x) + \deg q(x).$$

Thus, $\deg(f(q_1 - q_2)) > \deg f$, which is a contradiction. Therefore, $r_1(x) = r_2(x)$. Since f is monic and thus not a zero divisor, it follows that $q_1 = q_2$. ■

Note here that although division appears to be unique, it is only really unique in the commutative case. For noncommutative rings, we may have distinct left- and right-divisors of f. We will only consider the commutative case here, largely for convenience.

This is equivalent to the following: for all $g(x) \in R[x]$, there exists a unique $r(x) \in R[x]$ with $\deg r < \deg f$ such that

$$g(x) + (f(x)) = r(x) + (f(x)).$$

We can refine this observation. Note that any polynomial with degree less than d can be viewed as an element of the direct sum $R^{\oplus d}$. The map

$$\psi(r_1, \ldots, r_{d-1}) = r_0 + r_1x + \cdots + r_{d-1}x^{d-1}.$$

is clearly a *group* homomorphism. We also have that it is injective fairly trivially. We would like ψ to be a *ring* homomorphism, but the multiplicative structure of $R[x]$ and R^d differ. We can reconsile this by the following proposition.

Proposition 3.11

Let R be a commutative ring, and let $f(x) \in R[x]$ have degree d. Then the map $\phi : R[x] \to R^{\oplus d}$ defined by mapping $g(x)$ to its remainder when dividing by $f(x)$ induces an isomorphism of groups

$$\frac{R[x]}{(f(x))} \cong R^{\oplus d}.$$

Proof. By lemma 3.2, ϕ is well-defined. We also have that ϕ is surjective, because ψ as previously defined is a right-inverse:

$$(\phi \circ \psi)(r_0, \dots, r_{d-1}) = \phi(r_0 + r_1 x + \dots + r_{d-1}x^{d-1}) = (r_0, \dots, r_{d-1}),$$

as f cannot divide into $r(x)$. By corollary 2.2,

$$R^{\oplus d} \cong \frac{R[x]}{\ker \phi}.$$

Note that $\phi(g(x)) = 0$ if and only if the remainder is 0; equivalently, $f(x)q(x) = g(x)$ for some $q(x) \in R[x]$, and thus $g(x) \in (f(x))$. Therefore, $\ker \phi = (f(x))$. ■

This allows us to define multiplication in a new way on the direct sum $R^{\oplus d}$, by using this isomorphism of groups and extending it in the most natural way to an isomorphism of rings. For a simple example, let $f(x) = x - a$ have degree 1, and note that r must thus be a constant. Then note that ϕ is simply the evaluation map at a. We have

$$g(x) = (x - a)q(x) + r \implies g(a) = (a - a)q(x) + r = r.$$

This induces an isomorphism

$$\frac{R[x]}{(x - a)} \cong R,$$

which is in fact an isomorphism of rings. We can extend this idea to more complex examples, as for every monic polynomial, there is a brand new ring structure on $R^{\oplus d}$. For $d = 1$ these structures are all the same, but new structures arise as soon as $d = 2$. Here is a concrete example.

Let $f(x) = x^2 + 1_R$. We have by proposition 3.11 that

$$R \oplus R \cong \frac{R[x]}{(x^2 + 1_R)}.$$

We can analyze the multiplication operation induced on $R \oplus R$. Recall that each element of the ideal can be represented as a single remainder polynomial; in this case, that is a polynomial of degree one, of the form $r(x) = a_0 + a_1 x$. We have

$$(a_0 + a_1 x)(b_0 + b_1 x) = a_0 b_0 + (a_0 b_1 + a_1 b_0)x + a_1 b_1 x^2.$$

Dividing by $(x^2 + 1_R)$, we have

$$a_0 b_0 + (a_0 b_1 + a_1 b_0)x + a_1 b_1 x^2 = \left(x^2 + 1\right) a_1 b_1 + ((a_0 b_0 - a_1 b_1) + (a_0 b_1 + a_1 b_0)x).$$

Applying ϕ, we have

$$\phi((a_0 + a_1 x)(b_0 + b_1 x)) = (a_0 b_0 - a_1 b_1, a_0 b_1 + a_1 b_0).$$

This induces the multiplication structure

$$(a_0, a_1)(b_0, b_1) = (a_0 b_0 - a_1 b_1, a_0 b_1 + a_1 b_0)$$

on $R \oplus R$. One might recognize this as precisely the multiplication operation in $\mathbb{C}$, where an element of $\mathbb{C}$ is identified with a pair of real numbers $(a_0, a_1) \in \mathbb{R} \oplus \mathbb{R}$. That is,

$$\frac{\mathbb{R}[x]}{(x^2 + 1)} \cong \mathbb{C}$$

as rings. This is not suprising, as $x^2 + 1$ does not have real roots, and since the roots will show up as classes of x and $-x$ in the quotient, the construction should contain the roots. We will revisit these constructions to solve equations in chapters 5 and 7.

3.4.3 Prime and Maximal Ideals

We are still assuming that rings are commutative, since we will not use these definitions in noncommutative contexts.

Definition 3.18: Prime/Maximal Ideal

Let $I \neq (1)$ be an ideal of a commutative ring R. I is called a prime ideal if R/I is an integral domain, and I is called a maximal ideal if R/I is a field.

These definitions are equivalent to the following proposition, and it is a matter of preference whether to define these in terms of quotients or this proposition.

Proposition 3.12

Let $I \neq (1)$ be an ideal of a commutative ring R. Then

- I is prime if and only if for all $a, b \in R$,

$$ab \in I \implies (a \in I \vee b \in I).$$

- I is maximal if and only if for all ideals J of R,

$$I \subseteq J \implies (I = J \vee J = R).$$

Proof. The ring R is an integral domain if and only if for all $a + I, b + I \in R/I$,

$$ab + I = I \implies (a \in I \vee b \in I).$$

Wow, this is an equivalent statement.

From the exercises, we know a commutative ring is a field if and only if its only ideals are (0) and (1). Using this and the fact that there is a bijection between ideals containing I and ideals of R/I (proposition 3.9), we have that R is a field. ■

Note that all maximal ideals are prime, but the converse is not necessarily true. There is, however, a case where it holds.

Proposition 3.13

Let R be a ring with an ideal I. If R/I is finite, then I is prime if and only if I is maximal.

Proof. Follows immediately from proposition 3.3. ■

One example is that $\mathbb{Z}/n\mathbb{Z}$ is finite, so (n) is prime/maximal when n is prime. Attributing this uniqueness to $\mathbb{Z}$ is mildly misleading, however, as it holds for all PIDs.

Proposition 3.14

Let R be a PID, and let $I \neq (0)$ be an ideal of R. Then R is prime if and only if R is maximal.

Proof. Maximal ideals are always prime, so we show the converse. Let $I = (a)$ be prime, and suppose $I \subseteq J$ is an ideal of R. Note that J must be a principal ideal, so $J = (b)$. Since $(a) \subseteq (b)$, we have that $a \in (b)$, and thus $a = bc$ for some $c \in R$. But then either $b \in (a)$ or $c \in (a)$ since (a) is prime. If $b \in (a)$, then $(b) \subseteq (a)$. If $c \in (a)$, then $c = da$ for some $d \in R$, so $a = bc = bda$, and thus $bd = 1$. Therefore, b is a unit, so $1 \in (b) = R$. ■

Definition 3.19: Spectrum

The spectrum of a commutative ring R, denoted $\operatorname{Spec} R$, is the set of all prime ideals in R.

The commentary on the spectrum involves figures too advanced for me to bother copying, so I recommend reading the tailend of section III.4.3 in the book. We can draw the spectrum, where each level higher contains the lower levels I believe. Drawing the spectrum of $k[x]$ for k a field would look a lot like $\mathbb{Z}$; all the prime numbers at one level, all containing the nonmaximal ideal (0).

This picture is nice for algebraicly closed fields, like $\mathbb{C}$. Using this fact, it is shown in the exercises that the maximal ideals in $\mathbb{C}[x]$ are all the ideals of the form $(x - z)$. Drawing $\operatorname{Spec} \mathbb{C}$, we draw one level of prime/maximal ideals above the ideal (0), so we say $\mathbb{C}[x]$ has Krull dimension 1.

Definition 3.20: Krull Dimension

The Krull dimension of a ring R is the longest chain of inclusion of prime ideals in R.

For example, any field k has Krull dimension 0. More generally, we can show that $k[x_1, \dots, x_n]$ has Krull dimension n. The chain of length n is the chain

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \dots, x_n).$$

These ideals are shown to be prime in the exercises.

In the future, we will explore these ideals in more detail. It can be shown that the maximal ideals of $\mathbb{C}[x_1, \dots, x_n]$ are all of the ideals of the form $(x_1 - z_1, \dots, x_n - z_n)$, but this requires a deep result known as *Hilbert's Nullstellensatz*. We will return to this in chapter 7.

3.5 Modules Over a Ring

There are many parallels between Grp and Ring, but there are a few issues in Ring. Notably, ideals and kernels need not be rings, cokernels behave strangely, and even if one allows ideals to be subrings and removes the requirment of an identity, there is no large class of substuctures (like Ab in Grp) where all subrings are ideals.

Modules will fix these issues. If R is a ring and I is an ideal of R, then R, I, and R/I are all modules over R. Moreover, the category of R-modules is much better behaved than Ring, meaning kernels and cokernels do what we expect. In fact, the category Ab is equivalent to the category $\mathbb{Z}$-Mod of $\mathbb{Z}$-modules, and for each new ring R, we will get a new well-behaved category of modules. These categories are all examples of *abelian categories*, which we will learn about later.

3.5.1 Definition of (Left-)R-Module

In short, R-modules are abelian groups endowed with an action of R on the abelian group. Note that we were able to define group actions as a group homomorphism $\sigma : G \to \operatorname{Aut}_{\mathsf{C}}(A)$, and we said G is acting on the object A.

Recall that $\operatorname{End}_{\mathsf{Ab}}(G)$ can be viewed as a *ring*, so we can give an analogous definition of an action of a ring on an abelian group. A left-action of a ring R on an abelian group M is simply a ring homomorphism

$$\sigma : R \to \operatorname{End}_{\mathsf{Ab}}(M).$$

An equivalent way to represent this is with a function $\rho : R \times M \to M$ defined by $(r, m) \mapsto \sigma(r)(m)$. Note that since σ is a ring homomorphism, we have

- $\rho(1, m) = m$
- $\rho(r, m + n) = \rho(r, m) + \rho(r, n)$
- $\rho(r + s, m) = \rho(r, m) + \rho(s, m)$
- $\rho(rs, m) = \rho(r, \rho(s, m))$

We usually just say rm instead of $\rho(r, m)$. The above bullet points then become

- $1m = m$,
- $r(m + n) = rm + rn$,
- $(r + s)(m) = rm + sm$,
- $(rs)m = r(sm)$.

Definition 3.21: Left-R-module

A left R-module structure on an abelian group M consists of a map $\cdot : R \times M \to M$ called *scalar multiplication*, defined by $(r, m \mapsto \sigma(r)(m)$, where $\sigma : R \to \operatorname{End}_{\mathsf{Ab}}(M)$ is a ring homomorphism.

Right R-modules are defined analogously, with right R-actions. Equivalently, we can define a right R-module as a left R°-module, where R° is the opposite ring obtained by reversing the order of multiplication. If R is commutative, these become the same ring. We will write *module* to denote *left-module*.

3.5.2 The Category R-Mod

The book requests that I prove the following statements to familiarize myself with modules. First, for all $m \in M$, $0m = 0$. It suffices to show the additive identity of $\operatorname{End}_{\mathsf{Ab}}(M)$ is the trivial homomorphism $m \mapsto 0$, since a ring homomorphism is a group homomorphism, and $0 \mapsto 0$ under a group homomorphism. We have

$$(\alpha + 0)(x) = \alpha(x) + 0(x) = \alpha(x) + 0 = \alpha(x).$$

Therefore, 0 is the additive identity of $\mathrm{End}_{\mathsf{Ab}}(M)$. Next, the book wants us to show $(-1)m = -m$. Since σ is a ring homomorphism, we have that -1_R will map to the additive inverse of $1_{\mathrm{End}_{\mathsf{Ab}}(M)}$; that is, $-1_{\mathrm{End}_{\mathsf{Ab}}(M)}$. Note that $1_{\mathrm{End}_{\mathsf{Ab}}(M)} = \mathrm{id}_M$, so the additive inverse would be the map α such that

$$(\mathrm{id}_M + \alpha)(x) = 0 \implies \alpha(x) = -\mathrm{id}_M.$$

Therefore, $(-1)m = -\mathrm{id}_M(m) = -m$.

The following proposition is also a way of familiarizing myself with modules or something.

Proposition 3.15

Every abelian group is a unique $\mathbb{Z}$-module.

Proof. Let G be an abelian group. A $\mathbb{Z}$-module structure on G is a ring homomorphism $\mathbb{Z} \to \mathrm{End}_{\mathsf{Ab}}(M)$. Since $\mathbb{Z}$ is initial in Ring, there exists precisely one such ring homomorphism. ■

Using our previous results, we can show the action of $n \in \mathbb{Z}$ on some element $a \in M$ is simply the ordinary multiple $na = \sum_{0<i\leq n} a$.

A homomorphism of R-modules is then a homomorphism of abelian groups which is compatible with the module structure. That is, a homomorphism $\phi : M \to N$ is a homomorphism of R-modules iff for all $m, n \in M$ an $r \in R$,

- $\phi(m + n) = \phi(m) + \phi(n)$,
- $\phi(rm) = r\phi(m)$.

It should be clear that composition of homomorphisms R-modules yields a homomorphism of R-modules, and the identity map is a homomorphism.

Definition 3.22: R-Mod

The category R-Mod is the category who's objects are (left) R-modules and morphisms are homomorphisms of R-modules.

If $R = k$ is a field, R-modules are called k-vector spaces. We will denote the category of k-vector spaces "k-Vect". Morphisms in k-Vect are called *linear maps*. The study of k-Vect, and when possible, R-Mod is called Linear Algebra. We will study these in chapters 6 and 8.

Any ring homomorphism $\alpha : R \to S$ can be used to define the R-module where $(S, +)$ is taken as the underlying abelian group and multiplication is defined as $\rho(r, s) = \alpha(r)s$, where the operation on the right is *multiplication in* S. For example, taking $\mathrm{id}_R : R \to R$ makes R an R-module. We usually just write rs instead of $\alpha(r)s$.

We can make the ring structure in S and the module stucture induced by α even more closely related by requiring R be commutative and making α map to the center of S; the set of elements of S that commute with all other elements of S. That means $\alpha(r)s = s\alpha(r)$. This makes this both a left- and a right-R-module, and additionally, the ring operation in S is compatable with the R-module structure in that

$$(r_1s_1)(r_2s_2) = (r_1r_2)(s_1s_2).$$

This makes multiplication in S 'R-bilinear'. We can thus move the action of R at will through elements of S. These examples are important, so we give them their own name

Definition 3.23: R**-algebra**

Let R be a commutative ring. An R-algebra is a ring homomorpism $\alpha : R \to S$ such that $\alpha(S)$ is contained within the center of S. A homomorphism of R-algebras is a ring homomorphism $S \to P$ that is also a R-module homomorphism $S \to P$.

Equivalently, an R-algebra is an R-module S with compatable ring structure; that is, we can move the actions of R about however we desire. An R-algebra S is called a division algebra if S is a division ring.

We then get a notion of R-algebra homomorphism that preserves both the ring AND module structure; an R-algebra homomorphism $\phi : S \to K$ is a ring homomorphism $S \to K$ sch that $\phi(rx) = r\phi(x)$ for $r \in R$, $x \in S$. R-algebras then form a category R-Alg.

If S is commutative, the center condition is unnecessary so the situation is more simple. We call such a structure a commutative R-algebra. Commutative R-algebras form a subcategory of R-Alg, which is equivalent to the 'coslice category' CRing^R of morphisms $\phi : R \to L$ for L a commutative ring (CRing is the category of commutative rings), since each morphism $\phi : R \to L$ determines a commutative R-algebra on L.

Also note that $\mathbb{Z}$-Alg is the same as Ring, as for every ring R there exists exactly one ring homomorphism $\mathbb{Z} \to R$, and thus exactly one $\mathbb{Z}$-algebra R.

The polynomial rings $R[x_1, \ldots, x_n]$ are, together with their quotients, commutative R-algebras (for R commutative, otherwise it isn't an R-algebra). The class of examples where $R = k$ is an algebraicly closed field are a particularly important class of examples that are used in classical affine algebraic geometry. See Chapter 7.2.3 for more discussion.

The trivial group $\{0\}$ is both initial and final in R-Mod, and we call such an object a "zero-object". As in the other categories we have seen, an isomorphism in R-Mod is a bijective homomorphism. This is the same as in Ab, and as we will see, they are similar in many other ways as well.

If R is commutative, there are even more similarities. We generally don't write $\mathrm{Hom}_{R\text{-}\mathsf{Mod}}(M, N)$, and instead write $\mathrm{Hom}_R(M, N)$. Since M, N are abelian groups,

$$\mathrm{Hom}_R(M, N) \subseteq \mathrm{Hom}_{\mathsf{Ab}}(M, N)$$

as sets. We have seen the operation of pointwise addition makes $\mathrm{Hom}_{\mathsf{Ab}}(M, N)$ into an abelian group, and it follows fairly immediately that $\mathrm{Hom}_R(M, N)$ is also an abelian group under the same operations. That is, $\mathrm{Hom}_R(M, N) \in \mathsf{Obj}(\mathsf{Ab})$. We can also identify a natural action of R on this group. For all $r \in R$ and $\phi \in \mathrm{Hom}_R(M, N)$, define the map $r\phi : M \to N$ as $(r\phi)(m) = r \cdot \phi(m)$. If R is commutative, we have that this new function is also a R-module homomorphism, as for all $a \in R$ and $m \in M$,

$$(r\phi)(am) = r\phi(am) = (ra)(\phi(m)) = (ar)(\phi(m)) = a((r\phi)(m)).$$

Thus, we have a very natural action of R on $\mathrm{Hom}_R(M, N)$ for all R-modules M, N, and thus we have $\mathrm{Hom}_R(M, N)$ as an R-module.

If R is *not commutative*, then $\mathrm{Hom}_R(M, N)$ is only an abelian group. We will find more structure in chapter 8 when we discuss *bimodules.*

3.5.3 Submodules and Quotients

We can construct things very quickly in R-Mod, since R-Mod is basically a modified version of Ab.

A submodule N of M is a subgroup $N \subseteq M$ such that for all $r \in R$, $n \in N$, $rn \in N$. More simply, N is an R-module and the inclusion $N \hookrightarrow M$ is a R-module homomorphism.

Interestingly, when R is viewed as a left-R-module, the submodules of R are precisely the left-ideals of R.

As usual, the image and kernel of a R-module homomorphism are submodules of their respective modules. Also note that if r is in the center of R and M is any R-module, then rM is a submodule of M. If I is any left-ideal of R, then

$$IM = \left\{ \sum_i r_i m_i \mid r_i \in R \wedge m_i \in M \right\}$$

is a submodule of M. Finally, if N is a submodule of M, then it is a subgroup of the abelian group $(M, +)$, and since all subgroups of abelian groups are normal, we can define the quotient M/N as an abelian group.

We would like to see this quotient as a module, so we would like the canonical projection $\pi : M \to M/N$ to be an R-module homomorphism. This forces

$$r(m+N) = r\pi(m) = \pi(rm) = rm + N$$

for all $m \in M$. That is, we need to define an action of R on M/N by

$$r(m+N) \coloneqq rm + N.$$

This trivially defines an R-module structure on M/N. The R-module M/N is called the quotient of M by N.

If R is a ring and I is an ideal of R, then R, I, and R/I are all R-modules. Additionally, if R is commutative, then R and R/I are R-algebras.

If R is not commutative and I is only a left-ideal, then the quotient R/I is not defined as a ring, and is defined as a module with the R-action given by left multiplication. This reconciles our requirement that ideals be two-sided.

We now have our theorems concerning homomorphisms, once again.

Theorem 3.3

Let N be any submodule of an R-module M. Then for all R-module homomorphisms $\phi : M \to P$ such that $N \subseteq \ker \phi$, there exists a unique homomorphism of R-modules $\widetilde{\phi} : M/N \to P$ such that the diagram

$$\begin{tikzcd} M \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & P \\ & M/N \arrow[ur, "\widetilde{\phi}"'] & \end{tikzcd}$$

commutes. We have once again that $\widetilde{\phi}(m+N) = \phi(m)$.

Proof. By theorem 2.1, this holds for groups, and all R-module homomorphisms are also group homomorphisms. We need only show the homomorphism is a homomorphism of R-modules. Let $m+N \in M/N$ and $r \in R$. We have

$$\widetilde{\phi}(rm+N) = \phi(rm) = r\phi(m) = r\widetilde{\phi}(m+N).$$

■

Since we can quotient by every submodule N, and the kernel of the canonical projection $M \to M/N$ is N, we have that every submodule is a kernel. Recall that we also have that all kernels are submodules, so all substructures are in fact kernels, in the same exact way as Ab. In other terms, every monomorphism in R-Mod is a kernel.

3.5.4 Canonical Decomposition and Isomorphism Theorems

Theorem 3.4

Every R-module homomorphism $\phi : M \to M'$ can be decomposed as follows:

$$\begin{tikzcd} M \arrow[r, two heads, "\pi"'] \arrow[rrr, bend left, "\phi"] & M/N \arrow[r, "\sim", "\widetilde{\phi}"'] & \phi(M) \arrow[r, hook, "\iota"'] & M' \end{tikzcd}$$

Proof. An R-module homomorphism is also a group homomorphism and this theorem already holds for groups by 2.2, so we need only show $\widetilde{\phi}$ is a homomorphism of R-modules. By theorem 3.3, this follows. ∎

Corollary 3.2 (First Isomorphism Theorem)

Let $\phi : M \to M'$ be a surjective ring homomorphism. Then

$$M' \cong \frac{M}{\ker \phi}.$$

Proof. Since $\phi(M) = M'$, this follows by theorem 3.4. ∎

Once again, if M is an R-module and $N \subseteq M$ is a submodule, then there is a bijection between submodules P of M containing N, and submodules of M/N.

Proposition 3.16 (Third Isomorphism Theorem)

Let N be a submodule of an R-module M, and let P be a submodule of M containing N. Then P/N is a submodule of M/N, and

$$\frac{M/N}{P/N} \cong \frac{M}{P}.$$

Proof. Consider the map $\phi : M/N \to M/P$ defined by $m + N \mapsto m + P$. Note that for all $m \in M$ and $r \in R$,

$$\phi(rm + N) = rm + P = r\phi(m + N).$$

Therefore, ϕ is an R-algebra homomorphism, since we have already verified it is a group homomorphism. We have $\ker \phi = P/N$, so

$$\frac{M/N}{P/N} \cong \frac{M}{P}.$$

∎

Finally, we have a second isomorphism theorem of modules.

Proposition 3.17 (Second Isomorphism Theorem)

Let N, P be submodules of an R-module M. Then

- $N + P$ is a submodule of M;
- $N \cap P$ is a submodule of M;

$$\frac{N + P}{N} \cong \frac{P}{N \cap P}.$$

Proof. maaan i dont feel like proving this ugjh I hate thiis theorem ∎

More generally, for any collection $\{N_\alpha\}_{\alpha \in A}$ of submodules of M,

$$\sum_{\alpha \in A} N_\alpha \text{ and } \bigcap_{\alpha \in A} N_\alpha$$

are submodules of M.

3.6 Products, Coproducts, etc. In R-Mod

In this section, we will explore in some sense what it means for R-mod to be *well-behaved.*

3.6.1 Products and Coproducts

Just like in Ab, products and coproducts exist in R-Mod, and moreover finite products and coproducts coencide. A direct sum $M \oplus N$ of R-modules can be given multiplicative structure via the map

$$r(m, n) = (rm, rn).$$

Note that this comes with the canonical projections

$$\pi_M : M \oplus N \to M, \qquad \pi_N : M \oplus N \to N$$

defined by $(m, n) \mapsto m$ and $(m, n) \mapsto n$. We also have the maps

$$i_M : M \to M \oplus N, \qquad i_N : N \to M \oplus N$$

defined by $m \mapsto (m, 0)$ and $n \mapsto (0, n)$.

Proposition 3.18

The direct sum $M \oplus N$ satisfies the universal properties of both the product and coproduct in R-Mod.

Proof. *Product:* Let P be an R-module, and let $\phi_M, \phi_N : P \to M, N$. The unique map $\phi_M \times \phi_N$ is required for the diagram

$$\begin{tikzcd}
 & & M \\
P \arrow[r, "\phi_M \times \phi_N"] \arrow[urr, bend left, "\phi_M"] \arrow[drr, bend right, "\phi_N"'] & M \oplus N \arrow[ur, "\pi_M"] \arrow[dr, "\pi_N"] & \\
 & & N
\end{tikzcd}$$

to commute. Also note that since it is a product, it is an R-module homomorphism, and is the unique one making the diagram commute fairly trivially.

Coproduct: Let P be an R-module, and let $\psi_M, \psi_N : M, N \to P$. Define $\psi_M \oplus \psi_N : M \oplus N \to P$ as the map

$$\begin{aligned}
(\psi_M \oplus \psi_N)(m, n) &= (\psi_M \oplus \psi_N)(m, 0) + (\psi_M \oplus \psi_N)(0, n) \\
&= (\psi_M \oplus \psi_N) \circ i_M(m) + (\psi_M \oplus \psi_N) \circ i_N(n) \\
&= \psi_M(m) + \psi_N(n).
\end{aligned}$$

This definition is forced by the commutativity of the diagram

$$\begin{tikzcd}
M \arrow[dr, "i_M"] \arrow[drr, bend left, "\psi_M"] & & \\
 & M \oplus N \arrow[r, "\psi_M \oplus \psi_N"] & P \\
N \arrow[ur, "i_N"] \arrow[urr, bend right, "\psi_N"'] & &
\end{tikzcd}.$$

Additionally, it is clearly a homomorphism of R-modules. It is unique by the commutative diagram, so we have that $M \oplus N$ is both a product and a coproduct in R-Mod. ■

We will write $M \oplus N$ to denote both the product and coproduct in R-Mod, since $M \times N$ will be seen again in chapter 8 in the category of bilinear maps, and will not be interpreted as the product of M and N. Finally, note that infinite products and coproducts do not coencide in R-Mod.

3.6.2 Kernels and Cokernels

R-Mod has a 0 object, its Hom sets are abelian groups, and it has finite products and coproducts. A category with these is called an *additive category*, and since it also has kernels and cokernels, this makes it into an *abelian* category. We will review these definitions in chapter 9, but for now, we will review kernels and cokernels in R-Mod.

Monomorphisms and epimorphisms behave as we expect in R-Mod, and for very simple reasons, unlike the complicated reasoning in Grp and the strange behavior in Ring. Recall our universal properties for kernels and cokernels. Let $\phi : M \to N$ be a R-module homomorphism. Then $\ker \phi$ is final with respect to the property of factoring R-module homomorphisms $\alpha : P \to M$ such that $\phi \circ \alpha = 0$:

$$
\begin{tikzcd}
P \arrow[r, "\alpha"] \arrow[rr, bend left, "0"] \arrow[dr, "\exists ! \overline{\alpha}"'] & M \arrow[r, "\phi"] & N \\
 & \ker \phi \arrow[u, hook] &
\end{tikzcd}
$$

Cokernels are then initial with respect to the property of factoring R-module homomorphisms $\beta : N \to P$ such that $\beta \circ \phi = 0$:

$$
\begin{tikzcd}
M \arrow[r, "\phi"] \arrow[rr, bend left, "0"] & N \arrow[r, "\beta"] \arrow[d, two heads, "\pi"] & P \\
 & \operatorname{coker} \phi \arrow[ur, "\exists ! \overline{\beta}"] &
\end{tikzcd}
$$

Proposition 3.19

Kernels and cokernels exist in R-Mod. Moreover,

- The following are equivalent:
  - ϕ is a monomorphism;
  - $\ker \phi$ is trivial;
  - ϕ is injective;
- The following are equivalent:
  - ϕ is an epimorphism;
  - $\operatorname{coker} \phi$ is trivial;
  - ϕ is surjective.

Proof. The book doesn't prove any of these, but I want to prove them from scratch. First, we show kernels and cokernels exist in R-Mod. Using the notation in the diagrams above, Let $\alpha : P \to M$ be a homomorphism of R-modules such that $\phi \circ \alpha = 0$. Note that this forces $\alpha(P) \subseteq \ker \phi$, so define $\overline{\alpha} : P \to \ker \phi$ as the map $m \mapsto \alpha(m)$. Since α is a R-module homomorphism, we have trivially that $\overline{\alpha}$ is a R-module homomorphism.

Recall that we require not just that the diagram to commute, but that $\iota \circ \overline{\alpha} = \alpha$, since the morphism $\iota : \ker \phi \to M$ needs to be final. We then trivially have that $\overline{\alpha}$ is the unique morphism from $\alpha \to \iota$, and thus $\ker \phi$ satisfies the universal property for kernels in R-Mod.

Next, we show $\operatorname{coker} \phi$ exists in R-Mod. For $\beta \circ \phi = 0$, we require that $\phi(M) \subseteq \ker \beta$. By theorem 3.3, since $\phi(M)$ is a submodule of N and $\phi(M) \subseteq \ker \beta$, there exists a unique R-module homomorphism $\overline{\beta} : N/\phi(M) \to P$ such that $\beta = \overline{\beta} \circ \pi_{\phi(M)}$, where $\pi_{\phi(M)}$ is the canonical projection $N \to N/\phi(M)$. Therefore, $\pi_{\phi(M)}$ is initial in the category, so we define $\operatorname{coker} \phi := N/\phi(M)$.

Now we show the statements concerning monomorphisms hold. Let ϕ be a monomorphism. Recall a monomorphism is a morphism $\phi : M \to N$ such that for all $\alpha, \beta : P \to M$,

$$\phi \circ \alpha = \phi \circ \beta \implies \alpha = \beta.$$

Consider the maps

$$\ker \phi \overset{0}{\underset{\iota}{\rightrightarrows}} M \xrightarrow{\phi} N$$

where ι is the inclusion map. Note that $\iota(\ker \phi) = \ker \phi$ when viewed as a submodule of M, and $0_M \in \ker \phi$ by previous logic, so we have $\phi \circ \iota = \phi \circ 0$. Therefore, $\iota = 0$ since ϕ is a monomorphism, and thus $\iota(\ker \phi) = \{0\}$. Thus, $\ker \phi = \{0\}$ is trivial.

Now suppose $\ker \phi$ is trivial, and let $m, n \in M$ such that $\phi(m) = \phi(n)$. We have

$$\phi(m) = \phi(n) \implies \phi(m - n) = 0_N \implies m - n \in \ker \phi \implies m - n = 0 \implies m = n.$$

Therefore, ϕ is injective.

Now suppose ϕ is injective. Since it is injective, and since all R-module homomorphisms are set functions, we have that ϕ is a monomorphism in R-Mod since it is a monomorphims in Set.

Finally, recall statements concerning epimorphisms. Suppose $\phi : M \to N$ is an epimorphism. Recall that this implies for all R-module homomorphism $\alpha, \beta : N \to P$, we have

$$\alpha \circ \phi = \beta \circ \phi \implies \alpha = \beta.$$

We wish to show $\operatorname{coker} \phi$ is trivial. Consider the dual maps

$$M \xrightarrow{\phi} N \overset{0}{\underset{\pi}{\rightrightarrows}} \operatorname{coker} \phi$$

where π is the canonical map $m \mapsto m + \phi(M)$. Note that for all $m \in M$, $\phi(m) \in \phi(M)$ by definition, so we have

$$(\pi \circ \phi)(m) = m + \phi(M) = 0_{\operatorname{coker} \phi}.$$

Therefore, $\pi = 0$ since ϕ is an epimorphism, and thus $\pi(N) = \{0\}$. Therefore, $\operatorname{coker} \phi$ is trivial.

Now suppose $\operatorname{coker} \phi$ is trivial. It follows that for all $n \in N$, we have $n + \phi(M) = \phi(M)$. It follows that $n \in \phi(M)$, and thus $N \subseteq \phi(M)$. Therefore, ϕ is surjective.

Now suppose ϕ is surjective. Since ϕ is an epimorphism in Set, and all R-module homomorphisms are morphisms in Set, it follows that ϕ is an epimorphism in R-Mod. ■

I also want to remark that for any R-module M, the submodules of M are precisely the R-module homomorphisms $M \to \_$, and the quotients of M by these submodules are precisely the cokernels of R-module homomorphisms $\_ \to M$. To see this, first note that we have already shown cokernels are quotient modules and kernels are submodules. Now note that for any submodule $N \subseteq M$, the kernel of the map $M \to M/N$ is N, so submodules are kernels. Then note that the cokernel of the inclusion map $N \hookrightarrow M$ is $M/\iota(N) = M/N$, so quotient groups are cokernels.

One last thing to remember is that in Set, a function was monic or epi iff it had a left or right inverse, respectively. We have been told not to expect this in general categories, and it does not hold in R-Mod.

3.6.3 Free Modules and Free Algebras

The universal property of free R-modules is modeled after the one for groups. The goal is to find an R-module that contains a given set A with no special properties in the most efficient way possible. The universal property goes as such:

Given a set A, a free R-module on A, denoted $F^R(A)$, is a R-module together with a function of sets $j : A \to F^R(A)$ such that for all R-modules M and functions of sets $f : A \to M$, there exists a unique R-module homomorphism $\phi : F^R(A) \to M$ such that the diagram

$$\begin{tikzcd} F^R(A) \arrow[r, "\phi"] & M \\ A \arrow[u, "j"] \arrow[ur, "f"'] & \end{tikzcd}$$

commutes. Equivalently, $(j, F^R(A))$ is initial with respect to this category. We have shown that $F^R(A)$ is unique up to isomorphism since it is initial, and the function j is necessarily injective. That is, if it exists.

Given any set A, define the (infinite) direct sum $N^{\oplus A}$ of an R-module N as

$$N^{\oplus A} := \{\alpha : A \to N \mid \alpha(a) \neq 0 \text{ for only finitely many } a \in A\}.$$

You can think of it as $\alpha(a) = 0$ almost everywhere; $\alpha(a) \neq (0)$ on a set of measure 0. We can thus treat each function as a finite collection of corresponding points. We can give it an R-module structure by defining $(r\alpha)(a) = r(\alpha(a))$, using the R-module structure on N.

If $N = R$, we can define a map $j : A \to R^{\oplus A}$ by sending $a \mapsto j_a$, where $j_a : A \to R$ is the map

$$j_a(x) = \begin{cases} 1, & x = a \\ 0, & x \neq a \end{cases}.$$

This identifies every element with a function in the expected way.

Proposition 3.20

$$F^R(A) \cong R^{\oplus A}.$$

Proof. We need to show $R^{\oplus A}$ is initial with respect to the given universal property. First, I actually want to show $R^{\oplus A}$ is an R-module, because why not. First, we show it is abelian. Note that $-1 \in R$ since R is abelian, so there exists a map

$$-j_a = \begin{cases} -1, & x = a \\ 0, & x \neq a \end{cases}.$$

Note that the 0 map $a \mapsto 0$ is the identity, and $j_a - j_a = 0$ fairly trivially. Associativity follows from R, so we have our abelian group $(R^{\oplus A}, +)$. Now note that the given definition of scalar multiplication is fairly trivially a ring action, so we have our R-module.

Now we show that the construction is initial. In order for $\phi \circ j = f$, we must have $\phi(j_a) = f(a)$ for all $a \in A$. Commutativity of the diagram forces this definition to be unique. We also require it be a morphism of R-modules, so first we need to figure out how to represent elements of $R^{\oplus A}$. Note that with our definition of multiplication,

$$rj_a(x) = \begin{cases} r1_R, & x = a \\ 0, x \neq a \end{cases}.$$

Note here that $r1_R$ denotes the ring action, not the ring multiplication. Moreover, we can then represent any element $\alpha \in R^{\oplus A}$ as a linear combination

$$\alpha = \sum_i r_i j_i,$$

where for all i such that $\alpha(i) \neq 0$, we have $\alpha(i) = (rj_i)(i) = r1_R$. We can now define our R-module homomorphism.

Let $\phi : R^{\oplus A} \to M$ be the map

$$\phi\left(\sum_i r_i j_i\right) = \sum_i r_i f(i).$$

Note that we require $j_i \mapsto f(i)$ to make the diagram commute, and we also require it map the more general elements in this specific way to make it a R-module homomorphism. This forces it to be unique (this can be shown rigorously but idc enough to write it out), and it is by definition a morphism of R-modules. Therefore, $R^{\oplus A} \cong F^R(A)$. ■

Definition 3.24: Free R-Module

Let R be a ring, and let A be a set. Then the free R-module on the set A is the R-module $R^{\oplus A}$.

We thus generally denote a free R-module as the set $R^{\oplus A}$, and if A is finite with order n, we generally write $R^{\oplus n}$. This is all analogous to Ab, which makes perfect sence since Ab is really just $\mathbb{Z}$-Mod.

Now we consider commutative R-algebras instead of R-modules. We will specifically consider the finite case here. Let $A = \{1, \ldots, n\}$. We denote by $R[A]$ the polynomial ring $R[x_1, \ldots, x_n]$, and define the function $j : A \to R[A]$ by $j(i) = x_i$.

Proposition 3.21

$R[A]$ is a free commutative R-algebra on the set A.

Proof. We wish to show for every commutative R-algebra S and function of sets $f : A \to S$, there exists a unique R-algebra homomorphism $\phi : R[A] \to S$ such that the diagram

$$\begin{tikzcd} R[a] \arrow[r, "\phi"] & S \\ A \arrow[u, "j"] \arrow[ur, "f"'] & \end{tikzcd}$$

commutes. That is, $\phi \circ j = f$.

Recall that for any fixed ring homomorphism $\alpha : R \to S$ of commutative rings, then for all $s \in S$ there exists a unique ring homomorpism $\overline{\alpha} : R[x] \to S$ that extends α by sending $x \mapsto s$. This is a fairly trivial statement, but turns out to complete the proof. We already have a fixed homomorphism $\alpha : R \to S$, as a commutative R-algebra is a R-module where the action is given by left multiplication under the homomorphism α. We can apply this extension principle n times, extending $x_i \mapsto f(i)$. This is the same logic as in our free R-module definition, but applied to a polynomial ring. It is evident that this is a ring homomorphism by this fact. ■

Definition 3.25: (Finite) Free R-Algebra

Let A be a finite set, and let R be a commutative ring. Then the free R-algebra on the set A is the R-algebra $R[A]$.

We showed earlier that for all sets A of order n and functions $j : A \to R$ for R a commutative ring, there exists a unique ring homomorphism $\phi : \mathbb{Z}[x_1, \ldots, x_n] \to R$ such that $\phi \circ j = f$, where $j(i) = x_i$. We can now realize that this also follows from the fact that $\mathbb{Z}[x_1, \ldots, x_n]$ is a free algebra in $\mathbb{Z}$-Alg, which is equivalent to the category CRing.

3.6.4 Submodule Generated by a Subset; Noetherian Modules

Let M be an R-module, and let $A \subseteq M$ be a submodule. By the universal property of free modules, there is a unique homomorphism of R-modules $\phi_A : R^{\oplus A} \to M$, where we take the map $f : M \to M$ to be the identity. We call the image $\phi_A\left(R^{\oplus A}\right) \subseteq M$ the *submodule generated by A in M*, and denote it $\langle A \rangle$. If A is finite, we generally denote it $\langle a_1, \ldots, a_n \rangle$.

Definition 3.26: Submodule Generated by a Set

Let M be an R-module, and let $A \subseteq M$ be a subset. We say that the submodule of M generated by A, denoted $\langle A \rangle$, is the image of the unique R-module homomorphism $R^{\oplus A} \to M$.

We also have the equivalent definition of

$$\langle A \rangle := \left\{ \sum_{a \in A} r_a a \mid r_a \neq 0 \text{ for only finitely many } a \in A \right\}.$$

Clearly, $\langle A \rangle$ is the smallest submodule in M containing A.

We call the module M *finitely generated* if $M = \langle A \rangle$ for a finite set A. That is, if there exists a surjective R-module homomorphism $R^{\oplus n} \twoheadrightarrow M$ for some n. One incredible result in algebra is the classification of all finitely generated modules over a PID, which will be covered in chapter 6. This will also classify all finitely-generated abelian groups, since $\mathbb{Z}$-Mod is the same as Ab.

Finitely generated R-modules are extremely important, but also not very well-behaved. For example, consider $R = \mathbb{Z}[x_1, \ldots, x_n]$, and note that $\mathbb{Z}[x_1, \ldots, x_n] = (1)$. However, the ideal

$$(x_1, x_2, \ldots)$$

is *not finitely generated*. There exist finitely-generated R-modules that have submodules that are not finitely generated.

Definition 3.27: Noetherian Module

A R-module M is a Noetherian module if all submodules of M are finitely generated as an R-module.

It can be seen that a ring R is Noetherian if and only if it is a Noetherian module over itself. That is, if the action of R on R is left-multiplication. We will study Noetherian modules in more depth later, but for now we will see some consequences of the definition.

Proposition 3.22

Let M be an R-module with a submodule $N \subseteq M$. Then M is Noetherian if and only if N AND M/N are Noetherian.

Proof. If M is Noetherian, then N is Noetherian because all submodules of N are submodules of M. To show a subgroup of M/N is Noetherian, it suffices to show the image of a Noetherian module under a R-module homomorphism is Noetherian, as $\pi : M \to M/N$ is a R-module epimorphism.

Let M be Noetherian, and let $\phi : M \to N$ be a R-module epimorphism (for convenience). Let $P \subseteq N$ be a submodule. We will first show $\phi^{-1}(P)$ is a submodule of M. It is a subgroup by group theory, so we need to only show it is closed under scalar multiplication. Let $r \in R$ and $m \in \phi^{-1}(P)$. It follows that

$$\phi(rm) = r\phi(m) \in rP,$$

since $\phi(m) \in \phi^{-1}(P)$. However, since P is a submodule, $rP \subseteq P$, since P must be closed under scalar multiplication. Thus, $rm \in \phi^{-1}(P)$. Therefore, all submodules of N have submodule preimages under ϕ. Let $P \subseteq N$ be a submodule. Since $\phi^{-1}(P)$ is a submodule of M, $\phi^{-1}(P)$ is finitely generated. That is, there exists a finite subset $A = \{a_1, \ldots, a_n\} \subseteq \phi^{-1}(P)$ such that $\langle A \rangle = \phi^{-1}(P)$. Let $p \in P$. It follows that there exists at least one $m \in \phi^{-1}(P)$ such that $\phi(m) = p$. Note that for all such p, there exist $\{r_a\}_{a \in A} \subseteq R$ such that

$$m = \sum_{a \in A} r_a a.$$

We have $|\phi(A)|$ is finite, and we also have that

$$p = \phi\left(\sum_{a \in A} r_a a\right) = \sum_{a \in A} r_a \phi(a) = \sum_{b \in \phi(A)} r_b b.$$

Note that r_b may not equal r_a, since $\phi : A \to \phi(A)$ may not necessarily be injective, so there may be duplicate as. However, the point is that A finitely generates P, so N is Noetherian.

For the converse, let N and M/N be Noetherian. Let P be a submodule of M. We must show P is finitely generated. Since $P \cap N$ is a submodule of N and N is Noetherian, we have $P \cap N$ is finitely generated. By propsition 3.17, we have

$$\frac{P}{P \cap N} \cong \frac{P + N}{N}.$$

Therefore, $P/(P \cap N)$ is isomorphic to a submodule of M/N, and is thus Noetherian, since M/N is Noetherian. Moreover, $P/(P \cap N)$ is finitely generated. This implies that P is finitely generated by the exercises. ■

Corollary 3.3

Let R be a Noetherian ring, and let M be a finitely-generated R-module. Then M is Noetherian.

Proof. Since M is finitely generated, the unique homomorphism $R^{\oplus n} \to M$ is surjective. By theorem 3.2, we have that M is isomorphic to a quotient of $R^{\oplus n}$. By proposition 3.22, it suffices to show $R^{\oplus}$ is Noetherian.

We use proof by induction. The base case is trivial, so suppose $R^{\oplus n-1}$ is Noetherian. Note that the map $R^{\oplus n-1} \to R^{\oplus n}$ given by $(r_1, \ldots, r_{n-1}) \mapsto (r_1, \ldots, r_{n-1}, 0)$ is an injective ring homomorphism that allows us to view $R^{\oplus n-1}$ as a submodule of $R^{\oplus n}$. The kernel of the surjective ring homomorphism $(r_1, \ldots, r_n) \mapsto r_n$ is $R^{\oplus n-1}$, so by corollary 3.2, we have

$$\frac{R^{\oplus n}}{R^{\oplus n-1}} \cong R.$$

Since R and $R^{\oplus n-1}$ are Noetherian, by proposition 3.22, $R^{\oplus n}$ is Noetherian. ■

3.6.5 Finitely Generated vs. Finite Type

If S is an R-algebra, then it can be finitely generated as both an R-module AND an R-algebra. Since these are not equivalent, we call S *finite* if it is finitely generated as an R-module, and also give the following definition.

Definition 3.28: Finite Type

A commutative R-algebra S is *of finite type* if it is finitely generated as an R-algebra; that is, there exists a surjective R-algebra homomorphism $R[x_1, \dots, x_n] \to S$ for some finite n.

Equivalently, S is finite if there exists a submodule M of $R^{\oplus n}$ such that $S \cong R^{\oplus n}/M$, and S is of finite type if there exists an ideal I of $R[x_1, \dots, x_n]$ such that $S \cong R[x_1, \dots, x_n]/I$. Recall here that morphisms in R-Alg are both R-module and ring homomorphisms, hence the ideal requirement. It shuold be clear that all finite such S are of finite type, but the converse is not true in general.

There are special cases where they are the same, however. One version of *Hilbert's Nullstellenatz* says that if k, K are finite fields with $k \subseteq K$, then K is of finite type over k *if and only if* it is finite over k. We will prove this, along with many other examples, later.

Hilbert also showed in *Hilbert's Basis Theorem* that if R is Noetherian as a ring (and thus a R-module) and S is a finite type R-algebra, then S is also Noetherian as a ring, and thus, a S-module. This is one immediate consequence of the theorem, which will be proven in chapter 5.

3.7 Complexes and Homology

Oftentimes, instead of dealing with a single module, we have to deal with an entire sequence of modules. For example, a reall manifold of dimension d has a homology group for all dimensions 0 to d. The language of *homological algebra* will allow us to deal with sequences of modules. We will study this further in chapter 9, but will introduce it here.

3.7.1 Complexes and Exact Sequences

Definition 3.29: Chain Complex

A *chain complex*, or for simplicity, a *complex* is a sequence of R-modules and R-module homomorphisms

$$\cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots$$

such that for all i, $d_i \circ d_{i+1} = 0$. We denote a complex as $(M_\bullet, d_\bullet)$, or for simplicity, $M_\bullet$.

Tails of a complex can be infinite in both directions. We generally also omit tails of 0. Other conventions are also used. For example, having the indicies increase instead of decrease yields what is known as a cochain complex, and the homology of this complex is called a cohomology. We will use this convention in chapter 9, but for now, the differences are immaterial.

The homomorphisms d_i are called *differentials*, due to important applications in geometry. Note that the requirement $d_i \circ d_{i+1} = 0$ is equivalent to the requirement that $\operatorname{im} d_{i+1} \subseteq \ker d_i$. The book continues with a very nice picture that I don't feel like recreating. The idea behind a homology is to 'measure' the difference between the image of d_{i+1} and the kernel of d_i. We say a complex is *exact* at M_i if it has no homology there; that is,

$$\operatorname{im} d_{i+1} = \ker d_i.$$

For example, if M_i is trivial, then the complex is necessarily exact at M_i, since $d_{i+1}(M_{i+1}) = \{0\} = \ker d_i$. A complex is called an *exact sequence* if it is exact at all of its modules.

Proposition 3.23

A complex
$$\cdots \longrightarrow 0 \longrightarrow L \xrightarrow{\alpha} M \longrightarrow \cdots$$
is exact at at L if and only if α is a monomorphism. Additionally, a complex
$$\cdots \longrightarrow M \xrightarrow{\beta} N \longrightarrow 0 \longrightarrow \cdots$$
is exact at N if and only if β is an epimorphism.

Proof. (1) Since the trivial homomorphism $0 \to L$ has an image of $\{0\}$, for it to be exact at M we require $\ker\alpha = \{0\}$, and thus α is a monomorphism.

(2) For it to be exact at N, we require the image of β to be the kernel of the trivial morphism $N \to 0$, and thus $\operatorname{im}\beta = N$. Equivalently,
$$\frac{N}{\beta(N)} = \{0\}\,.$$
Therefore, $\operatorname{coker}\beta$ is trivial. ∎

Definition 3.30: Short Exact Sequence

A short exact sequence is an exact complex of the form
$$0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

The only extra information here that was not included in the proposition is that the sequence, to be exact at M, must have $\alpha(L) = \ker\beta$. By corollary 3.2, we have
$$N \cong \frac{M}{\ker\beta} = \frac{M}{\alpha(L)} = \operatorname{coker}\alpha.$$

We can then identify L with a submodule $\alpha(L)$ of M, since α is necessarily mono, and then we can identify N with the quotient $M/\alpha(L)$, since $\beta$is necessarily epi.

Short sequences are apparently common in nature. For example, every homomorphism $\phi : M \to M'$ gives rise to the short sequence
$$0 \longrightarrow \ker\phi \xhookrightarrow{\iota} M \xrightarrow{\phi} \phi(M) \longrightarrow 0$$

This observation allows us to break long complexes into many short exact sequences, which simplifies some proofs.

$$
\begin{tikzcd}
 & & 0 \arrow[dr] & & 0 & & \\
 & & & \operatorname{im} d_{i+1} = \ker d_i \arrow[ur] \arrow[dr] & & & \\
M_{i+2} \arrow[rr, "d_{i+2}"] \arrow[dr] & & M_{i+1} \arrow[rr, "d_{i+1}"'] \arrow[ur] & & M_i \arrow[rr, "d_i"] \arrow[dr] & & M_{i-1} \\
 & \operatorname{im} d_{i+2} = \ker d_{i+1} \arrow[ur] \arrow[dr] & & & & \operatorname{im} d_i = \ker d_{i-1} \arrow[ur] \arrow[dr] & \\
0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0
\end{tikzcd}
$$

This large diagram illustrates this idea. The diagonals are short exact sequences of the exact form we just considered, and there is one for each differential.

3.7.2 Split Exact Sequences

One important class of short exact sequences arises when considering the direct sum $M_1 \oplus M_2$. More specificly, we consider the projection $M_1 \oplus M_2 \to M_2$, and have the exact sequence

$$0 \longrightarrow M_1 \xrightarrow{i_{M_1}} M_1 \oplus M_2 \xrightarrow{\pi_{M_2}} M_2 \longrightarrow 0$$

We then identify M_1 with the kernel of the projection. We say this sequence 'splits'. More generally, a short exact sequence *splits* if it is isomorphic to one of these sequences in that there is a commutative diagram

$$
\begin{tikzcd}
0 \arrow[r] & M_1 \arrow[r, "\phi"] \arrow[d, "\mathrm{id}"', "\sim"] & N \arrow[r, "\psi"] \arrow[d, "f"', "\sim"] & M_2 \arrow[r] \arrow[d, "\mathrm{id}"', "\sim"] & 0 \\
0 \arrow[r] & M_1 \arrow[r, "i"] & M_1 \oplus M_2 \arrow[r, "\pi"] & M_2 \arrow[r] & 0
\end{tikzcd}
$$

in which all vertical maps are isomorphisms. As we will see in the exercises, we actually only need there to exist a R-module homomorphism $N \to M_1' \oplus M_2'$ for the isomorphisms to be forced. Note that we have that $f\phi$ is the natural injection i_{M_1}, and ψf^{-1} is the natural projection π_{M_2}.

Splitting sequences allows us to answer an important question; that is, what is the condition for a homomorphism having a left/right inverse? This must be stronger than being mono/epi, and split sequences allow us to find the condition.

Proposition 3.24

Let $\phi : M \to N$ be an R-module homomorphism. Then

- ϕ has a left-inverse if and only if the sequence

$$0 \longrightarrow M \xrightarrow{\phi} N \longrightarrow \operatorname{coker}\phi \longrightarrow 0$$

splits.

- ϕ has a right-inverse if and only if the sequence

$$0 \longrightarrow \ker\phi \longrightarrow M \xrightarrow{\phi} N \longrightarrow 0$$

splits.

Proof. We begin by proving the statement for left-inverses. Suppose the sequence splits. Then $N \cong M \oplus M'$ for some module $M' \cong \operatorname{coker}\phi$. Since the sequence is exact, because it splits, we have that ϕ is necessarily a monomorphism. We can then identify with the embedding map $i : M \to M \oplus M'$. More precisely, if f is the isomorphism $N \to M \oplus M'$, then we have $f\phi = i$. We then have a left inverse given by the projection $\pi_M : M \oplus M' \to M$. More precisely, $\pi_M f \phi = \mathrm{id}_M$.

Now for the converse. Suppose ϕ has a left inverse ψ:

$$\begin{tikzcd} 0 \arrow[r] & M \arrow[r, "\phi"] \arrow[dr, "\mathrm{id}"'] & N \arrow[d, "\psi"] \\ & & M \end{tikzcd}$$

First, we will show that $N \cong M \oplus \ker\psi$. The isomorphism $\eta : M \oplus \ker\psi \to N$ is given by $(m, k) \mapsto \phi(m) + k$. Since ϕ is a homomorphism of R-modules, we have trivially that this is a homomorphism. The inverse $\zeta : N \to M \oplus \ker\psi$ is given by $n \mapsto (\psi(n), n - (\phi\psi)(n))$. Note that

$$\psi(n - (\phi\psi)(n)) = \psi(n) - (\psi\phi)\psi(n) = \psi(n) - \psi(n),$$

and thus $n - (\phi\psi)(n) \in \ker\psi$. Since ψ is a homomorphism of R-modules, this is also trivially a homomorphism of R-modules. Now we verify they are indeed inverses.

$$\zeta\eta(m, k) = \zeta(\phi(m) + k) = (\psi(\phi(m) + k), \phi(m) + k - (\phi\psi)(\phi(m) + k)).$$

For the first component, since $k \in \ker\psi$,

$$\psi(\phi(m) + k) = (\psi\phi)(m) + \psi(k) = m.$$

For the second component,

$$\phi(m) + k - (\phi(\psi\phi)(m) + \psi(k)) = \phi(m) + k - \phi(m) = k.$$

Now we check the other direction.

$$\eta\zeta(n) = \eta(\psi(n), n - (\phi\psi)(n)) = (\phi\psi)(n) + n - (\phi\psi)(n) = n.$$

Therefore, $M \oplus \ker\psi \cong N$, provided ψ is a left-inverse to ϕ.

There are many left-inverses to ϕ, and we are at our leisure to choose one such that $\ker\psi \cong \operatorname{coker}\phi$, which will indeed make the diagram split. Note that

$$\operatorname{coker}\phi = \frac{M \oplus \ker\psi}{M}.$$

We wish to find $\ker\psi$ such that

$$\ker\psi \cong \frac{M \oplus \ker\psi}{M}.$$

Since ϕ is necessarily a monomorpism, we must have $\ker\psi \cap M = \{0\}$, where we view M as a submodule of N via the injection $i : M \to M \oplus \ker\psi$ defined by $m \mapsto (m, 0)$. However, for all $r \in M \oplus \ker\phi \setminus M$, we can define whatever we desire. I propose we define $r \mapsto 0$. This makes sense, as everything else must be in the kernel of ψ, so it is actually the only possible choice. Consider the R-module homomorphism $\ker\psi \to M \oplus \ker\psi/M$ given by $k \mapsto (0, k) + M$. First, note that this is the same as the inclusion map $\ker\psi \hookrightarrow M \oplus \ker\psi$ composed with the canonical injection, so it is trivially a homomorphism. Next, consider the inverse $(m, k) + M \mapsto k$. This is trivially an inverse, so it suffices to show it is well-defined.

Let $(m_1, k_1) + M = (m_2, k_2) + M$. Note that since the m terms all drop out, and in fact $(m_1, k) + M = (m_2, k) + M$ for all $m_1, m_2 \in M$, we can completely ignore the m terms without loss of generality, and simply write $(0, k_1) + M = (0, k_2) + M$. It follows that

$$(0, k_1 - k_2) \in M.$$

However, since our inclusion map $M \hookrightarrow M \oplus \ker\psi$ was defined as $m \mapsto (m, 0)$, we have that $k_1 - k_2 = 0$. Therefore, the map is well-defined, and $\ker\psi \cong \operatorname{coker}\phi$. I have no idea why the book didn't include this, since it is *not* trivial. It might have been neater to show this using universal properties in retrospect, but oh well.

Now we prove the statement for right inverses, which is left completely as an exercise. Suppose the sequence splits. Then $M \cong \ker\phi \oplus N$, and we can view ϕ as the projection $\ker\phi \oplus N \to N$. The canonical injection $i : N \to \ker\phi \oplus N$ is then an inverse. This argument can be formalized in the same say I formalized the previous argument, using the explicit isomorphism.

Now suppose ϕ has a right-inverse ψ:

$$\begin{tikzcd}
0 \arrow[r] & N \arrow[r, "\psi"] \arrow[dr, "\mathrm{id}"'] & M \arrow[d, "\phi"] \\
 & & N
\end{tikzcd}$$

We will show that $M \cong \ker\phi \oplus N$. Let $\eta : \ker\phi \oplus N \to M$ as the map $(k, n) \mapsto k + \psi(n)$. By the same logic as previously, this is a R-module homomorphism. Let $\zeta : M \to \ker\phi \oplus N$ be the mao $m \mapsto (m - (\psi\phi)(n), \phi(n))$. This is an inverse function by the same proof as previously. Therefore, $M \cong \ker\phi \oplus N$. Therefore, the sequence splits fairly trivially. ■

Because of proposition 3.24, we call left-invertible R-module homomorphisms '*split monomorphisms*', and likewise we call right-invertible R-module homomorphisms '*split epimorphisms*'.

We will discuss split exact sequences again in chapter 4 in the context of Grp, and then to modules and abelian categories in chapters 8 and 9.

3.7.3 Homology and the Snake Lemma

Definition 3.31: Homology

The i-th homology of a complex

$$M_\bullet : \cdots \xrightarrow{d_{i+2}} M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1} \xrightarrow{d_{i-1}} \cdots$$

of R-modules is the R-module

$$H_i(M_\bullet) \coloneqq \frac{\ker d_i}{\operatorname{im} d_{i+1}}.$$

We then have

$$H_i(M_\bullet) = 0 \iff \operatorname{im} d_{i+1} = \ker d_i \iff M_\bullet \text{ is exact at } M_i.$$

That is, $H_i(M_\bullet)$ is the measure of the failiure of $M_\bullet$ to be exact at M_i. In fact, homology can be thought of as a *vast* generalization of kernel and cokernel. The particular case where $M_\bullet$ is the complex

$$0 \longrightarrow M_1 \xrightarrow{\phi} M_0 \longrightarrow 0$$

is exact if and only if $H_1(M_\bullet) \cong \ker\phi$ and $H_0(M_\bullet) \cong \operatorname{coker}\phi$.

We will now discuss how two short exact sequences yield one long exact sequence. We will come back to this construction later in the book in a more general manner in chapter 9. This is also important in algebraic topology apparently. The simple form we will learn about is known as the snake lemma.

Lemma 3.3 (The Snake Lemma)

Let the following diagram of two short exact sequences commute:

$$\begin{tikzcd}
0 \arrow[r] & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] & M_0 \arrow[r, "\beta_0"] & N_0 \arrow[r] & 0
\end{tikzcd}$$

Then there exists an exact sequence

$$0 \longrightarrow \ker\lambda \longrightarrow \ker\mu \longrightarrow \ker\nu \xrightarrow{\delta} \operatorname{coker}\lambda \longrightarrow \operatorname{coker}\mu \longrightarrow \operatorname{coker}\nu \longrightarrow 0$$

We will skip the proof because it is a notational mess, and no real benefit comes out of stating the proof itself. It's better to investigate the definitions of the various morphisms. We will possibly return to it in chapter 9, when we will be dealing with things purely catgorically, and can prove it without appealing to elements of sets.

Most of the homomorphisms are induced in straightforward ways, which I will contemplate when I return to these notes tomorrow. One thing to note is that we could have written the complex in ur statment as

$$\begin{tikzcd}
0 \arrow[r] & H_1(L_\bullet) \arrow[r] & H_1(M_\bullet) \arrow[r] & H_1(N_\bullet) \arrow[dll, rounded corners, "\delta"] & \\
& H_0(L_\bullet) \arrow[r] & H_0(M_\bullet) \arrow[r] & H_0(N_\bullet) \arrow[r] & 0
\end{tikzcd}$$

where each $M_\bullet$ is the complex

$$0 \longrightarrow M_1 \xrightarrow{\mu} M_0 \longrightarrow 0$$

This is the same as our previous note on how all homomorphisms can be represented as a complex.

A popular version of the snake lemma does not assume α_1 is injective and β_0 is surjective. This means the commutative diagram of exact sequences looks like

$$\begin{tikzcd}
 & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] & M_0 \arrow[r, "\beta_0"] & N_0 &
\end{tikzcd}$$

The lemma will then instead state there exists an exact sequence

$$\begin{tikzcd}
\ker\lambda \arrow[r] & \ker\mu \arrow[r] & \ker\nu \arrow[r, "\delta"] & \operatorname{coker}\lambda \arrow[r] & \operatorname{coker}\mu \arrow[r] & \operatorname{coker}\nu .
\end{tikzcd}$$

We will now consider the homomorphism δ, since it is the heart of the proof. Here is the full diagram of the snake lemma.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \ker\lambda \arrow[r] \arrow[d] & \ker\mu \arrow[r] \arrow[d] & \ker\nu \arrow[d] \arrow[ddll, rounded corners, to path={ -- ([xshift=2ex]\tikztostart.east) |- ([yshift=-2ex]\tikztostart.south) -| ([xshift=-4ex]\tikztotarget.west) -- (\tikztotarget)}] & \\
0 \arrow[r] & L_1 \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & M_1 \arrow[r, "\beta_1"] \arrow[d, "\mu"] & N_1 \arrow[r] \arrow[d, "\nu"] & 0 \\
0 \arrow[r] & L_0 \arrow[r, "\alpha_0"] \arrow[d] & M_0 \arrow[r, "\beta_0"] \arrow[d] & N_0 \arrow[r] \arrow[d] & 0 \\
 & \operatorname{coker}\lambda \arrow[r] \arrow[d] & \operatorname{coker}\mu \arrow[r] \arrow[d] & \operatorname{coker}\nu \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

We will show here that elements of $\ker\nu$ can be mapped to $\operatorname{coker}\lambda$ through the diagram, and that the map we obtain is well-defined. We want to choose an element $a \in \ker\nu$, and map it along the diagram as follows:

$$\begin{tikzcd}
 & 0 \arrow[d, dotted, no head] & 0 \arrow[d, dotted, no head] & 0 \arrow[d, dotted, no head] & \\
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & a \arrow[d] & \\
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & c \arrow[r, "\beta_1"] \arrow[d, "\mu"] & b \arrow[r, dotted, no head] \arrow[d, dotted, no head, "\nu"] & 0 \\
0 \arrow[r, dotted, no head] & e \arrow[r, "\alpha_0"] \arrow[d] & d \arrow[r, dotted, no head, "\beta_0"'] \arrow[d, dotted, no head] & * \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
 & f \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Now we will explain the diagram. Let $a \in \ker\nu$. Our goal is to show a can be mapped into $\operatorname{coker}\lambda$. Since $\ker\nu$ can be viewed as a submodule of N_1 under the inclusion map $\ker\nu \hookrightarrow N_1$, we have $a \mapsto b \in N_1$. Since the complexes are exact, we have by proposition 3.23 that β_1 is an epimorphism, and is thus surjective.

There then exists at least one $c \in M_1$ such that $\beta(c) = b$. We will claim for now that we can map $b \mapsto c$ and this operation is in fact well defined; we will show this later. We then very naturally have $\mu(c) = d \in M_0$ simply from the diagram itself. Now note that since the diagram is commutative, every subdiagram must also be commutative by definition. We then have that $(\beta_1 \circ \nu)(c) = \mu(\beta_0(c))$, and thus $\nu(b) = \beta_0(d)$. This is the image on the spot marked $*$. However, recall that b was the inclusion of a in N_1, and $a \in \ker \nu$. Therefore, $\beta_0(d) = 0$, and $d \in \ker \beta_0$. Since rows are exact, $\ker \beta_0 = \operatorname{im} \alpha_0$. Therefore, there exists $e \in L_0$ such that $\alpha_0(e) = d$. We claim once again we can define the map $d \mapsto e$, and will justify later. Finally, let $f \in \operatorname{coker} \lambda$ be the image of the natural projection $\pi : L_0 \to L_0/\lambda(L_1) \cong \operatorname{coker} \lambda$. We then have our map $e \mapsto f$, and we claim that we can define the snaking homomorpism δ such that $a \mapsto f$. To summarize:

- $a \mapsto b$ under the inclusion $\ker \nu \hookrightarrow N_1$;
- Since rows are exact, $\exists c \in M_1$ such that $\beta_1(c) = b$. Map $b \mapsto c$;
- $c \mapsto d$ under μ;
- $d \in \ker \nu$, so since rows are exact, $\exists e \in L_0$ such that $\alpha_0(e) = d$. Map $d \mapsto e$;
- $e \mapsto f$ under the natural projection $\pi : L_0 \to \operatorname{coker} \lambda$.

We need to show our two steps with preimages are valid. The step where we map $d \mapsto e$ is valid because since rows are exact, by proposition 3.23, α_0 is monic, and thus an injection, so $\exists! e \in L_0$ such that $e \mapsto d$ under α_0.

However, we made a choice in mapping $b \mapsto c$, so we must to show that the choice itself does not matter; that is, choosing a different c does not impact the value of $\delta(a)$. This will make δ well-defined. We perform another diagram chase to justify this, because of course. THe relevant part of the above diagram is as follows:

$$\begin{tikzcd}
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head, "\alpha_1"] \arrow[d, dotted, no head, "\lambda"] & c \arrow[r, "\beta_1"] \arrow[d, "\mu"] & b \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
0 \arrow[r, dotted, no head] & e \arrow[r, "\alpha_0"] \arrow[d] & d \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} & \\
 & f \arrow[r, dotted, no head] & {} & &
\end{tikzcd}$$

Suppose we choose a new c' mapping to b:

$$\begin{tikzcd}
0 \arrow[r, dotted, no head] & {} \arrow[r, dotted, no head, "\alpha_1"] & c' \arrow[r, "\beta_1"] & b \arrow[r, dotted, no head] & 0
\end{tikzcd}$$

Since β_1 is an R-module homomorphism,

$$0 = b - b = \beta_1(c') - \beta_1(c) = \beta_1(c' - c).$$

Thus, $c' - c \in \ker \beta_1$. By exactness, $c' - c \in \operatorname{im} \alpha_1$, and thus there exists $g \in L_1$ such that $\alpha_1(g) = c' - c$:

$$\begin{tikzcd}
0 \arrow[r, dotted, no head] & g \arrow[r, "\alpha_1"] & (c' - c) \arrow[r, "\beta_1"] & 0 \arrow[r, dotted, no head] & 0.
\end{tikzcd}$$

Since columns form exact sequences, we have that $\lambda(g) \in \ker \pi$, where π is the canonical map $L_0 \to \operatorname{coker} \lambda$.

Thus, g dies in $\operatorname{coker} \lambda$:

$$
\begin{tikzcd}
0 \arrow[r, dotted, no head] & g \arrow[r, "\alpha_1"] \arrow[d, "\lambda"] & (c' - c) \arrow[r, "\beta_1"] \arrow[d, dotted, no head, "\mu"] & 0 \arrow[r, dotted, no head] \arrow[d, dotted, no head] & 0 \\
0 \arrow[r, dotted, no head] & \lambda(g) \arrow[r, dotted, no head, "\alpha_0"] \arrow[d] & {} \arrow[r, dotted, no head] \arrow[d, dotted, no head] & {} & \\
& 0 \arrow[r, dotted, no head] & {} & &
\end{tikzcd}
$$

Since the diagram commutes, and sicne α_1, α_0 are injective, we have that $\mu(c) = \alpha_0(e)$. We then have that $\mu(c' - c) = (\alpha_0 \lambda)(g)$, and thus

$$\mu(c') = \mu(c' - c) + \mu(c) \mapsto e + \lambda(g).$$

Since $e \mapsto f$ and $e + \lambda(g) \mapsto f + 0$, we have that δ is well-defined.

The rest of the proof is left as an exercise, and I may try it tomorrow. The snake lemma makes various proofs very trivial. For example, consider the following corollary.

Corollary 3.4

In the same situation as the snake lemma, if μ is surjective and ν is injective, then λ is surjective and ν is an isomorphism.

Proof. If μ is surjective, then $\operatorname{coker} \mu = 0$, and if ν is injective, then $\ker \nu = 0$. By the snake lemma, there exists an exact sequence

$$0 \longrightarrow \ker \lambda \longrightarrow \ker \mu \longrightarrow 0 \longrightarrow \operatorname{coker} \lambda \longrightarrow 0 \longrightarrow \operatorname{coker} \nu \longrightarrow 0.$$

Since the sequence is exact, we have $\operatorname{coker} \lambda, \operatorname{coker} \nu = 0$. ■

Chapter 4

Groups, Second Encounter

In this section, we will return to groups in a less general manner. We will focus largely on finite group theory, which is a suprisingly important topic and has applications in important fields such as Galois theory, which we will see in chapter 7.

4.1 The Conjugation Action

4.1.1 Action of Groups on Sets, Reminder

We showed previously that every transitive action of a group G on a set S is isomorphic to the action of left multiplication on the set of left cosets $G/\operatorname{Stab}_G(a)$, for any $a \in S$. Also recall that isomorphisms of group actions are equivariant bijections $\phi : S_1 \to S_2$ such that $\phi(ga) = g\phi(a)$. This fact applies to every set G acts transitively on, and thus for any finite orbit O, we were able to show $|O| = [G : \operatorname{Stab}_G(a)]$, and thus $|O|$ divides $|G|$.